# CMF 3D-QSAR

The continuous molecular fields approach, as a program

Desktop application, library and command-line tool
for the Continuous Molecular Fields approach to 3D-QSAR
User manual

`github.com/igbaskin/cmf-qsar`

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# 1 What this program is

CMF 3D-QSAR is a desktop application, a Python library and a command-line tool implementing the **Continuous Molecular Fields** approach to three-dimensional structure–activity modelling. The method was introduced by Zhokhova, Baskin and co-workers (Dokl. Chem. **429** (2009) 273; J. Comput.-Aided Mol. Des. **27** (2013) 427; continuous indicator fields: J. Comput.-Aided Mol. Des. **29** (2015) 233; co-fields: Future Med. Chem. **11** (2019) 2701), and the program follows those papers closely enough that their published statistics can be reproduced from the same data — `docs/benchmark.md` records that comparison set by set.

## 1.1 The idea in one page

A molecule is represented not by a table of descriptors and not by values on a lattice, but by **continuous functions of space**. Each field is a sum of Gaussians centred on the atoms,

$$X_f(\mathbf{r}) = \sum_l w_{lf} \exp\left(-\frac{\alpha_f}{2} |\mathbf{r} - \mathbf{r}_l|^2\right)$$

where $w_{lf}$ is the weight of atom $l$ in field $f$ — its partial charge for the electrostatic field, its Crippen logP contribution for the hydrophobic field, 1 for an indicator field of a given atom type, or any number the file carries per atom — and $\alpha_f$ is the attenuation factor that sets how far the field reaches.

Two molecules are compared by the **overlap integral** of their fields, which for Gaussians has a closed form:

$$K_f(A,B) = \int X_{fA}(\mathbf{r})\, X_{fB}(\mathbf{r})\, \mathrm{d}\mathbf{r} = \left(\frac{\pi}{\alpha}\right)^{3/2} \sum_l \sum_m w_{lf} w_{mf} \exp\left(-\frac{\alpha}{4} |\mathbf{r}_l - \mathbf{r}_m|^2\right)$$

No grid is ever evaluated for this: the kernel is a sum over atom pairs, computed exactly. The kernels of the individual fields are mixed with coefficients $h_f$ into one kernel, and that kernel goes into a kernel method — ridge regression, a support vector machine, a Gaussian process. Three consequences follow, and they are what distinguishes the approach from CoMFA:

- **Nothing is discretised.** There is no lattice spacing to choose, no alignment of grids, no variable selection among tens of thousands of columns, and no truncation at a cut-off distance.
- **The model is a function of space again.** A kernel ridge model predicts $y(X) = \Sigma_j a_j K(X, X_j) + b$, and because the kernel is an integral of a product, the whole sum can be rewritten as the overlap of the molecule's field with one function, the **co-field** $C_f(\mathbf{r}) = \Sigma_j a_j X_{fj}(\mathbf{r})$. The co-field is the program's map: it plays the role of CoMFA's coefficient contour, but it is the model rather than a projection of it.
- **The decomposition is exact.** Because the prediction is bilinear in the fields, it can be split without approximation into contributions of training molecules, of field types, of regions of space, or of individual atoms.

## 1.2 What you can do with it

| Task | How |
|---|---|
| Build a 3D-QSAR regression model on an aligned congeneric series | Model panel, or `cmf_cli.py build` |
| Work without an alignment | topomer fields, the rotation-invariant autocorrelation kernel, or the pose-free count kernel |
| Classify actives against inactives, or screen with actives alone | binary (SVC, ν-SVC, Laplace GP) and one-class (1-SVM, centroid, k-NN, GP, kernel PCA) tasks |
| See what the model says about space | co-fields as isosurfaces, molecular fields, overlap maps, Gaussian-cube export |
| See what the model says about a molecule | training-molecule contributions, atom contributions, Shapley values and interaction indices |
| Turn a model into design instructions | pharmacophore extraction from co-field lobes, field-difference maps against a reference ligand |
| Use the protein | fetch or load a pocket, draw it behind the fields, score field complementarity, dock |
| Average over conformations | conformational ensembles (4D) and multi-instance learning |
| Map a data set | kernel GTM chemical-space maps with activity, class and density landscapes |

## 1.3 How this manual is arranged

*How a prediction is made* follows one prediction from the atoms to the number, which is the method in full. *Installation and first run* and *The application window* get you running and name every part of the window. *Preparing the data*, *Before you model*, *Building a model* and *Reading the model* are the working cycle: prepare the data, check what it can support, fit a model, read it. *Classification and virtual screening*, *Conformations*, *Working without an alignment* and *Protein pockets* are the specialised routes, and *Three series, worked through* applies the tools to three series from start to finish. Then come the complete *Command-line reference*, the library (*Using it as a library*), *The example data sets*, and *When something goes wrong*, which collects the things that go wrong. The appendices hold the field parameterisation, the file conventions, a glossary of the terms this manual uses in a particular sense, and the bibliography. In the typeset manual every chapter and section carries a number; the numbers are assigned when the PDF is built, so this text refers to chapters by their titles.

Conventions: *Menu → Item* names a menu command, `monospace` names a file, an option or a piece of code. Equations are quoted in the notation of the papers, with their equation numbers.

# 2 How a prediction is made

This chapter follows one number — the activity predicted for one molecule — from the atoms it starts in to the value that appears in the table. Nothing here is a sketch: every formula is the one the program evaluates, and the worked example at the end was computed by running it.

The chain has six links:

1. each molecule becomes a set of **fields**, one per field type, each a sum of Gaussians on its atoms;
2. every pair of molecules gets a **field kernel** per field type, the overlap integral of those two fields, computed in closed form;
3. each field kernel is **normalised** and **scaled**;
4. the field kernels are **mixed** into one kernel matrix;
5. the learner turns that matrix and the activities into **dual coefficients**;
6. a new molecule is pushed through steps 1 to 4 against the training set, and its prediction is the **weighted sum of its kernel values** with those coefficients.

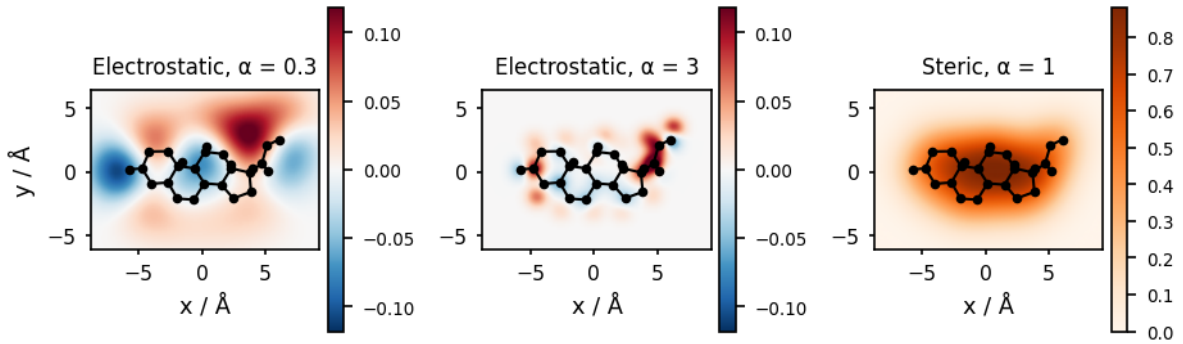
## 2.1 The fields

Atom $l$ of a molecule carries a weight $w_{lf}$ in field $f$ — its partial charge in the electrostatic field, its Crippen logP contribution in the hydrophobic field, an Abraham-type increment in the hydrogen-bond fields, the Tripos van der Waals well depth in the steric field, 1 in an indicator field for atoms of that type and 0 elsewhere, or a number read from the file in a `prop:` field. Appendix A lists them.

The field itself is the sum of Gaussians those weights define,

$$X_f(\mathbf{r}) = \sum_l w_{lf} \exp\left(-\frac{b_{lf}}{2}|\mathbf{r} - \mathbf{r}_l|^2\right)$$

with width $b_{lf} = \alpha_f$ for every field except the steric one, where it is scaled by the atom's van der Waals radius, $b_{lf} = \alpha_f / R_l^2$, so that a large atom spreads its bulk over a correspondingly larger region.




*The electrostatic field of cortisol in the mean plane of the steroid series at α = 0.3 and at α = 3, and its steric field at α = 1; red is positive, blue negative, and the skeleton of the molecule is drawn over the map. A small attenuation spreads the Gaussian of every atom over its neighbours, a large one keeps each atom's contribution to itself. Drawn by `docs/make_figures.py`, as are the other computed figures of this manual.*

There is no grid. The field is never evaluated anywhere in order to make a prediction; it is evaluated only when a map has to be *drawn*.

## 2.2 The overlap integral

Two Gaussians have an overlap integral in closed form, so the inner product of two whole fields is a double sum over atom pairs:

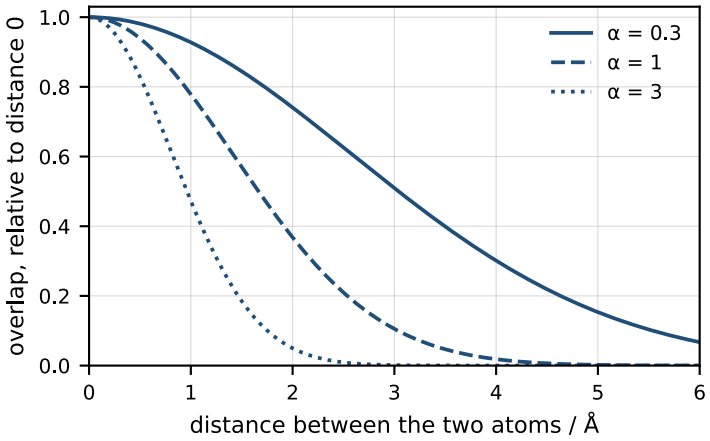
$$K_f(A,B) = \int X_{fA}(\mathbf{r}) X_{fB}(\mathbf{r})\,\mathrm{d}\mathbf{r} = \sum_{l\in A}\sum_{m\in B} w_{lf} w_{mf} \left(\frac{2\pi}{b_{lf}+b_{mf}}\right)^{3/2} \exp\left(-\frac{b_{lf} b_{mf}}{2(b_{lf}+b_{mf})} |\mathbf{r}_l - \mathbf{r}_m|^2\right)$$

When every atom shares one width, $b_{lf} = b_{mf} = \alpha$, this collapses to the form the papers quote,

$$K_f(A,B) = \left(\frac{\pi}{\alpha}\right)^{3/2} \sum_l \sum_m w_{lf} w_{mf} \exp\left(-\frac{\alpha}{4} |\mathbf{r}_l - \mathbf{r}_m|^2\right)$$

and the general form is what the steric field needs, since its widths differ atom by atom. A unit test checks both against numerical quadrature of the fields on a fine grid.

Two things follow immediately. The cost of a kernel is the number of atom pairs, not the number of grid points, which is why a 300-molecule set takes seconds. And the attenuation α is the only thing that decides how far apart two atoms may be and still contribute: large α demands close matching, small α lets distant atoms see each other, and in the limit α → 0 the geometry drops out altogether and the kernel becomes a product of atom counts — which is the alignment-free baseline of *Working without an alignment*.




*The overlap of the Gaussians of two atoms of the same width against the distance between them, relative to its value at distance zero, exp(−αd²/4), for three attenuations. At α = 3 two atoms 2 Å apart no longer see each other; at α = 0.3 their overlap is still 74 % of the maximum.*

## 2.3 Normalisation and scaling

Two corrections stand between the raw integrals and a usable kernel, and they do different jobs.

**Cosine normalisation** (*Normalise kernels*) divides each field kernel by the norms of the two fields,

$$\hat{K}_f(A,B) = \frac{K_f(A,B)}{\sqrt{K_f(A,A)\,K_f(B,B)}}$$

so that every molecule has unit self-similarity in every field -- the normalised kernel of a molecule with itself is 1 by construction -- and a large molecule cannot look important merely for being large.

**Field scaling** (*Scale fields*) then multiplies each field kernel by a constant $c_f$ chosen so that its mean diagonal over the training set is 1. It equalises the fields' contributions before they are mixed, which matters when one field's raw magnitudes dwarf another's — an electrostatic kernel built from partial charges and a steric kernel built from well depths are not naturally on the same scale. With cosine normalisation on, every diagonal is already 1 and $c_f$ = 1; the two options are therefore alternatives in practice, and unticking both gives the raw-kernel convention of the papers.

The scaling constants are computed **on the training set and then stored**: a prediction must not rescale anything according to the molecules being predicted, or two test molecules would be scored on different scales.

## 2.4 Mixing the fields

The joint kernel is a weighted sum over field types,

$$K(A,B) = \sum_f h_f c_f \hat{K}_f(A,B)$$

with the mixing coefficients $h_f$ left at 1 by default, edited by hand in the field table, chosen by coordinate descent on the cross-validated error, or set by centred kernel alignment. Since each normalised field kernel has unit diagonal, the joint self-kernel of any molecule is $\Sigma_f h_f c_f$ — five, for the five default fields with unit coefficients, which is the 5 on the diagonal of the matrix in the worked example below.

## 2.5 Fitting: what is learned from the training set

Kernel ridge regression with a bias term solves one linear system (Eq. 12 of the 2013 paper):

$$(\mathbf{K} + \lambda \mathbf{I})\,\mathbf{a} + b\,\mathbf{1} = \mathbf{y}, \qquad \mathbf{1}^{\mathrm{T}}\mathbf{a} = 0$$

The outcome is one **dual coefficient** $a_j$ per training molecule and one bias $b$. The last row of the system is the constraint $\Sigma_j a_j = 0$, which is what makes the bias meaningful: the coefficients describe only *departures* from a constant level, and $b$ carries the level itself. The ridge parameter λ adds to the diagonal, which keeps the system solvable when molecules are nearly identical — as they are in a congeneric series, where the off-diagonal kernel values run above 0.97 — and controls how closely the model follows the training activities.

The other learners differ only in how **a** and $b$ are obtained, never in what they are: ε-SVR and ν-SVR get them from a quadratic program in which most $a_j$ come out exactly zero, C-SVC and ν-SVC the same for class labels, and the Laplace Gaussian-process classifier from the posterior mode of a latent function. In every case the fitted model is the pair (**a**, $b$) over the training molecules.

## 2.6 Predicting a new molecule

For a molecule the model has never seen, the program:

1. builds its fields **with the same field set** — the same α, the same charge model, the same parameterisation, the same atom typing;

2 computes the per-field overlap integrals between its atoms and the atoms of every training molecule, giving one row $K_f(x, j)$ per field;

3 normalises that row with the molecule's own self-kernels and the **stored** training self-kernels, and multiplies by the **stored** scales $c_f$;

4 mixes those normalised, scaled fields into one row $k(x)_j$ with the mixing coefficients $h_f$, which is the joint kernel $K(x, j)$ of *Mixing the fields* above;

5 returns

$$\hat{y}(x) = \sum_j a_j\, k(x)_j + b = \mathbf{k}(x)^{\mathrm{T}}\mathbf{a} + b$$

That is the whole prediction: a similarity-weighted combination of the training activities, where "similarity" is the overlap of molecular fields and the weights were fixed when the model was fitted. Molecules with $a_j = 0$ — most of them, after an SVM fit — contribute nothing, which is why the *Contributions* tab can list the handful that actually make a given prediction.

## 2.7 The same prediction, as a field

Because the kernel is an integral of a product and the prediction is linear in it, the sum over training molecules can be moved inside the integral. Writing $s_j = 1/\sqrt{(K_f(j,j))}$ for the normalisation of training molecule $j$, define the **co-field**

$$C_f(\mathbf{r}) = \sum_j a_j\, s_j X_{fj}(\mathbf{r})$$

and the prediction becomes

$$\hat{y}(x) = \sum_f \frac{h_f c_f}{\sqrt{K_f(x,x)}} \int X_{fx}(\mathbf{r})\, C_f(\mathbf{r})\, \mathrm{d}\mathbf{r} + b$$

The model is therefore a *function of space*, one per field type, and a molecule is scored by how much of its own field lies in the positive regions of that function and how much in the negative ones. This is the identity that makes the pictures of *Reading the model* the model rather than an illustration of it, and the division by the molecule's own field norm is the only thing the normalisation adds to the picture. It also explains what co-fields cannot survive: a nonlinear kernel layer, which is not an integral of a product, and the autocorrelation kernel source, which has no frame to draw in.

## 2.8 A worked example

Four molecules from `examples/benzamides_synthetic.sdf` as the training set, the five default fields, α = 0.5, cosine normalisation, λ = 0.1, and the fifth molecule as the one to predict. The numbers below are the program's own, printed from the objects it fits.

The joint kernel of the training set, the sum of the five normalised field kernels, so the diagonal is 5:

| | parent | 4F | 4'F | 4Cl | y |
|---|---|---|---|---|---|
| parent | 5.0000 | 2.8362 | 3.2195 | 2.5368 | 5.585 |
| 4F | 2.8362 | 5.0000 | 2.3310 | 3.9647 | 5.230 |
| 4'F | 3.2195 | 2.3310 | 5.0000 | 2.8882 | 6.471 |
| 4Cl | 2.5368 | 3.9647 | 2.8882 | 5.0000 | 5.355 |

Solving the bordered system gives

```
a = [-0.1358, -0.0581, +0.3487, -0.1549]      sum(a) = 0      b = 5.7124
```

The test molecule 4'Cl is then compared with each training molecule, field by field:

```
                        parent      4F      4'F     4Cl
electrostatic           0.0878  0.6028  0.1009  0.3915
steric                  0.9793  0.9824  0.9863  0.9579
hydrophobic             0.8936  0.9412  0.9443  0.8437
H-bond acceptor         0.2600  0.9895  0.1927  0.6539
H-bond donor            0.5815  1.0000  0.4424  0.7379
                        ------  ------  ------  ------
k(x)  (their sum)       2.8022  4.5159  2.6667  3.5849
```

and the prediction is that row times the coefficients, plus the bias:

```
y(x) = 2.8022(-0.1358) + 4.5159(-0.0581) + 2.6667(0.3487) + 3.5849(-0.1549) + 5.7124
     = 5.4444
```

which is exactly what *Pred (test)* shows for that molecule. Two things can be checked directly in this example rather than taken on trust. The steric and hydrophobic rows are nearly constant across the training set — these molecules differ by one halogen — so almost all the discrimination comes from the electrostatic and hydrogen-bond fields, which is visible in the spread of their rows. And the prediction, 5.44 against an observed 7.21, is poor, because four training molecules and no hyper-parameter search is not a model: the arithmetic is the lesson here, not the accuracy.

## 2.9 What changes for the other tasks

| Model | What the prediction is |
|---|---|
| Kernel ridge | $\hat{y} = \mathbf{k}^{\mathrm{T}}\mathbf{a} + b$, as above |
| Gaussian process | the same mean, plus the variance $\sigma^2(x) = K(x,x) - \mathbf{k}^{\mathrm{T}}(\mathbf{K}+\lambda\mathbf{I})^{-1}\mathbf{k}$, which the *Align by prediction confidence* tool minimises |
| ε-SVR, ν-SVR | the same formula; most $a_j$ are zero |
| C-SVC, ν-SVC | the same formula, read as a decision value: its sign is the class, its magnitude the margin |
| Laplace GP classifier | the same formula gives the latent mean, which is squashed to a probability and averaged over the latent variance |
| One-class methods | a score built from the same kernel row, but also from the molecule's own self-kernel: distance to the centroid, to the k-th nearest active, to the principal subspace, or the 1-SVM decision value |
| Nonlinear kernel layer | the joint kernel is passed through $\exp(-\gamma\, d^2)$ or an arc-cosine function first; the prediction formula is unchanged, the co-field identity is lost |
| Autocorrelation source | the kernel row comes from the rotation-invariant distance profiles instead of the overlap integrals; everything downstream is identical |

| Model | What the prediction is |
|---|---|
| Surface source | for molecules over a surface at $z = 0$: the in-plane distance profiles resolved by the heights of both atoms, invariant in the plane and to rotation about the normal only; heights and tilts are kept, no in-plane alignment is needed (command line: `--kernel surface --surface-z ZMIN ZMAX ZSTEP`) |
| Periodic source | for crystals: the distance profiles summed over the neighbour list of the cell with its images, per atom, background removed, cutoff tapered; invariant to the choice of cell, origin and orientation (command line: `build-crystals`) |
| Two endpoints | the same formula once per endpoint, with coefficients tied through a task correlation; the difference of the two coefficient vectors gives the selectivity co-field |

## 2.10 What has to match between fitting and predicting

Because the kernel row is computed against the stored training atoms with the stored constants, a prediction is only meaningful if the new molecule was prepared the way the training molecules were. The saved `.cmf` model therefore carries the field set, the α values, the charge model, the parameterisation, the normalisation and scaling flags, the mixing coefficients, the training atoms and the coefficients — everything in steps 1 to 5 — and reuses them. It also carries a record of how the training molecules were prepared: the version of the program that wrote it, the date, how many molecules, where their partial charges came from (the file, or computed), whether they carried explicit hydrogens, and the box their atoms occupy. Before any prediction the program compares the new molecules with that record and reports what does not match:

- **The chemistry behind the weights.** Charges read from a file for the training set and Gasteiger charges for the new molecules are two different electrostatic fields under one name. A model fitted on file charges therefore **refuses** a molecule that carries none, rather than falling back to Gasteiger charges silently; `cmf_cli.py predict --allow-mismatch` overrides the refusal, and the application asks. Supply the charges (a MOL2 charge column or a `partial_charge` atom property) or build the model with computed charges.
- **Explicit hydrogens.** The donor field sits on the hydrogens and the steric field counts them, so a set with hydrogens and a set without are two different sets of fields. A mismatch is a **warning**.
- **The frame.** The overlap integral is not invariant to rotation or translation, so a molecule predicted in a different frame from the training set scores low on every field and is predicted towards the mean. A molecule whose centroid lies more than 4 Å outside the box of the training atoms is reported as a **warning**.
- **The pose inside the frame.** A molecule can be in the right place and still be turned the wrong way, and a rotation about its own centre moves no centroid, so the box test above sees nothing. The check that does see it compares the molecule with the *shape* the training set occupies rather than with its box: the share of its atoms that come within 1.5 Å of some training atom, against the share this training set asks of its own molecules — the fifth percentile of the same quantity measured on each training molecule against the others, less a small slack, so that a series of close analogues gets a strict rule, exactly as the applicability domain's threshold does. Measured on the eight Sutherland sets, turning every held-out molecule 90° about its own centre is caught **95 % of the time, for a false-positive rate of 2.6 %** on the molecules left as they were. If the new molecules are not in the frame, align them (*Preparing the data*) or use a frame-free kernel source (*Working without an alignment*).

- **Per-atom property fields** that are zero on a molecule, because it carries no value of that property, are a warning; a file written by another version of the program, or by a version that recorded no provenance, is a note.

The findings appear in the log of the application and on the console of the command-line tool, each marked `[refuse]`, `[warn]` or `[note]`. A model file written by a newer version of the program than the one reading it is not opened.

# 3 Installation and first run

## 3.1 Where the program lives

The repository is github.com/igbaskin/cmf-qsar — private, so the link asks for a GitHub account with access:

```
git clone https://github.com/igbaskin/cmf-qsar.git
cd cmf-qsar
```

Everything this manual describes is in that checkout: the `cmfqsar` package with the command-line tool inside it (`cmfqsar/cli.py`), the two launchers (`cmf_app.py`, `cmf_cli.py`), the example data under `examples/`, the benchmark scripts under `benchmarks/` with their recorded results, the tests, and this manual with the script that typesets it.

## 3.2 Requirements

Python 3.10 or newer. The package is declared in `pyproject.toml`, so from the checkout

```
pip install -e .[gui]
```

installs the library with RDKit (chemistry), scikit-image (marching cubes for the isosurfaces), scikit-learn (the SVM solvers), NumPy, SciPy and joblib, and with the `gui` extra PySide6 (the interface) and pyqtgraph with PyOpenGL (the 3D view); it also puts the commands `cmf-app` and `cmf` on the path. The library alone, `pip install -e .`, needs no Qt. The program is pure Python and needs no compilation step; it has been used on Windows, Linux and macOS.

Further extras, only needed for particular work:

- `docs` — reportlab and matplotlib (which supplies the DejaVu fonts) to typeset this manual;
- `dev` — pytest, for the test suite;
- `examples` — `dimorphite_dl`, protonation states when importing SMILES for a given pH;
- a local ORCA or VASP installation if you intend to compute per-atom quantum properties to use as fields (see *Preparing the data* and appendix A); these are programs, not packages.

`pip install -r requirements.txt` installs the working checkout, `.[gui,docs,dev]`. `requirements-lock.txt` pins the exact versions the benchmark tables were produced with: `pip install -r requirements-lock.txt` and then `pip install --no-deps -e .` reproduces that environment. `CITATION.cff` says how to cite the software and the papers.

## 3.3 Starting the application

```
cmf-app
```

or `python cmf_app.py` from the checkout, which takes the same arguments; or directly with a data set, which skips two dialogs:

```
cmf-app examples/benzamides_synthetic.sdf --activity pActivity --test-fraction 0.25
```

The command-line tool is `cmf` once installed and `python cmf_cli.py` from the checkout; the rest of this manual writes the latter, and the two are the same program.

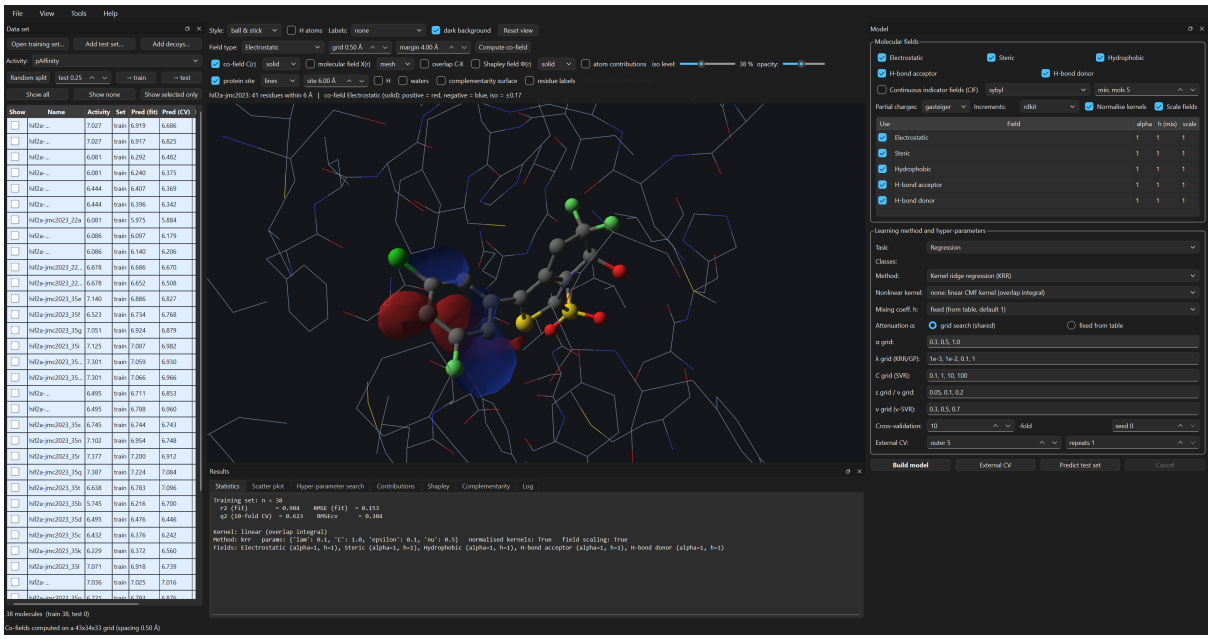
If the 3D view stays black, the OpenGL driver is the usual cause; see *When something goes wrong*.

## 3.4 A first model in five minutes

1. *File → Open training set…* and choose `examples/benzamides_synthetic.sdf`. It holds 109 aligned N-phenylbenzamides with a synthetic activity, so you know the answer the model should find. Pick `pActivity` as the activity property.
2. In the data-set table, press *Random split* and set a quarter of the molecules aside as a test set.
3. In the Model panel leave the five physico-chemical fields ticked, leave the method at kernel ridge regression, and press **Build model**.
4. Read the *Statistics* tab: $q^2$ should be about 0.87 and $R^2_{pred}$ about 0.84 — the noise ceiling of that set is 0.94, so the model has recovered nearly all of the signal that was put into it.
5. Press **Compute co-field**, choose *Electrostatic*, and move the *iso level* slider. The red and blue lobes that appear are the pseudo-receptor zones the activity was generated from.

Nothing in that sequence is specific to the example: it is the working cycle of the program.

# 4 The application window



*The HIF-2α series of the Merck free-energy benchmark, 38 ligands: a kernel ridge model with $q^2 = 0.62$, its electrostatic co-field drawn as ± isosurfaces, and the binding site of the reference crystal structure, which the data set brings with it and which needs no superposition.*

The window is a 3D view with three dockable panels around it. The **3D view** and its control strip occupy the centre and are always there; the **Data set** panel docks on the left, the **Model** panel on the right inside a scroll area, and the **Results** panel across the bottom under the view. All three are ordinary dock widgets: drag one by its title bar to another edge, drop it on another to stack them as tabs, or pull it off entirely to float it on a second monitor. The layout is remembered between sessions, and *View → Restore default layout* puts it back when an experiment goes wrong.

At the bottom, the **status bar** shows what the program is doing, with a progress bar on the right during a long task. Every long task — a grid search, a co-field, an alignment, a conformer generation — runs in a worker thread, so the window stays responsive and the *Cancel* button in the model panel stops the work rather than the application.

## 4.1 The Data set panel

Three buttons at the top load structures: **Open training set…**, **Add test set…** and **Add decoys…**, the same actions as the first three entries of the *File* menu. Beside them the *Activity* selector names the SD property the activities are read from; changing it re-reads the column without reloading the file.

The table has one row per molecule and seven columns:

| Column | Contents |
|---|---|
| Show | the tick that decides whether this molecule is drawn in the 3D view |
| Name | the molecule's title from the file |
| Activity | the experimental value, from the chosen property |

| Column | Contents |
|---|---|
| Set | train, test or decoy |
| Pred (fit) | the model's prediction for the molecule it was fitted on |
| Pred (CV) | its prediction when it was in the held-out fold |
| Pred (test) | its prediction as a test molecule |
| AD | its similarity to the training set, when the applicability domain was computed |

An eighth column, **Chemotype**, appears once the chemotypes have been found. Rows are selected singly or in blocks, and the selection drives everything else: the co-field overlap, the atom contributions, the Shapley decomposition and the complementarity numbers all refer to the molecule currently selected.

**Sorting.** Clicking a column header sorts by it, a second click reverses the order, and a third restores the order of the file. Molecules with no value in that column are always placed at the bottom, whichever direction is in force, so an unpredicted molecule never hides among the predictions. The *Show* ticks, the selection and the 3D view follow the molecule and not the row, so sorting never disturbs what is on screen.

**Splitting.** *Random split* assigns a fraction to the test set; the → *train* and → *test* buttons move the selected rows by hand. A `subset`, `set` or `split` tag in the file is honoured on loading, which is how the published splits of the example sets arrive without any clicking.

**Showing.** *Show all*, *Show none* and *Show selected only* control the view in bulk. With one molecule shown, the maps in the viewer refer to it; with all of them shown, the co-field sits in the middle of the series as it should.

A line under the table reports the state of the data set — how many molecules, how many in each subset, which activity property — and a note in brackets records any transformation that has been applied, for instance *[topomer substituents of …]*, so that a changed data set never passes for the original.

## 4.2 The Model panel

Everything that defines a model, in the order one normally works down it. The panel scrolls; on a small screen nothing is hidden, it is merely below.

**Molecular fields.** Five tick boxes for the physico-chemical fields — Electrostatic, Steric, Hydrophobic, H-bond acceptor, H-bond donor — then *Continuous indicator fields (CIF)* with its typing scheme (`sybyl`, `element`, `element+aromatic`) and the minimum number of training molecules an atom type must occur in to be given a field. Below them, *Per-atom property fields* lists the per-atom properties the loaded file carries, each of which can be turned into a field of its own.

*Partial charges* chooses between Gasteiger, MMFF94 and the charges stored in the file; *Increments* chooses the parameterisation (`rdkit`, `tripos`, `tripos-legacy`). Two tick boxes set the kernel conventions: *Normalise kernels* for cosine-normalised field kernels, *Scale fields* to rescale every field kernel to unit mean diagonal before mixing — untick it for the raw-kernel convention of the papers.

*Conformations* switches between one conformation per molecule and an ensemble, with the weighting (uniform or Boltzmann), kT, a cap on the number of conformations, and the multi-instance settings (the softness β, and *one conformation* for the hard selection). A label beside

them reports how many conformations the data set currently holds.

A table at the foot of the group lists the fields that will actually be built, with the columns **Use**, **Field**, **alpha**, **h (mix)** and **scale**: per-field attenuation and mixing coefficients are editable there, which is how a field is given more or less weight than the others without leaving the panel.

**Learning method and hyper-parameters.** A form, row by row:

| Row | What it sets |
|---|---|
| Task | regression, binary classification, or one-class (virtual screening) |
| Classes | for classification: a threshold on the activity, with *active below* for data where small is potent, or a class property from the file |
| Method | the learner; the list changes with the task — KRR, GP, ε-SVR, ν-SVR for regression; C-SVC, ν-SVC, Laplace GP for classification; 1-SVM, centroid, k-NN, one-class GP and kernel PCA for one-class work |
| Nonlinear kernel | none (the linear CMF kernel), a Gaussian layer, or one or two arc-cosine layers |
| Mixing coeff. h | fixed from the table, optimised by coordinate descent, or set by centred kernel alignment |
| Attenuation α | a single value, or the grid below |
| Kernel from | overlap integrals of the fields, which need one frame; the rotation-invariant autocorrelation, which does not; or the surface autocorrelation for poses over a surface at $z = 0$, which needs only the normal (command line) |
| α grid, λ grid, C grid, ε/ν grids | the search ranges, as comma-separated lists |
| Cross-validation | the number of folds and the shuffling seed |
| External CV | the outer folds and the number of repeats for the nested procedure |
| Probe fields | GRID-like fields for the water, methyl, carbonyl, amine and cation probes, and the number of Gaussians per atom |
| Computed descriptors | molecular weight, clogP, TPSA and rotatable bonds, computed by RDKit; all off by default |
| Descriptors (per molecule) | which numeric molecule properties enter as non-field descriptors, and the block's mixing coefficient |
| Validation | *shuffles* for y-randomisation, *bootstrap* for confidence intervals, and the *applicability domain* tick — all three off by default, all three run as part of **Build model** rather than from a button of their own |

Four buttons close the panel: **Build model** runs the grid search, the final fit, the internal cross-validation and the test prediction; **External CV** runs the nested procedure and reports $q^2ex$; **Predict test set** applies the current model to the test molecules; **Cancel** stops whatever is running.

## 4.3 The 3D view

The view itself is an OpenGL scene: drag to orbit, the wheel to zoom, middle-drag (or shift-drag) to pan. *Reset view* reframes the whole scene, *Tools → Center view on current molecule* reframes the selected one, and *File → Save 3D view as image…* writes what you see to PNG.

The control strip above the view is four rows, and everything on it redraws immediately:

- **Molecules.** Drawing style (ball & stick, sticks, lines, space fill), *H atoms*, *Labels* (none, element + number, atom type, partial charge), *dark background*, *Reset view*.
- **Maps.** *Field type* chooses which field the maps refer to; **Compute co-field** evaluates the co-fields on a grid around the training set; then the maps themselves — *co-field C(r)*, *molecular field X(r)*, *overlap C·X*, *Shapley field Φ(r)*, *atom contributions*, *difference map*, *pharmacophore* — each with its own solid-or-mesh style where that applies. After a two-endpoint fit an *Endpoint* box joins the row and chooses whose co-field the maps show: either endpoint, or their difference, the selectivity co-field.
- **Levels.** *iso level*, the isosurface threshold as a percentage of the largest absolute value rather than an absolute number, so the surfaces stay comparable when the field changes; and *opacity*.
- **Protein site.** *protein site* draws the residues within the chosen distance of the molecules on screen, as lines or sticks, with *H*, *waters* and residue labels optional, and *complementarity surface* paints the molecule's surface by how well it complements the pocket.

Co-field colours are fixed, so that maps made in different sessions can be compared at a glance:

| Field | Positive lobe | Negative lobe |
|---|---|---|
| Electrostatic | red | blue |
| Steric | green | yellow |
| Hydrophobic | orange | light blue |
| H-bond acceptor | magenta | dark red |
| H-bond donor | cyan | purple |
| Indicator field | red: an atom of that type is favoured here | blue: it is disfavoured here |

## 4.4 The Results panel

Twelve tabs, each filled by the work that produces it:

| Tab | What it holds |
|---|---|
| Statistics | every figure of merit, for the fit, the cross-validation, the test set and the external CV; the y-randomisation, bootstrap and applicability-domain results follow at the bottom, under a *Validation* heading, when they were asked for |
| Scatter plot | observed against predicted, by subset, with the molecule under the cursor named |
| ROC | the receiver-operating curves of a classification model, per subset |
| Scores | the ranked decision values or calibrated probabilities: rank, molecule, score, set, class |
| Enrichment | the enrichment curve against a decoy set, with EF at 0.5, 1, 5 and 10 %, BEDROC and RIE |
| Map | the kernel-GTM chemical-space map, with the activity, class and density landscapes, and a control for how the molecules themselves are drawn over it |
| Hyper-parameter search | the whole grid — α, the other parameters, RMSEcv and $q^2$ — so that an optimum at the edge of the grid is visible rather than hidden |

| Tab | What it holds |
|---|---|
| Contributions | two tables for the selected molecule: the training molecules that make its prediction, with `a_j` and `a_j K(X, X_j)`, and its own atoms with their contributions |
| Shapley | the Shapley decomposition, over atoms, substituents of the common scaffold or field types, with second-order interactions |
| Complementarity | the field complementarity of the molecules and of the co-field with the pocket, by sign agreement, cosine or correlation |
| Chemotypes | the Murcko scaffolds found in the set, their sizes, and the leave-one-chemotype-out result beside the alignment-free baseline |
| Log | everything the program did, with timings and warnings — including the names of the molecules that fell outside the applicability domain — the first place to look when a result surprises you |

## 4.5 The menus

**File** — *Open training set…* (Ctrl+O), *Add test set…*, *Add decoys (background set)…*, *Import activities from CSV…*; *Load protein structure (PDB/mmCIF)…*, *Fetch protein from the PDB…*, *Forget protein structure*; *Load model…* (Ctrl+L), *Save model…* (Ctrl+S); *Export predictions (CSV)…*, *Export aligned data set (SDF)…*, *Export current co-field (Gaussian cube)…*, *Export current Shapley field (Gaussian cube)…*, *Save 3D view as image…*; *Quit* (Ctrl+Q).

**View** — show or hide each panel, *Results panel spans full width* for when the tables and plots matter more than the view, and *Restore default layout*.

**Tools** — *Generate conformations (4D)…*; *Dock into the pocket…*; *Align data set to reference…*; *Align by prediction confidence (GP variance)…* and its cross-validated form for the training set; *Replace molecules by their topomer substituents…*; *Field difference from a reference molecule…*; *Extract pharmacophore from the co-field…*; *Model several endpoints (selectivity co-field)…*; *Chemical space map…*; *Align decoys to the training set…*; *Unalign (randomise poses)…*; *Center view on current molecule*; *Quantum-chemical properties with ORCA…* and *Save the data set with its properties (SDF)…*; and the submenu *Heterogeneous sets (several chemotypes)* with its four numbered steps.

**Help** — *About CMF 3D-QSAR / method summary*, which names the equations of the papers that each part of the program implements.

Every *Tools* entry opens a dialog that states what it is about to do and on which molecules, and each one runs in the background with the progress bar and the *Cancel* button live.

# 5 Preparing the data

## 5.1 What the program reads

| Format | Notes |
|---|---|
| `.sdf`, `.sd`, `.mol` | 3D coordinates used as they are; hydrogens are added with coordinates if missing |
| `.mol2` | Sybyl atom types and partial charges are read and can be used directly; a tolerant fallback reader handles blocks RDKit refuses |
| `.smi`, `.csv`, `.tsv`, `.txt` | 2D input: each molecule is embedded with ETKDG and minimised with MMFF94, which means **the poses are arbitrary** until you align them |
| `.pdb`, `.mmcif` | protein structures only, read by a deliberately permissive reader |

A CSV needs a column of SMILES; a column called `name` or the first non-SMILES column names the molecules, and every other column is kept as a molecule property, so activities and metadata travel with the file.

## 5.2 Activities

Three ways, in order of convenience:

1. **An SD property.** The loader guesses the activity property and the dialog lets you change it. Anything numeric will do: `pIC50`, `pKi`, `pActivity`, `CBG_pK`.
2. **A separate table** (*File → Import activities from CSV…*, `--activity-file`): molecule name, activity, and optionally a third column with the split (1 = train, 2 = test, or the words).
3. **A class property** for classification tasks, or a threshold on the activity.

A per-molecule tag named `subset`, `set` or `split` is honoured automatically, which is how the published splits of the example sets arrive.

## 5.3 Alignment

CMF compares fields in one frame of reference. If your molecules came from a superposition, from a crystallographic series or from a docking run, that frame already exists and you should leave it alone. If they came from SMILES, they have no frame at all and you must make one — or use one of the frame-free routes of *Working without an alignment*, which is often the better answer.

The tools, in increasing order of sophistication:

- *Tools → Align data set to reference…* — rigid superposition on the maximum common substructure with a reference molecule, optionally refined by maximising the CMF kernel similarity, which is the alignment criterion proposed in the 2013 paper.
- *Tools → Align by prediction confidence…* — minimises the Gaussian-process posterior variance $\sigma^2(x) = K(x,x) - \mathbf{k}(x)^{\mathrm{T}}(\mathbf{K} + \lambda\mathbf{I})^{-1}\mathbf{k}(x)$ of a molecule under an existing model, which moves it into the model's applicability domain. It uses no activity value, so it is valid for prospective prediction. The program refuses to move training molecules, because the optimum there is degenerate: superimposing everything maximises mutual similarity and destroys the signal.

- *Tools → Align decoys to the training set…* — the same idea for a background set, so that no decoy is rejected merely for having been posed badly.
- *Tools → Unalign (randomise poses)…* — the control experiment. It gives every molecule a random rigid pose, which leaves internal geometry and self-kernels untouched and scrambles only the mutual arrangement. Randomise, realign, and see how much comes back; the program remembers the poses from before and reports the recovery.

## 5.4 Judging an alignment, and aligning flexibly

Everything in this manual says the alignment decides more than the kernel, the learner or the fields. What it could not say until recently is whether one alignment method is better than another, because the obvious measure cannot tell you: realigning changes the geometry $q^2$ is computed on, so a method is rewarded for making the data look like the model, and $q^2$ and the external $R^2$pred have been seen to move in opposite directions under exactly that operation.

The criterion has to come from outside the model, and the FEP benchmark sets supply one: every ligand of `examples/openff` is co-crystallised in the same frame, so the pose a method *should* have found is known. `benchmarks/run_poses.py` fits a model on training molecules in their crystal poses, destroys the poses of the held-out ones, and asks each method to recover them without ever seeing a held-out activity. Two numbers come back for every method — how near the crystal pose it got, measured over heavy atoms, symmetry-aware and **without superposition**, and what the model then predicts — and both are reported between two fixed references: the crystal poses as the ceiling and the scrambled poses as the floor.

That last detail is what makes the protocol work. Superposing before measuring would report 0.00 Å for a molecule placed in the wrong site: on a ligand displaced by 1 Å, `CalcRMS` returns 1.0 and `GetBestRMS` returns 0.0.

### What the methods are worth

Measured this way over sixteen FEP sets, with the held-out **conformations regenerated** rather than read from the crystal — which is the case a new molecule actually presents:

| method | median RMSD | within 2 Å | $R^2$pred | conformer limit |
|---|---|---|---|---|
| crystal poses (the ceiling) | 0.00 Å | 100 % | +0.334 | 0.00 Å |
| scrambled (the floor) | 6.12 Å | 1 % | −0.696 | 1.73 Å |
| substructure (MCS) | 1.94 Å | 54 % | −0.391 | 1.73 Å |
| MCS + rigid confidence | 2.27 Å | 45 % | −0.396 | 1.73 Å |
| MCS + flexible confidence | 1.41 Å | 74 % | +0.065 | 1.20 Å |
| MCS + flexible, 5 conformer starts | **0.86 Å** | **88 %** | **+0.264** | **0.71 Å** |

Three things follow. **Freeing the torsions makes the confidence criterion work**: with a conformation to fix, the rigid search of the 2019 paper is worse than not refining at all, which is the failure the protocol was built to expose; the same objective with the rotatable bonds free is the best method available. The **conformer generator is what makes it work** — five generated starting conformations improve the median pose by 0.55 Å, better on 15 of 16 sets, and the combination beats plain substructure alignment on 16 of 16. And the **displacement guard has little left to prevent**: with these starts no set is worse than 1.52 Å, a 1 Å guard costs 0.6 Å, and 2 Å is

indistinguishable from none.

What none of it does is replace knowing the conformation: +0.264 against the crystal poses' +0.334. The conformational residue is 0.71 Å, and that is where the remaining error comes from.

## Inside the pocket

Both criteria above know only half of the problem. The confidence criterion knows the series and not the receptor, so nothing stops it putting a molecule where no molecule could go; the docking score of `cmfqsar/docking.py` knows the receptor and not the series, so nothing makes the pose it finds comparable with the poses the model was fitted on. The item that joins them reads the complementarity as a **prior** over poses and the confidence as the **likelihood**, and takes the pose that maximises the product:

$$\min_{p} \sigma^2(p) - w\,[\,\mathrm{complementarity}(p) - \mathrm{clash}(p)\,]$$

over the six rigid-body parameters $p$, with the same displacement guard. At $w = 0$ this is the confidence alignment exactly; as $w$ grows the receptor takes over.

```
cmf predict model.cmf new.sdf --align-variance --align-pocket complex.pdb --pocket-weight 1
```

Judged by the protocol above on 4 FEP sets with a receptor structure (mcl1, p38, thrombin, tnks2), the held-out poses scrambled and recovered:

| method | median RMSD | within 2 Å | $R^2_{pred}$ |
|---|---|---|---|
| crystal poses (the ceiling) | 0.00 Å | 100% | +0.335 |
| scrambled (the floor) | 5.03 Å | 0% | -0.965 |
| substructure (MCS) | 0.16 Å | 100% | +0.229 |
| confidence alone | 0.30 Å | 59% | -1.166 |
| MCS + confidence | 0.10 Å | 100% | +0.286 |
| pocket alone | 3.93 Å | 31% | -0.805 |
| pocket + confidence | 2.50 Å | 46% | -0.831 |
| MCS + pocket + confidence | 0.13 Å | 100% | +0.265 |

Per set, the comparison that carries the answer — the receptor added to the confidence criterion, with and without a substructure frame to start from:

| set | confidence alone | pocket + confidence | MCS + confidence | MCS + pocket + confidence |
|---|---|---|---|---|
| mcl1 | 0.29 Å | 0.31 Å | 0.17 Å | 0.22 Å |
| p38 | 0.32 Å | 3.73 Å | 0.09 Å | 0.13 Å |
| thrombin | 5.00 Å | 7.58 Å | 0.10 Å | 0.14 Å |
| tnks2 | 0.21 Å | 1.26 Å | 0.07 Å | 0.09 Å |

**The receptor term is not worth its cost.** From a scrambled start the confidence criterion alone finds the pose on three of the four sets, and adding the complementarity makes it worse on two of them and better on none. After a substructure frame every refinement ends within a tenth of an

ångström of the others, and the predictions differ by less than the sets differ among themselves. The joint objective also costs about ten times as much, because the pocket term has no analytic gradient and the search falls back on Powell.

**The problem is the pocket term, not the search method.** At weight zero the joint aligner is the confidence objective optimised by a different method, and it recovers the pose as well as the gradient search does (0.29 Å on p38 against 0.32 Å). Turning the weight up from there is what makes it fail:

| pocket weight | p38 | tnks2 |
|---|---|---|
| 0 (the confidence criterion alone) | 0.29 Å | 0.07 Å |
| 0.25 | 3.52 Å | 0.11 Å |
| 1 | 3.73 Å | 1.26 Å |
| 4 | 3.63 Å | 1.32 Å |

On p38 any amount of the receptor destroys the recovery; on tnks2 it degrades it mildly. The reason is visible in what complementarity is: a weak prior over six dimensions. A ligand that fits a site fits it in many placements, so the term adds a broad, multi-modal surface to an objective that was already multi-modal, and its optimum is not where the crystal pose is — not with a scoring function that has no desolvation, no entropy and no torsional search.

The proper use of what is implemented is therefore narrow. `--align-pocket` is for a pose that must be defensible **to the receptor** as well as to the model: it will not improve on the confidence criterion, and it will tell you when the confident pose is one the protein would not allow.

Nothing here is a docking program: there is no desolvation, no entropy and no torsional search, and the internal geometry is never touched. What the receptor adds is a term that knows where the protein atoms are, and the measurement says where that helps and where it does not.

### Using it

In the application, *Tools → Align by prediction confidence…* gained a **turn the rotatable bonds as well** tick box, with the number of starting conformations and the strain allowance beside it. From the command line:

```
cmf predict model.cmf new.sdf --align-variance --align-flexible --align-conformers 5
```

The force field acts as a prior on the conformation — it may rise ten kcal/mol above where it started before being penalised, one-sided so that the aligner does not quietly become a conformer optimiser — and the displacement limit now applies to the **rigid motion only**, since a turning bond is not a translation and counting it as one would suppress exactly what this option adds.

## 5.5 Charges and per-atom properties

The electrostatic field needs charges: Gasteiger (the default), MMFF94, or **read from the file**, which is what you want with the Sutherland sets and their MNDO ESP-fit charges, or with anything you computed yourself.

Beyond that, any per-atom number can become a field of its own. `prop:<name>` reads the atomic weights from a property of the molecule — a quantum charge, a Fukui index, a local softness, an atomic polarisability, even a per-atom embedding — and that field then behaves exactly like a

built-in one, with its own attenuation, its own mixing coefficient, its own co-field, atom contributions and Shapley values.

Two conventions are read, in this order:

1. a property set on **every atom**, which RDKit writes to SDF as `atom.dprop.<name>` and reads back automatically;
2. a **molecule property holding one number per atom**, separated by spaces, commas or newlines, which is how most quantum-chemistry packages dump such a table.

The hydrogens are the usual source of error. A file without 3D coordinates is embedded on loading and hydrogens are added, so a list written for the heavy atoms alone no longer matches the atom count. The program says so rather than silently mis-assigning: `--prop-missing error` (the default) stops, `--prop-missing zero` gives the unlisted atoms zero weight. The next section shows the whole route from an ORCA run to fields and descriptors, from inside the program or from the command line.

## A number that belongs to an atom, not to the molecule

A field is built from a property every atom has. A **target** of that shape is a different object, and the program now carries one (plan item LP1): a carbon shift, an acidity, a local reactivity or a site's energy is a number per atom, and a table of them differs from a per-atom field in the way that decides everything downstream -- **most atoms have no value at all**, and the atoms that do are not independent samples of each other, because they share a molecule.

`cmfqsar/atomtargets.py` holds that data model, and nothing more: there is no model over atoms yet, and the item that adds one needs this first.

- **Reading keeps the blanks.** `read_atom_target` gives one number per atom and *not a number* where the atom has none. The two conventions of the previous section are read, with a value on the atoms themselves taking precedence, and in a molecule property a blank field between two commas is an atom that was not measured -- `1.2, , 3.4` is three atoms of which the middle one has nothing. `n/a`, `-`, `none` and an empty field all mean the same. A list as long as the heavy atoms is spread over them, leaving the hydrogens unmeasured; any other length is an error naming the molecule and both counts.
- **The index remembers the molecule.** `AtomTargets` is the list of measured (molecule, atom) pairs with their values; a fit iterates over it, and the molecule each sample came from is kept beside it because every honest split needs it.
- **The folds hold out whole molecules.** `AtomTargets.folds` splits the molecules, never the samples. Two shifts of one molecule, one in the training set and one in the test set, would let a model see the molecule it is being tested on, and a per-atom $q^2$ obtained that way would not mean anything. This is the grouped-fold rule of the chemotype splits, with the molecule as the group.
- **A prediction goes back on its atom.** `write_atom_predictions` writes an SDF whose atoms carry the measured value and the predicted one, in RDKit's own `atom.dprop.<name>` spelling, so a predicted shift sits on the carbon it belongs to and every program that reads an SDF can see it.

```
cmf atoms shifts.sdf --list                      # what here could be a target
cmf atoms shifts.sdf --target shift --out with_atoms.sdf
```

`cmf atoms` is the inspection the data model deserves before a model exists: it says how many atoms were measured in how many molecules, which molecules carry the most, and what the folds would

look like, and it writes the file back in the canonical form. Listing is a looser rule than the one for fields on purpose -- a property with gaps cannot be a field but is exactly what a target looks like.

## 5.6 Quantum-chemical properties with ORCA

*Tools → Quantum-chemical properties with ORCA...* runs ORCA over the loaded molecules and brings two kinds of number back, to the two places the model keeps them.

**Per atom**, and so available as `prop:` fields: Mulliken, Löwdin, Hirshfeld and CHELPG charges, the HOMO and LUMO population of every atom, and — when the N+1 and N−1 states are computed as well — the condensed Fukui indices $f^+$, $f^-$, $f^0$, the dual descriptor $f^+ - f^-$, the local softnesses and the local electrophilicity. **Per molecule**, and so available as descriptors: the total energy, the HOMO and LUMO energies and their gap, the dipole moment, and from the ionised states the ionisation potential, the electron affinity, the chemical hardness, the softness, the chemical potential and the electrophilicity index.

That split is the purpose of the feature. A Fukui index is a statement about one atom, so it belongs in a field, where a co-field can then say *where* a nucleophilic centre helps; the gap is a statement about the molecule, so it belongs in the descriptor block, where it cannot pretend to have a position.

The dialog asks for

- **the ORCA program** — found by itself at the standard location, in `$ORCA_EXE` or on the PATH, and otherwise given here;
- **the level of theory** — the ORCA keyword line, which may be edited: `r2SCAN-3c` (fast, with its own basis) is the one to start with, B3LYP-D3/def2-SVP the usual hybrid, and anything else ORCA accepts as a single point will do;
- **which molecules** — all of them; only those that do not carry the properties yet, which is the default, so a set extended with new compounds costs only the new ones; the selected ones; or the training set;
- **the Fukui indices and the global reactivity numbers**, which need the N+1 and N−1 states and so cost three calculations per molecule instead of one;
- **optimising the geometry first**, off by default — see the warning below;
- **how many molecules at a time**, and, where the ORCA build has MPI, how many cores per job. The Windows build is the serial one: it has no MPI helpers, `%pal` fails outright, and the way to use a many-core machine is to run several molecules at once, which is what the first box is for. The dialog disables *cores per job* when it sees a serial build;
- **memory per core**, a **time limit** per job, after which that molecule is given up on rather than holding up the set, and the **working directory**.

Before you press *Run* the dialog says what the run will cost: the number of molecules times the number of ORCA jobs each, and the wall time that comes to at the chosen parallelism. After the first run that estimate is no longer a guess — the time every job took is recorded in the working directory, and the dialog quotes the measured mean. For scale: a molecule the size of methanol at `r2SCAN-3c` takes a few seconds per job, a drug-sized one at B3LYP-D3/def2-SVP a couple of minutes, so a hundred of the latter with their Fukui indices is a few hours with four running at a time.

The dialog also states the charge and multiplicity it read off the structures — the formal charge of the molecule, and the lowest spin state its electron count allows — so an unintended anion or an odd electron is visible before the hours are spent, not afterwards in a column of nonsense.

Three things protect a long run. Every job is **cached** under a key over the molecule, its charge, the level of theory and the geometry, so *Cancel* — which kills the ORCA processes themselves, not merely the bookkeeping — loses only the jobs that were running, and starting again continues from where it stopped. A molecule that **fails** is named in the log and skipped, never fatal to the set. And the working directory keeps every input and output, so any number can be traced back to the calculation that produced it.

When the run finishes, the new quantities appear as tick boxes — the per-atom ones in *Per-atom property fields*, the per-molecule ones in *Descriptors (per molecule)* — and the program offers to write the annotated set as an SDF. Write it: the file carries the properties with it, as atom property lists and molecule properties, so opening it later brings them all back without running ORCA again. *Tools → Save the data set with its properties (SDF)…* does the same at any time.

> **About the geometry.** *Optimise the geometry first is off by default and should normally stay off. ORCA optimises each molecule on its own, which moves every atom: a series that was superimposed is not superimposed afterwards, and a model trained on the old poses no longer refers to the molecules in memory. With the box off, the properties are computed exactly where the molecules stand, which is what a field model needs. If you do optimise, align the set again before building anything.*

The same computation runs from the command line as `examples/orca_atom_properties.py` (`--help` lists its options), which reads an SDF and writes the annotated one. The script and the dialog share the module `cmfqsar/orca.py`, so the numbers are the same either way.

## 5.7 Can a Fukui function be a field?

A `prop:` field puts one isotropic Gaussian on each atom, weighted by a number: a charge, a condensed Fukui index. That is a statement about *how much* and says nothing about *where*. The quantity a condensed Fukui index is condensed from is a function of position, $f(\mathbf{r}) = \rho_{N+1}(\mathbf{r}) - \rho_N(\mathbf{r})$, ORCA will write it on a grid, and `cmfqsar/cdft.py` asks whether that function can be brought into the program's algebra the way the probe fields were — by fitting several Gaussians per atom, which keeps the overlap of two fields a closed-form sum over atom pairs and so keeps the co-field and the atom contributions.

**It cannot, and the reason is a node.** Fitted against the valence part of the map on the four corrosion series with finished quantum runs, the relative error of an isotropic fit is 84%. A Fukui function is essentially a frontier-orbital density: on acrolein it is $\pi^*$, three quarters of its squared magnitude lies more than 0.5 Å off the molecular plane, and there is a node *in* the plane. A function that is positive everywhere and centred on a nucleus cannot make a node.

### Giving every atom a direction

The smallest extension that can make one is to multiply the Gaussian by a linear factor. With the width convention this program uses, atom $l$ then carries

$$s_l(\mathbf{r}) = e^{-b|\mathbf{r}-\mathbf{R}_l|^2/2}, \qquad p_l^u(\mathbf{r}) = (r_u - R_{l,u})\, e^{-b|\mathbf{r}-\mathbf{R}_l|^2/2}$$

for $u = x, y, z$: one s and three p functions per atom per width. A p function is odd about its own centre, so it has a node through the nucleus and a lobe on either side, and three Cartesian components span every direction — a lobe along a bond, along a lone pair or perpendicular to a ring plane are the same three numbers in different proportions, and nothing has to be oriented by hand.

The overlaps stay closed-form, which is the only reason this is the extension worth trying. With $p = a + b$, $\mathbf{d} = \mathbf{B} - \mathbf{A}$ and $S = (2\pi/p)^{3/2}\, e^{-ab|\mathbf{d}|^2 / 2p}$ the four blocks are

$$\langle s_A|s_B\rangle = S, \qquad \langle p_A^u|s_B\rangle = \frac{b}{p} d_u S, \qquad \langle s_A|p_B^v\rangle = -\frac{a}{p} d_v S, \qquad \langle p_A^u|p_B^v\rangle = \left[\frac{\delta_{uv}}{p} - \frac{ab}{p^2} d_u d_v\right] S$$

and `cmfqsar/anisotropic.py` evaluates them. A kernel between two such fields is still a double sum over pairs of atoms, with sixteen terms per pair of widths instead of one.

## What it is worth, measured

The plan sets two conditions for keeping the extension: the fit error has to drop, and the model has to notice. Measured on 4 series, 101 molecules in all (`benchmarks/run_anisotropic.py`):

| series | n | field | error, s only | error, s and p | coefficients per atom |
|---|---|---|---|---|---|
| 12_c1ccoc1 | 27 | fukui_plus | 78% | **61%** | 4 → 16 |
| 12_c1ccoc1 | 27 | dual | 87% | **66%** | 4 → 16 |
| 13_c1c_nH_nn1 | 25 | fukui_plus | 79% | **61%** | 4 → 16 |
| 13_c1c_nH_nn1 | 25 | dual | 87% | **67%** | 4 → 16 |
| 161_c1ccc2scnc2c1 | 20 | fukui_plus | 81% | **61%** | 4 → 16 |
| 161_c1ccc2scnc2c1 | 20 | dual | 91% | **68%** | 4 → 16 |
| 7_c1nncs1 | 29 | fukui_plus | 80% | **63%** | 4 → 16 |
| 7_c1nncs1 | 29 | dual | 91% | **69%** | 4 → 16 |

**The fit improves, and by a lot**: 84% to 64% on average, the best case reaching 61% (fukui_plus on `13_c1c_nH_nn1`). The kernel the fit produces also agrees better with the one the exact field produces, +0.920 to +0.977 in correlation. The diagnosis was right: what the isotropic basis was missing is a direction.

| series | field | five fields | exact grid | s alone | s and p alone | five + s | five + s and p |
|---|---|---|---|---|---|---|---|
| 12_c1ccoc1 | fukui_plus | +0.220 | +0.052 | +0.103 | +0.072 | +0.250 | +0.235 |
| 12_c1ccoc1 | dual | +0.220 | +0.028 | +0.057 | +0.012 | +0.272 | +0.241 |
| 13_c1c_nH_nn1 | fukui_plus | +0.654 | +0.636 | +0.685 | +0.684 | +0.672 | +0.681 |
| 13_c1c_nH_nn1 | dual | +0.654 | +0.575 | +0.505 | +0.567 | +0.649 | +0.664 |
| 161_c1ccc2scnc2c1 | fukui_plus | +0.689 | +0.638 | +0.636 | +0.627 | +0.658 | +0.655 |
| 161_c1ccc2scnc2c1 | dual | +0.689 | +0.522 | +0.556 | +0.573 | +0.596 | +0.624 |
| 7_c1nncs1 | fukui_plus | -0.037 | +0.121 | +0.090 | +0.027 | -0.012 | -0.024 |
| 7_c1nncs1 | dual | -0.037 | -0.031 | +0.045 | -0.021 | -0.001 | -0.022 |

**And the model does not notice.** Averaged over the 8 series-field pairs the p functions change $q^2$ of the field alone by **-0.017** (2 of 8 better), and of the field added to the five ordinary fields by **-0.004**.

The two measurements disagree for the reason the isotropic fit disagreed with itself: a kernel is an inner product between molecules, and an error that falls the same way for every molecule cancels in it. The p functions recover the π lobes above and below a ring — and every molecule of a congeneric series has a ring in the same place, so every field improves by about the same amount and the distances between the molecules stay where they were. What a model needs is not a better field but a field that differs between molecules in a way activity follows.

So the extension is implemented, tested against quadrature, and **not wired into the ordinary field machinery**, because by the plan's own test it has not earned its four-fold cost. It is there for the case that would earn it: a series whose members differ in the *shape* of a frontier orbital rather than in where their atoms are.

# 6 Before you model: `cmf diagnose`

Every expensive mistake recorded in this manual was visible before the model was built, in a number that takes seconds to compute. The corrosion benzenes were modelled with five fields and an automatic frame, and atom-type counts — no geometry at all — beat them. The ACE set is predicted as well by a handful of two-dimensional descriptors as by the whole 3D model. A thrombin model in the exposure study scored +0.059 with its attenuation pinned to the top of the grid, which is what a mis-specified model looks like from outside. None of these is subtle; they are simply never asked, because asking means running four baselines by hand.

```
cmf diagnose data.sdf --charges file
```

The command answers five questions and prints the numbers behind each.

**Is there anything to model?** The size of the set and the spread of the activity. Twenty molecules across half a log unit cannot support a conclusion, whatever $q^2$ comes out, and the command says so and stops.

**Can a frame exist at all?** The number of chemotypes, the size of the common Bemis–Murcko framework, and whether the molecules are actually superimposed. That last test is asked of the shared framework atoms rather than of the centroids, because `EmbedMolecule` leaves every molecule centred on the origin: a set of independently generated conformations has a centroid spread of 0.02 Å and looks perfectly aligned while every molecule points a different way. The test is also allowed to give no answer. A framework that is a small symmetric ring matches a dozen ways and two rings near the origin may coincide by luck, while a framework that is a small part of the molecule may match a *different* copy in each one — measured that way an author-aligned Sutherland set read as 9 Å apart. So a coincidence within half an ångström is a verdict on its own, a large deviation is a verdict only when the framework is a fair share of the molecule, and otherwise the answer is "this cannot tell — look at it".

**Does the frame carry signal?** The aligned model against the two routes that need no frame: the atom-type count kernel, which has thrown the geometry away entirely, and the rotation-invariant field autocorrelation, which keeps it without a pose. If neither is beaten, the alignment adds nothing and the cheaper route is also the more defensible one.

**Does the geometry contribute anything?** The same comparison against descriptors alone — what the file carries plus the handful the program computes.

**Was the model specified sensibly?** Whether the chosen attenuation sits on the edge of its grid, which means the search wanted a value it was not offered.

A fifth route, the topomer rule, is scored where the molecules can be cut at their framework. It covers only those molecules, so the aligned model is refitted on the same ones for the comparison: a subset against the whole set is not a comparison, and it favours whichever route happens to drop the hard compounds.

## 6.1 Two things the command protects you from

**The activity as a descriptor.** Most files store the activity as a molecule property, so the descriptors available usually include the quantity being predicted. Given it, the descriptor baseline returns $q^2 = 1.000$ and the advice becomes "use descriptors" for every set in the world. Columns

named as the activity are dropped, and so is anything correlating with it above 0.95 — but the leaked information is rarely a plain copy. The corrosion files carry the potency as `activity`, again as `mwila_pooled_raw`, and again as two laboratory-adjusted variants; excluding the first two still left a descriptor baseline at $q^2 = 0.99$, because the remainder reconstruct it between them. No per-column test can see that, so the outcome is judged as well: a descriptor baseline reaching $q^2 \geq 0.95$ on a set of this size is not chemistry but the measurement itself, and the command says so, names the nearest columns with their correlations, and refuses to recommend descriptors.

**Searching harder and calling it a better model.** The five routes do not carry the same number of hyper-parameters — the field routes search six attenuations and five ridges, the others search ridges only — so scoring each at the best point of its own grid biases the comparison toward whichever route searched the most. Every route therefore chooses its hyper-parameters inside each training fold and is scored outside it. On ACE this costs the aligned model 0.05 and the topomer route 0.07 while leaving counts, descriptors and the invariant kernel where they were, which is the size and the shape of the bias it removes.

## 6.2 Reading the answer

Each comparison is reported with a paired bootstrap interval over the molecules, because on twenty to thirty molecules a difference below about 0.1 in $q^2$ is not a difference. The last line recommends a route, and it should be read as a prompt rather than a verdict: it says which route to take and which question the set has not answered, never that a model is good.

One wording deserves care. When the command recommends `invariant` it often means *the frame is not worth the trouble* rather than *the invariant kernel predicts better* — on DHFR the aligned model is nominally ahead, +0.659 against +0.641, and the recommendation follows the interval rather than the point estimate. And `aligned, with the frame checked` means the frame carries signal and the molecules really are superimposed, so a frame exists whoever made it; several Murcko scaffolds only mean the program could not have built it, not that it is wrong.

## 6.3 Does the advice work?

A tool that reports five baselines is useful whatever its last line says, but advice that has never been followed is only a rule its author liked. `benchmarks/run_diagnose.py --validate` measures it the only way that matters: hold out 30 % of a set, run the diagnosis on the remaining 70 % alone, build the model it recommends, and score that model on the held-out molecules against every other route — thirteen sets, three splits each.

The advice named the best of the five routes on **21 of 39 splits**, and averaged over all of them the route it chose fell **0.096** in $q^2$ short of the best available. Always taking the aligned model — the default this command exists to question — names the best route on 12 splits and costs 0.201. Aggregated per set, which is the correct unit because three splits of one set are not three measurements, following the advice costs **0.105 less in $q^2$ than always aligning** (95 % bootstrap interval −0.259 to +0.019), better on six sets, worse on four, indistinguishable on three.

The shape of that distribution matters more than its mean. The advice is **insurance, not an advantage**: where it loses it loses a little — 0.02 to 0.17 on AChE, BZR, THR and the thermolysin set, all of them well-aligned series where the aligned model was always going to be right — and where it wins it wins a lot, 0.44 to 0.70 on the three corrosion sets, where the automatic frame is meaningless and the aligned model collapses. The interval crosses zero, so on thirteen sets this is not a proven improvement in the average case. What it is, is insurance against the failure described at the start of this chapter, at a price of a few hundredths of $q^2$.

# 7 Building a model

## 7.1 Choosing the fields

The five physico-chemical fields — electrostatic, steric, hydrophobic, H-bond acceptor, H-bond donor — are the default and reproduce the published CMF models. The steric field is special: its width is scaled by the van der Waals radius of each atom ($\alpha/R^2_{vdW}$, Eq. 10–11 of the 2013 paper) rather than being shared.

**Continuous indicator fields** (CIF, 2015 paper) add one field per atom type, with weight 1 on atoms of that type and 0 elsewhere. Tick *Continuous indicator fields* and set the minimum number of training molecules an atom type must appear in (3 by default) to be given a field of its own. Three typing schemes are available: `sybyl` (the types of the original scripts, read from MOL2 or assigned), `element`, and `element+aromatic`. CIFs are the most descriptive field set the program has, and on several of the benchmark sets they match or beat the physico-chemical fields on their own.

**Six pharmacophore features instead of twenty atom types** (plan item D2). A Sybyl vocabulary describes what an atom *is*; a pharmacophore describes what it is *for*. `--pharm` adds one indicator field per feature — `donor`, `acceptor`, `aromatic`, `hydrophobe`, `cation`, `anion` — taken from RDKit's own feature definitions rather than from SMARTS written here, with weight one on every atom of a matching group: six atoms for a benzene ring, three for a carboxylate, because the character of a carboxylate is not on one of its atoms.

Two things follow. The fields are **dense** — a pharmacophore feature is carried by 70% of the molecules of a set on average, so each has something to say about almost every molecule, where a Sybyl field keyed to `N.am` is silent on whatever lacks an amide. And they **read as a pharmacophore with no translation step**: the co-field of `pharm:donor` is already a map of where a donor is wanted, so *A pharmacophore read out of the co-field* types its features as donors and acceptors rather than as field names.

`--pharm-min N` decides which features are used — one carried by fewer than N training molecules is left out, the rule `--cif-min` applies to atom types — and `--pharm-features donor,anion` names them explicitly. `--fields pharm:donor,steric` works too, for a field set assembled by hand.

### What they are worth, and where

Measured on two collections (`benchmarks/run_pharmfields.py`, with and without `--mittal`), each set on its own alignment with the same attenuation and ridge grids and the same ten-fold cross-validation as *Does an alignment help?*. The Sutherland sets cannot answer the question, because the smallest of the eight has 66 molecules and the claim is about series below forty; the 38 Mittal sets can, 15 of them being smaller than that.

| sets | how many | median n | five | five + pharm | pharm | five + cif | cif |
|---|---|---|---|---|---|---|---|
| Mittal, below 40 molecules | 15 | 29 | +0.342 | **+0.368** | +0.358 | +0.344 | +0.335 |
| Mittal, 40 and above | 23 | 92 | +0.299 | +0.305 | +0.228 | **+0.345** | +0.321 |
| Sutherland (none below 66) | 8 | 112 | +0.574 | +0.586 | +0.538 | **+0.597** | +0.580 |

**Below forty molecules the smaller vocabulary wins.** The six features add +0.027 to the five physico-chemical fields where the Sybyl indicators add +0.002, and on their own they reach +0.358 — better than the five fields themselves (+0.342) and than the Sybyl indicators (+0.335). **Above forty the order reverses**: the Sybyl indicators add +0.047 against +0.006, and the six features alone fall to +0.228. The Sutherland sets, all of them large, agree with the second half: +0.023 for the Sybyl indicators against +0.012 for the features.

### Why, which is not the usual explanation

The explanation usually given is the rarity of individual Sybyl types, and that is not what is happening, because the rare types were already being dropped: a pharmacophore feature is carried by 70% of the molecules and the median Sybyl type by 47%, but `--cif-min` keeps only the types in five molecules or more — 11 of 14 on these sets — and the median coverage of what is kept is 88%.

What does change with set size is how many molecules there are per field to estimate it from:

| sets | molecules per Sybyl field | molecules per pharmacophore field |
|---|---|---|
| Mittal, below 40 | 3.4 | 6.4 |
| Mittal, 40 and above | 8.9 | 20.9 |
| Sutherland | 11.8 | 28.5 |

At three or four molecules per field a vocabulary cannot be estimated, whatever each of its fields covers; at nine or twenty it can, and then the more descriptive vocabulary wins because it is more descriptive. So the rule to take away is about size, not about chemistry: `--pharm` **below about forty molecules,** `--cif` **above it**, and the six features whenever the model is going to be read as a pharmacophore, where they cost 0.040 $q^2$ on a large set and save the translation.

`prop:<name>` fields are added by naming them in the per-atom property row of the model panel, or in `--fields`.

## 7.2 GRID-like probe fields

Goodford's GRID describes a molecule by the energy a small probe — a water, a methyl, a carbonyl oxygen — would feel at every point around it. That is a field in exactly this program's sense, and it says something the built-in fields do not: not "there is hydrophobic character here" but "a probe placed here would be bound". *Probe fields* in the model panel (`--probes water,carbonyl`) adds one such field per probe. Five are defined:

| name | what it stands for | R (Å) | well (kcal/mol) | charge | hydrogen bonding |
|---|---|---|---|---|---|
| `water` | water, $OH_2$ | 1.70 | 0.15 | 0 | both, weakly |
| `methyl` | a methyl group | 2.00 | 0.10 | 0 | neither |
| `carbonyl` | a carbonyl oxygen | 1.60 | 0.20 | −0.40 | accepts |
| `amine` | an amine nitrogen | 1.75 | 0.16 | +0.40 | donates |
| `cation` | a generic +1 cation | 1.50 | 0.10 | +1.00 | neither |

Those four numbers are the whole definition of a probe: the radius and well depth enter the dispersion term through the usual combining rules, the charge the electrostatic term, and the hydrogen-bond character decides which atoms it is complementary to — a donor probe feels acceptor atoms and not other donors. The parameters sit in the usual GRID and Tripos ranges rather than being taken from a published GRID table, which matters if you ever want to compare numbers with GRID itself rather than shapes. Adding a probe is one entry in `PROBES` in probes.py; the fitting, the kernel, the drawing and both interfaces read that dictionary, so a new one appears everywhere at once.

The obstacle is arithmetic, not conceptual: a Lennard-Jones term is not a Gaussian, has no closed-form overlap, and diverges at the atom, so the integral of its square does not exist. The solution is to fit each atom's contribution by a few Gaussians, which keeps every kernel in the program a closed form over atom pairs. Three things had to be measured before that worked, and each one shaped what is implemented:

- **The repulsive wall cannot be represented.** Fitting a curve that is +30 kcal/mol at 2.5 Å and −0.13 at 3.5 Å with a handful of Gaussians produces coefficients of $\pm 10^{7}$ that cancel, and a worst-case error larger than the well it is meant to represent. The wall is a truncation artefact in every grid implementation anyway, and this program already includes it — the steric field *is* a Gaussian van der Waals envelope — so a probe field carries the **attraction** only.
- **A well at a radius is a shell**, and a shell is not a sum of Gaussians centred on the atom. Each term is therefore capped at its own optimum, which makes it monotone in $r$, and those a sum of Gaussians represents naturally.
- **The $r^{-6}$ tail is not Gaussian either**, and is negligible beyond 6 Å, so the energy is tapered to zero between 5 and 7 Å exactly as GRID's own cut-off does.

What comes out reproduces the attraction to about 15 % in the volume-weighted norm the kernel sees (23 % on a realistic worst case), and the kernel it implies agrees with numerical integration of the fields themselves to $4 \cdot 10^{-7}$ — a test asserts both. Each atom carries $K$ Gaussians (four by default), so a probe field costs $K^2$ passes over the distance matrix where an ordinary field costs one; three is the fast setting and five the accurate one.

**Measured against the fields it would join**, same molecules, same α and λ grids, 10-fold CV:

| set | n | the five fields | + water probe | probes only |
|---|---|---|---|---|
| GPB | 66 | +0.539 | +0.539 | +0.406 |
| THER | 76 | +0.538 | +0.543 | +0.428 |
| THR | 88 | +0.672 | **+0.699** | +0.665 |
| ACE | 114 | +0.647 | +0.648 | +0.607 |

So: adding a water probe to the five fields adds between nothing and 0.027 to $q^2$, and probe fields on their own are worse than the fields they would replace. That is the measured result and it was the expected one — CoMSIA replaced GRID-like energies with Gaussian similarity indices for much the same reason, and this program's fields are closer to CoMSIA than to GRID by design. What a probe field is actually *for* is interpretation: its co-field is drawn, exported and decomposed like any other, and it says where a water or a carbonyl oxygen would be bound, which is a statement that is easier to act on than "more acceptor character here".

Two limitations are worth stating plainly. The directional part of GRID's hydrogen-bond term — the angles at the donor and at the acceptor's lone pairs — is **not** represented; the radial part is, broadened to 0.6 Å because a sharper shoulder is not representable by the shared basis. And alignment by field similarity refuses to run on a probe field rather than silently ignoring it, because the gradient of the expansion is not implemented.

## 7.3 Weighting the fields by solvent exposure

A kernel that integrates over all space weights an atom buried inside a molecule as heavily as one on the outside, although a binding partner can only touch the second. That is the claim the surface methods are built on, and `cmfqsar/exposure.py` tests it in the simplest way: every atomic weight multiplied by its atom's relative solvent accessibility, which keeps the kernel in closed form and keeps the co-field.

**It does not help, and on the larger sets it slightly harms the model.** Across sixteen FEP sets the median paired gain is −0.002 in $q^2$ and the weighting wins on eight of sixteen — no better than chance — while the one large win is a model whose attenuation was fixed at the edge of the grid. On the eight Sutherland sets every variant is worse, and the softened one that looked best on the FEP sets has an interval entirely below zero. The effect also shrinks as the sets grow: on DHFR (361) and COX2 (282) the change is −0.004 and +0.003, and the movers are the two smallest sets. An effect that vanishes with more data is noise.

The reason is geometric and settles the general question. Measured by group — a heavy atom and its hydrogens counted as one, because atom by atom a methyl carbon is 93 % occluded by its own hydrogens and ethanol reads as two-thirds buried — the median group exposure is 0.37 to 0.42 and **0 to 3 % of groups are buried** in either family. A ligand of thirty heavy atoms has no interior. Surface-based representations are solving a problem that molecules of this size do not have.

## 7.4 Descriptors that are not fields

Everything above describes a molecule as a function of space. Some things are not functions of space and never will be: a measured logP, a molecular weight, the medium or temperature a series was recorded in, a count of rotatable bonds, an energy from a quantum calculation. A model that cannot see them has to pretend the activity depends on geometry alone, which for an adsorption potency or a permeability is simply false.

The *Descriptors (per molecule)* row of the model panel (`--descriptors A,B`) offers every numeric molecule property the file carries — and hides itself when the file carries none besides the activity, which is why it is invisible on most of the example sets; the computed descriptors of the next section are always there. Ticking some adds one more block to the kernel:

$$K(A,B) = \sum_f h_f c_f \hat{K}_f(A,B) + h_d c_d \, \mathbf{z}(A) \cdot \mathbf{z}(B)$$

where **z** holds the descriptors of a molecule, standardised with the **training set's** mean and standard deviation so that a quantity in kilojoules cannot outweigh one in log units, and $h_d$ is a mixing coefficient like a field's (the *h* box beside the row, `--descriptor-h`; zero switches the block off without removing it).

Three properties of the block are deliberate:

- **It is scaled, not cosine-normalised.** Dividing by the two norms, as the fields are divided, would send a molecule sitting at the training mean in every descriptor — an unremarkable

molecule — to a zero row and a zero diagonal, out of the model altogether. Instead the whole block is multiplied by a constant that puts its mean diagonal at 1, so ten descriptors do not outweigh the geometry merely by being ten.

- **The statistics belong to the training set** and are stored in the saved model. Standardising a test molecule by its own set would score two test sets on two different scales, which is the mistake the field scaling avoids as well.
- **The model stays linear in its inputs**, so the decomposition rules survive. A prediction splits into a field part and a descriptor part, and the descriptor part splits further into one number per descriptor, which is what the program reports.

A molecule missing a descriptor is refused by default (`prop_missing`), because a model that silently substitutes the mean is telling you nothing about that molecule while looking as confident as ever; with the `mean` policy it is substituted and the molecule simply contributes nothing through that descriptor. A descriptor that is constant over the training set is reported as inert rather than dividing by a spread of zero.

**The obvious danger, stated plainly.** A descriptor derived from the activity — the activity itself, a transform of it, a value fitted to it — will dominate the model and make the geometry look worthless, while $q^2$ goes up. The program cannot detect this, so it reports the mean absolute contribution of every descriptor beside the model statistics:

```
Non-field descriptors (mean |contribution| per training molecule):
  clogp                          0.0779
  mw                             0.0176
  tpsa                           0.0123
```

If one descriptor carries most of the prediction, the model is really a descriptor model with a 3D appearance, and it should be reported as one.

```
python cmf_cli.py build series.sdf --activity pIC50 --descriptors clogp,mw,tpsa
```

## Descriptors the program computes itself

A file that carries no numeric tags has nothing to offer the row above, so four descriptors are always available, computed by RDKit from the structure: `calc:mw` (molecular weight), `calc:clogp` (Wildman–Crippen logP), `calc:tpsa` (topological polar surface area) and `calc:rotatable` (rotatable bonds). They have a row of their own in the model panel, *Computed descriptors*, and **all four are off by default**.

Those four are shortcuts, not the whole list. **Any** of RDKit's descriptors — some two hundred of them — can be asked for by its own RDKit name, typed into the box at the end of that row (with completion) or passed on the command line: `calc:NumHAcceptors`, `calc:FractionCSP3`, `calc:BalabanJ`, `calc:LabuteASA`, `calc:qed`. Names are matched case-insensitively, and a name that does not exist is refused before the fit with the nearest ones that do — `calc:NumHDonor` answers *did you mean NumHDonors?*.

The default is deliberate, and so is the warning that belongs with it. These are two-dimensional quantities: they can improve a model that has a size or lipophilicity trend in it, and they can also carry a model that then gains nothing at all from its geometry. Measured on four Sutherland sets, same α and λ grids, 10-fold CV:

| set | n | five fields | + the four computed | the four alone |
|---|---|---|---|---|
| THR | 88 | **+0.671** | +0.634 | +0.620 |
| GPB | 66 | **+0.534** | +0.525 | +0.494 |
| ACE | 114 | +0.657 | +0.707 | **+0.712** |
| THER | 76 | **+0.509** | +0.496 | +0.480 |

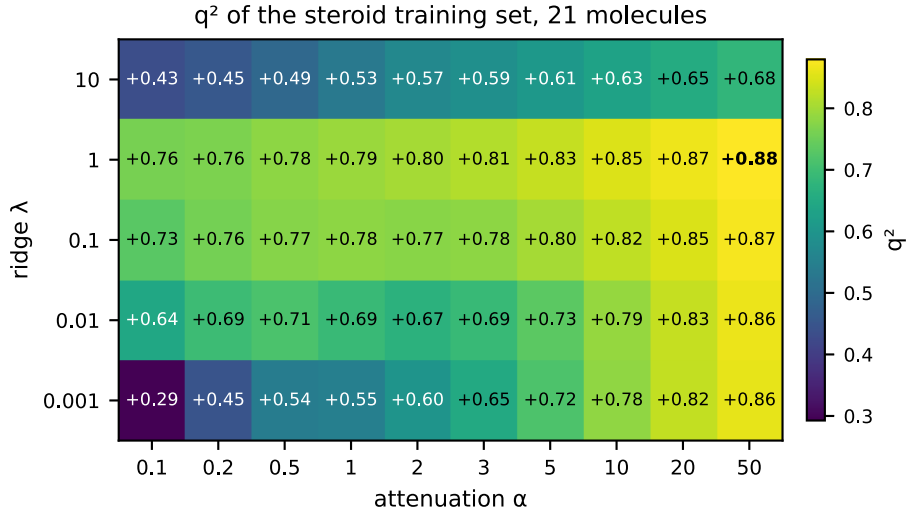
Read ACE carefully: four numbers that know nothing of the geometry predict better than the whole 3D model. That is not an argument for using them — it is a warning about that data set, and it agrees with what the alignment-free baseline already said about ACE elsewhere in this manual. On the other three sets they lower q² slightly.

So use them to *test* a model rather than to improve it: if adding them changes nothing, the geometry is carrying the model; if they take over, the 3D part was contributing nothing. The contribution table printed beside the statistics is what tells you which happened.

## 7.5 Attenuation, regularisation and the grids

Two hyper-parameters matter. α sets how far each field reaches: large α makes the fields sharp, so molecules must match closely, while small α makes them diffuse. λ (or *C*, ε, ν for the SVM learners) sets the regularisation. Both are chosen by cross-validated grid search, and the whole grid is shown in the *Hyper-parameter search* tab — look at it, because an optimum at the edge of the grid means the grid was too small.

```
python cmf_cli.py build data.sdf --alphas 0.1,0.3,0.5,1,2,5 --lams 1e-3,1e-2,0.1,1,10
```




*The cross-validated q² of the steroid training set, 21 molecules, over a grid of attenuation and ridge; the best point is in bold. The surface is flat along α: from α = 5 on, every doubling of α adds one or two hundredths, so the best point sits on the edge of the grid and a wider grid would move it further without changing the model much. The ridge decides more: the lowest row is what too little regularisation costs.*

Refinements, all optional: continuous Brent minimisation over λ as in the original R scripts (`--lam-search brent`), a Nelder–Mead search over α (`--alpha-search nelder-mead`), coordinate descent or centred kernel alignment for the mixing coefficients `h_f` (`--optimize-h`, `--mkl`), and a grid over the bandwidth of a nonlinear kernel layer (`--gammas`).

Two conventions decide how the field kernels are combined, and they change the numbers: *Normalise kernels* uses cosine-normalised field kernels, and *Scale fields* rescales each field kernel to unit mean diagonal before mixing. Untick the second to get the raw-kernel convention of the papers (`--raw-scale`).

## 7.6 How long the kernel takes, and the three settings that change it

A field kernel is a sum over atom pairs, so its cost is the number of atom pairs: the square of the set size times the square of the molecule size. A 200-molecule congeneric series costs under a second per attenuation and nothing has to be done about it. A library, a set of long molecules, or a grid search over six attenuations is a different matter, and three things were measured and changed (`benchmarks/run_speed.py` records all of them).

**The training kernel is computed once rather than twice.** $K_f(i,j) = K_f(j,i)$, so for a set against itself only the pairs with $j \geq i$ are evaluated and then mirrored. This is the largest of the three, because it removes half of the work rather than part of it, and it is exact; it also makes the matrix exactly symmetric, which it was not before, the two triangles having differed by round-off from summing the atom pairs in two different orders.

**The radius-scaled prefactor is a square root rather than a power.** The steric field gives every atom its own width, so $(2\pi/(b_l+b_m))^{3/2}$ differs for every atom pair, and a general power is a library call per element. Writing it as $z\sqrt{z}$ is the same number to fourteen digits and costs a third as much. Before this change that one expression was half the time of the whole kernel.

**The distances are shared across the attenuation grid.** The attenuation enters only through the exponential, so a grid search can compute each molecule's squared distances once and take six exponentials from them. This is the smallest of the three: the distance matrix is 3 % of a block.

Measured on 200 molecules of a congeneric series and 60 decapeptides, all six attenuations of the usual grid (0.1, 0.3, 0.5, 1, 2, 5):

| shortcut | COX2 (40.8 atoms each) | AMPH1 (168.2 atoms each) | still agrees to |
|---|---|---|---|
| as before | 14.3 s (1.0x) | 42.2 s (1.0x) | -- |
| + square root | 10.5 s (1.4x) | 38.7 s (1.1x) | 3e-16 |
| + symmetry | 4.6 s (3.1x) | 16.0 s (2.6x) | 1e-15 |
| + shared distances | 3.8 s (3.7x) | 15.3 s (2.8x) | 1e-15 |

The last column is the point: every shortcut is exact, and what is left is round-off from summing the atom pairs in a different order. End to end, a `grid_search` over the same six attenuations with five-fold cross-validation:

| shortcut | seconds | speed-up | alpha chosen | $q^2$cv |
|---|---|---|---|---|
| as before | 13.8 | 1.00x | 0.1 | +0.504175 |
| + square root | 9.9 | 1.40x | 0.1 | +0.504175 |
| + symmetry | 5.0 | 2.76x | 0.1 | +0.504175 |
| + shared distances | 4.1 | 3.40x | 0.1 | +0.504175 |

**3.4 times faster**, and the attenuation chosen and the cross-validated $q^2$ are identical to six decimal places.

Nothing here changes a number the program reports; the benchmark prints the chosen α and the cross-validated $q^2$ beside every timing for exactly that reason.

## The cutoff, which is off unless you ask

A Gaussian overlap falls below a fraction τ of its peak beyond a cutoff radius of $\sqrt{(-4 \ln \tau / b)}$, which at $\tau = 10^{-6}$ is 10.5 Å for a shared-width field at α = 0.5 and 19 Å for the radius-scaled steric field. Atom pairs further apart than that can be dropped, and `--cutoff auto` does so, deciding per field by sampling the real distances and building a neighbour list only when most pairs would go.

Whether this helps is a property of the molecules, not of the program:

| set | atoms per molecule | α | pairs kept, shared width | pairs kept, steric | exact sum | `--cutoff auto` | what auto did |
|---|---|---|---|---|---|---|---|
| COX2 | 40.8 | 0.5 | 98% | 100% | 0.95 s | 0.85 s (1.12x) | kept the exact sum |
| COX2 | 40.8 | 2 | 56% | 96% | 0.90 s | 0.91 s (0.99x) | kept the exact sum |
| COX2 | 40.8 | 5 | 24% | 67% | 0.84 s | 0.82 s (1.02x) | kept the exact sum |
| AMPH1 | 168.2 | 0.5 | 62% | 99% | 2.61 s | 2.64 s (0.99x) | kept the exact sum |
| AMPH1 | 168.2 | 2 | 14% | 52% | 2.75 s | 2.15 s (1.28x) | used the neighbour list |
| AMPH1 | 168.2 | 5 | 4% | 19% | 2.64 s | 1.21 s (2.18x) | used the neighbour list |

The two columns of surviving pairs explain the rest. The sampling that decides costs about a hundredth of a block, so `auto` is free when it declines, and on these sets it declines wherever the dense sum is faster. Forcing the list on with `--cutoff 10` or `--cutoff auto` where it does not pay can cost a factor of five, which is why it is not the default.

On a congeneric series superimposed in one frame the cutoff keeps almost every pair and `auto` refuses to build a list, so the sum stays exact. On molecules five times longer it is worth a factor at a strong attenuation. A cutoff makes the kernel an approximation — each dropped pair contributed less than a millionth of a peak overlap, and the measured error is about $10^{-7}$ relative — so it is off by default and every table in this manual was produced without it.

## The compiled kernel, if Numba is installed

The kernel above is NumPy, and NumPy is single-threaded. The same arithmetic written as one fused loop over the atom pairs and compiled with Numba can use the other cores: `cmfqsar/fastkernel.py` does that, threaded over the molecules, which is safe because every

molecule writes its own row of the kernel and nothing is shared.

Two things are worth knowing before reaching for it.

**Per core it is slower, not faster.** `np.exp` on an array is vectorised; the same calls written one at a time in a compiled loop are not, unless Intel's SVML is installed. Measured serially on 200 molecules, the NumPy path takes 0.67 s and the compiled loop 0.90 s. All of the advantage below is the threads, and on a machine with two cores there will be almost none.

**It is not the default.** Install it with `pip install numba` or `pip install -e .[fast]`, and ask for it with `--engine numba` (or `--engine auto`, which uses it where it is installed and falls back silently where it is not). The default everywhere except `screen` is `numpy`, because the two paths differ by about 2e-14 relative — the compiled loop is built with `fastmath` and may add the pairs in another order — and every table in this manual was produced with NumPy. That difference is the same order as the round-off between two orderings of the NumPy sum itself, but a default that depends on whether an optional package is installed is not something a table should rest on.

What it is worth, on 32 cores:

| set | atoms per molecule | NumPy | compiled | speed-up | differs by |
|---|---|---|---|---|---|
| COX2 | 40.8 | 0.50 s | 0.09 s | 5.6x | 4e-15 |
| AMPH1 | 168.2 | 1.99 s | 0.12 s | 16.6x | 2e-14 |

The long molecules gain more because each thread then has more work to amortise the cost of being started. End to end it is less, because the kernel stops being all of the time: a whole grid search over six attenuations falls from 2.9 s to 0.7 s (4.0 times), choosing the same attenuation and the same cross-validated $q^2$ to six decimals, and a screening run over 4166 molecules from 46 s to 12 s (3.8 times, 91 to 347 molecules a second) — where reading the file and building the fields are now the larger half of the work.

The compiled loop also takes a cutoff without a neighbour list, since inside the loop a cutoff is one comparison per pair; `screen --cutoff R --engine numba` is therefore worth trying on long molecules, where the NumPy neighbour list of the previous section is also worth building.

## 7.7 The learners

| Task | Methods | Key |
|---|---|---|
| Regression | kernel ridge regression with bias; Gaussian process (the same fit plus predictive variances); ε-SVR; ν-SVR; the ranking SVM, which fits the order and not the values | `krr`, `gp`, `svr`, `nusvr`, `ranksvm` |
| Binary classification | C-SVC; ν-SVC; Gaussian-process classifier by the Laplace approximation | `svc`, `nusvc`, `gpc` |
| One-class / screening | 1-SVM; distance to the centre of the actives; distance to the k-th nearest active; one-class Gaussian process; distance from the principal subspace (kernel PCA) | `ocsvm`, `ocmean`, `ocknn`, `ocgp`, `ockpca` |

Kernel ridge regression is the method of the papers and the default. The Gaussian-process view adds a posterior variance per prediction at no extra cost, which is what the confidence-based alignment of *Preparing the data* minimises. The Laplace GP classifier adds calibrated probabilities, a posterior variance and model selection by marginal likelihood — it scores no better than C-SVC on the benchmark sets (mean difference 0.001 in Matthews correlation), so it is not the default, but its probabilities are calibrated and its evidence picks hyper-parameters without holding anything out.

## 7.8 Fitting the order instead of the values: the ranking SVM

A regression model is asked for the activity and judged on the squared error, which assumes the numbers are exact. Often they are not. Potencies pooled from several papers carry an offset per source that no descriptor explains, a series measured in two assays carries two scales, and an IC50 read off the end of a dose–response curve carries a bound rather than a value. In all three the *order* survives what the values do not, and a model fitted to the values spends its capacity on the part that is noise.

`--method ranksvm` fits the order. For every pair of molecules whose activities differ it asks that the more active one score higher by a margin, and is penalised for every pair where that fails:

$$\min \tfrac{1}{2}\|w\|^2 + C\sum_p \xi_p \;\text{ with }\; w\cdot(X_i - X_j) \geq 1 - \xi_p$$

over the pairs $p = (i, j)$ with $y_i > y_j$ — an ordinary support-vector classification on the *difference* of two molecules' fields. The kernel of two such differences is a sum of four ordinary kernel entries,

$$K_{\text{pair}}((i,j),(k,l)) = K(i,k) - K(i,l) - K(j,k) + K(j,l)$$

so the whole thing is built from the kernel matrix the program already has, with no new fields and no new parameterisation.

**The fitted model is still a dual expansion over molecules.** The weight vector is $w = \Sigma_p\, a_p\, (X_{i_p} - X_{j_p})$, so collecting the terms gives one coefficient per *molecule* and the score of a new one is $K(x, \text{train})\,\mathbf{a}$, exactly as for every other learner here, with no bias. Everything that reads a linear CMF model — co-fields, atom contributions, Shapley values, the pharmacophore — keeps working. What changes is what they mean: the co-field of a ranking model says where more field *promotes* a molecule, not what it adds to the activity.

What is given up is the scale. A ranking score has no units and cannot be compared with an activity; only its order means anything. The program therefore reports Spearman's ρ, Kendall's τ and the share of pairs ordered correctly, and reports no $r^2$ or RMSE, which would be arithmetic on a quantity that does not exist. The hyper-parameter search selects on ρ for the same reason.

```
python cmf_cli.py build data.sdf --charges file --method ranksvm --Cs 0.1,1,10 \
    --rank-gap 0.5 --rank-groups paper
```

Three options shape the pairs. `--rank-gap` drops the pairs whose activities differ by less than it: a pair separated by less than the experimental error is a matter of chance, and asking a model to call it is how a ranking model learns noise. Half a log unit suits pooled literature potencies and zero suits one careful assay. `--rank-groups PROP` forms pairs only inside a group — the same paper, assay or series — which is what makes the method useful on pooled data, because an offset between two sources then cancels instead of being compared across. `--rank-pairs` caps the number of pairs: the solver works on the kernel *between pairs*, so the cost is quadratic in that

number, and the default of 4,000 already asks for 128 MB. The order of a set is heavily over-determined and a few thousand comparisons fix it, so the cap is a bound on cost rather than a loss of information.

### When it is worth it, measured

`benchmarks/run_ranking.py` asks the only question that matters — does fitting the order beat fitting the values — on the eight Sutherland sets with five random 70/30 splits each, every model choosing its own hyper-parameters on its own criterion inside the training part.

**On clean data it does not.** Where the activities are one assay's numbers and mean what they say, Spearman's ρ on held-out molecules is **+0.713 for kernel ridge and +0.689 for the ranking SVM**, ridge ahead on five of the eight sets. That is the expected answer and worth stating plainly: a regression model is already a ranking model, since the order of its predictions is all a ranking model ever produces, and it had the magnitudes to learn from as well. Fitting the order discards how far apart two molecules are, which on reliable data is information.

**On pooled data it does, but only with the groups.** The second experiment gives the same molecules to three synthetic sources — assigned *by scaffold*, because an offset that travels with the chemistry is what a model will learn as if it were an effect of the chemistry, while an offset assigned at random is simply averaged away — shifts each source by 1.5 standard deviations, and asks for the order of the true activity, which no model ever saw:

| | KRR | ranking | ranking by source |
|---|---|---|---|
| same-source pairs ordered correctly | 72.5 % | 71.6 % | **74.3 %** |
| Spearman ρ over all held-out pairs | +0.574 | +0.565 | **+0.663** |

A ranking model that is not told about the sources does no better than ridge: it is handed the same corrupted comparisons and learns them. Told which molecules share a source, it beats ridge by **0.089 in ρ**, on six of the eight sets, because the cross-source pairs — the only ones the offsets corrupt — are never formed. Note that it wins on the *global* order too, although it was never shown a cross-source comparison: it does not need one, since it fits a single scoring function over the fields and all that grouping did was stop it fitting an offset.

So the summary is narrow and useful: **the ranking SVM is for data whose numbers are not comparable across their sources, and its value is `--rank-groups`.** On a single clean assay, use kernel ridge.

## 7.9 Nonlinear kernel layers and multiple kernel learning

A layer on top of the CMF kernel makes the model more expressive than an overlap integral: a Gaussian layer, $\exp(-\gamma\, d^2)$ with $d^2 = K(i,i) + K(j,j) - 2K(i,j)$, or one or two arc-cosine layers, the kernel of an infinitely wide neural network (Cho & Saul 2009). The cost is interpretability: the model is then no longer linear in the field, so co-fields and atom contributions are unavailable. The similarity decomposition stays valid, and the **Shapley tab** decomposes the prediction instead, which is why it exists.

Multiple kernel learning chooses the mixing coefficients $h_f$ rather than leaving them at 1, either by coordinate descent on the cross-validated error, by centred kernel alignment (`--mkl align|alignf`), which is a closed-form, training-only criterion, or by the marginal likelihood of the mixture (`--mkl evidence`), which is the next section.

## 7.10 Which fields does this series need?

A sum of kernels is the kernel of a sum of independent processes,

$$f(x) = \sum_f f_f(x), \qquad f_f \sim \mathcal{GP}(0,\ h_f c_f K_f),$$

so $h_f$ is not a tuning parameter but the **prior variance of the part of the model that field f carries** — how much of the activity that field is allowed to explain. Put a prior on it and integrate, and the answer to *does this series care about the steric field at all* comes with an interval instead of a number:

```
python cmf_cli.py mixing model.cmf
```

What is sampled is each field's **share** of the kernel, written $h = F\,\mathrm{softmax}(u)$ so that the shares sum to one and can be compared with $1/F$, the share a field would have if all fields mattered equally. The prior is $u_f \sim \mathrm{N}(0,\ 1.5^2)$, independent and the same for every field, so it prefers no field over any other and only limits how far apart their shares can be drawn before the data are taken into account; the regularisation is sampled with them. The sampler is a slice sampler, which has no step size to tune.

Three readings come out, in increasing order of sharpness. The **shares with their intervals** say how much of the kernel each field carries. The list of fields **carrying less than an equal share** names those whose whole interval lies below $1/F$. And the **Bayes factor** against *this field does nothing* — the log evidence lost when the field is removed outright and the others are re-weighted — is the sharpest and the cheapest form of the question.

### What it is good for, measured

Eight Sutherland sets, five 70/30 splits each. First, whether weighing the fields predicts better at all — the weights set four ways on the training part, each scored on the molecules held out:

| weights | $R^2$pred |
|---|---|
| equal (h = 1) | +0.481 |
| centred kernel alignment, per field | +0.487 |
| centred kernel alignment, jointly | +0.485 |
| the evidence optimum | +0.485 |

The evidence optimum is +0.003 in $R^2$pred against equal weights, 95 % interval [-0.026, +0.035], which contains zero — the same lesson as the per-field attenuation: **the evidence takes whatever freedom it is given over the shape of the kernel, improves its own number, and leaves the prediction unchanged.** Choosing the weights is not how a CMF model is made better.

What the posterior is *for* is the other question, and that one can be checked. Each field's Bayes factor against *this field does nothing*, beside what actually happens to the held-out molecules when that field is removed outright:

| field | mean share | 90 % interval | Bayes factor | ΔR²pred without it |
|---|---|---|---|---|
| Electrostatic | 0.179 | [0.024, 0.432] | +1.4 | -0.032 |
| Steric | 0.337 | [0.091, 0.663] | +2.2 | -0.012 |
| Hydrophobic | 0.184 | [0.021, 0.460] | +0.7 | -0.006 |
| H-bond acceptor | 0.159 | [0.032, 0.385] | +1.6 | -0.018 |
| H-bond donor | 0.142 | [0.025, 0.349] | +1.5 | -0.003 |

Over all 200 (set, split, field) cases the Bayes factor and the cost of removing the field correlate at Spearman ρ **+0.121** (positive means the evidence was right). The two orderings barely agree: the Bayes factor says what the training activities can see, which is not the same as what the held-out molecules need, and it should be read as the first and not the second.

**The intervals are wide, and that is the answer rather than a failure of the sampler.** On a congeneric series the field kernels are strongly correlated: several quite different mixtures explain the activities about equally well, and a point estimate — the alignment's, the evidence optimum, or any other — is claiming a precision the data do not support. That is the argument for integrating rather than optimising here, and it is the reason `--mkl evidence` exists as an option rather than as a default.

## 7.11 Choosing α and λ without holding anything out

Everything above chooses hyper-parameters by cross-validation: ten fits and ten predictions per candidate, and the best held-out score wins. A kernel ridge model is the posterior mean of a Gaussian process, and a Gaussian process scores its own hyper-parameters without holding anything out. The marginal likelihood — the probability of the training activities with the weights integrated away — is one determinant and one quadratic form on the factorisation the fit needs anyway, so a candidate costs **one factorisation instead of ten**:

```
python cmf_cli.py build data.sdf --charges file --select evidence
```

Two adjustments are made to it, and both matter. The bias of the program's KRR is a constant mean in the Gaussian process, so it is **marginalised under a flat prior** rather than fitted and then scored, which would count it twice. And the amplitude of the kernel — unknowable, since the kernels are cosine-normalised and the activities arrive in whatever units the assay used — is **profiled out** at its closed-form optimum, so the criterion depends on α and λ alone and its ranking does not move when the activity is rescaled.

**Measured, it chooses as well and costs a quarter as much.** Over the eight Sutherland sets, five random 70/30 splits each, with both criteria searching the same grid on the training part and both choices scored on the held-out molecules:

| | R²pred |
|---|---|
| chosen by cross-validation | +0.471 |
| chosen by the evidence | +0.479 |
| paired difference | **+0.007**, 95 % interval [−0.013, +0.028], better on 6 of 8 sets |

The two are indistinguishable in quality and the evidence is four times faster at selecting. The detail worth more than either number is that **they agree on the point of the grid in 0 to 60 % of splits** and reach the same predictive quality anyway: the surface is flat near its optimum, so what a search costs matters more than which point it selects. That is also what makes a finer grid — a separate α per field, say — affordable for the first time, and the next section uses the saved time for exactly that and reports what it gained, which was nothing.

The evidence is not a cross-validated score and does not pretend to be. It answers *how probable are these activities under this model*, not *how well would this model predict a molecule it has not seen*, and the two can disagree when the kernel is misspecified, which for molecular fields is the normal case. On these sets they do not. It is available for regression with `krr` or `gp`; the support-vector methods and the classifiers have no such quantity and are refused rather than silently cross-validated.

## One width per field

One α for five fields is a convention, not a result. Electrostatics is long-ranged and a hydrogen-bond donor is nearly a point; there is no reason the same attenuation suits both. The protocol of the papers shared it because the computational cost required it — a grid over one width is five kernels, a grid over one width per field is $5^5$ of them — and the evidence removes that objection twice over. The per-field kernels are independent, so every combination is assembled from the matrices the shared grid already built, and each candidate costs one factorisation. A coordinate descent over the widths, starting from the best shared one, then costs a few hundred:

```
python cmf_cli.py build data.sdf --select evidence --alpha-search per-field
```

It reports the width of each field and what the extra freedom gained, in the units it was selected in. Fields whose kernel does not depend on α at all — the probe fields, whose widths come from their own expansion — are left alone rather than searched in vain.

**Measured, the extra freedom is real and gains nothing.** Eight Sutherland sets, five 70/30 splits each, the shared width chosen by the evidence on the training part and the descent started from it:

| | |
|---|---|
| log evidence gained by the per-field widths | **+2.6**, on every one of the 40 splits |
| widths moved away from the shared one | 3.5 of 5 on average, and at least one on every split |
| $R^2$pred against the shared width | **−0.001**, 95 % interval [−0.020, +0.018], better on 4 of 8 sets |
| the same count per split | ahead on **20 of 40** |
| correlation of the evidence gained with the $R^2$pred gained | **−0.07** |

The descent does what it was asked: it moves most of the widths, improves its criterion every time, and settles on widths a chemist would find reasonable — the steric field sharpest (mean α 2.33), the donor field widest (1.16). The molecules held out cannot tell the difference, and the gain in evidence carries no information about whether a split improved.

This is the same flat surface the previous section found in a different way. There, two different criteria picked different shared widths and predicted equally well; here, one criterion is given five widths instead of one, moves them, and predicts equally well. Around its optimum the attenuation

does not decide much, and five of them decide no more than one — which is worth knowing, because it says the search is not what decides the accuracy of a CMF model. The alignment is (see *Alignment*, and the third part of `docs/benchmark.md`). Use `--alpha-search per-field` to ask the question of a new series, not as a default.

## 7.12 Validation

- **k-fold cross-validation** (10 by default) gives $q^2$ and RMSEcv. Folds are shuffled unless you ask for the interleaved folds of the original scripts (`--cv-no-shuffle`).
- **A test set** gives $R^2$pred and RMSEpred. Both variance conventions are reported, because they differ for sets whose test activities span a narrow range — the steroid benchmark is the classic example, where the ten test affinities cover 1.9 log units and $R^2$pred computed against the test variance is negative while the predictions are good.
- **External (nested) cross-validation** (`--nested`, *External CV*) wraps the whole model selection in an outer loop and reports $q^2$ex and RMSEcvex. This is the number to quote: the ordinary $q^2$ has seen the hyper-parameter choice.
- **y-randomisation**, a **bootstrap** interval and the **applicability domain** are the three sections below.
- **Leave-one-chemotype-out** folds, for a set of several scaffolds, never leave a molecule's own scaffold-mates in the training part. Random folds overstate the quality of such sets badly — by 0.11 to 0.47 $q^2$ on the five chemotype benchmarks here — and the program reports both, together with an alignment-free fragment baseline that the aligned model has to beat.

## 7.13 Is it chance? y-randomisation

A cross-validated $q^2$ says how well the model predicts; it does not say whether a model of the same shape could have reached that number from activities that mean nothing. In a congeneric series, where kernel values above 0.97 are normal, that is a real possibility, and it is the first thing a referee asks about a 3D-QSAR model.

The **shuffles** box of the model panel's *Validation* row (`--y-scramble N`) shuffles the activities N times, refits and cross-validates each time, and reports where the real $q^2$ falls among the scrambled ones — the mean and spread of the shuffled values, the largest of them, how many standard deviations above them the real one lies, and the permutation p-value.

```
python cmf_cli.py build data.sdf --alphas 0.5 --lams 0.1 --y-scramble 100
```

Two properties of the test are worth knowing, and the program states both rather than letting the number mislead:

- **A permutation test cannot report a p below 1/(N+1).** Ten shuffles cannot go below p = 0.091, which is above every conventional threshold *however good the model is*. The program says so — "no shuffle came close, but 10 of them cannot support a p below 0.091" — instead of reporting a chance-correlation warning that the data do not support. Use 19 shuffles for $p < 0.05$ and 99 for $p < 0.01$.
- **The hyper-parameters are held fixed** by default, so the test asks whether the *fit* can be reached by chance. Tick **redo the search** beside the shuffles box (`--y-scramble-select`) for the stricter question — whether the whole procedure, model selection included, can be. It matters: on the steroid set with a 3 x 3 grid, the best scrambled $q^2$ rises from −0.08 to **+0.32** once the

search is allowed to pick the grid point that best fits the noise, while the real value stays at +0.78. The kernels do not depend on the activities, so they are computed once and the extra cost is one cross-validation per grid point per shuffle.

## 7.14 Where does the model apply?

A kernel method always returns a number. For a molecule unlike anything in the training set, that number is the training mean presented as a prediction, and nothing in q² or R²pred reveals it.

The **applicability domain** tick of that row (`--domain`) measures how close each molecule outside the training set is to it, **in the kernel the model itself uses**: the cosine similarity of their fields, averaged over the *k* nearest training molecules (3 by default). The threshold is not an arbitrary number — it is taken from the training set's own spread, the 5th percentile of the same quantity computed among the training molecules, so a series of close analogues gets a tight domain and a diverse set a loose one. Molecules below it are listed by name in the log, their similarity appears in the data-set table's **AD** column — which sorts like any other, so the extrapolations can be brought to the top — and the exported predictions carry both the value and a yes/no verdict, so that an extrapolation stays marked after it leaves the program.

On the steroid benchmark, nine of the ten test compounds fall inside and one — 4-pregnene-3,11,20- trione — falls outside. On BZR, training on the largest chemotype and scoring the others, every molecule of the training chemotype is inside and 29 of 40 molecules from other chemotypes are outside, which is the behaviour one wants from a domain estimate.

The field-space distance is reported alongside the similarity, because the two can disagree: a molecule that is a scaled-up version of a training molecule points the same way (high cosine) but sits far away (large distance).

## 7.15 Is the whole library inside? The two-sample test

The applicability domain answers the question one molecule at a time. A screening run asks a different one — is this library drawn from the same population as the training set, or is the whole exercise an extrapolation? — and averaging a per-molecule verdict does not answer it: two sets can have identical nearest-neighbour similarities and still occupy different regions.

Every set of molecules has a **mean field**, the mean of its feature maps, which for CMF is itself a function of space. The maximum mean discrepancy (Gretton *et al.*, JMLR **13** (2012) 723) is the distance between two of them:

$$\mathrm{MMD}^2(X,Y) = \overline{K(x,x')} - 2\,\overline{K(x,y)} + \overline{K(y,y')}$$

— three averages over blocks of the kernel matrix and nothing else. The unbiased estimator leaves the self-similarities out of the two within-set terms, which is why it can come out slightly negative: that is how "no difference at all" looks once the bias has been removed, not a defect. The null distribution comes from permuting the labels, which costs three block means per shuffle because the kernel is computed once.

```
python cmf_cli.py compare library.sdf --model model.cmf --permutations 999
python cmf_cli.py compare seriesA.sdf seriesB.sdf --charges file
```

With `--model` the library is compared with that model's training set, in that model's own fields, reusing the cross-kernel the prediction computes anyway. The p-value is never zero: it is one plus

the number of shuffles that matched or beat the observed value, over one plus the number of shuffles, because a finite number of them cannot support a claim of impossibility.

**The witness says where the difference lies**, which is the useful form of a p-value. It is the function whose expectation separates the two sets — a molecule's mean kernel value against the first set minus its mean against the second — evaluated at every molecule: most negative for the compounds that make the library unlike the training set, most positive for those most typical of it. The command prints both ends and `--out` writes the whole column. On the Sutherland benzodiazepine set against the COX2 one, the witness names nordazepam and delorazepam as the most typical benzodiazepines and the COX2 sulphonamides as the least — without being told that there are two targets.

### Is the test any good?

A test is useless unless it is calibrated as well as powerful, and `benchmarks/run_structure.py` measures both halves. Twenty random half-splits of each of the eight Sutherland sets, where there is nothing to find: **5.6 % are called significant at the 5 % level**, which is what a correctly calibrated test does and the permutation construction guarantees. All twenty-eight pairs of *different* targets, where there is everything to find: **100 % are called significant, median z = +171**. The test is calibrated where there is nothing and saturated where there is.

## 7.16 How certain is that number?

A difference of 0.02 in $q^2$ is quoted all over the benchmarks in this repository. Some of those differences are real and some are the particular molecules they were computed over.

The **bootstrap** box of that row (`--bootstrap N`) resamples the (observed, predicted) pairs N times and reports a percentile confidence interval for $q^2$ and for $R^2_{pred}$. It is the uncertainty of the *statistic* given these predictions, which is exactly what decides whether two settings differ. On the 21 training steroids, $q^2 = 0.768$ carries a 95 % interval of [0.47, 0.89] — wide enough to show that a difference of a hundredth means nothing.

`--bootstrap-refit N` adds Cramer's bootstrap as well: resample the training set with replacement, refit, and score the molecules that were left out of the bag. That one measures the variability of the fit rather than of the statistic, and it is the more pessimistic of the two. The kernel is never recomputed — a resample is an index into the matrix — so a hundred refits cost about as much as one cross-validation.

## 7.17 How certain is one prediction? Conformal intervals

The bootstrap says how firm a figure of merit is. It says nothing about the molecule in front of you, and neither does anything else the program has offered so far: the Gaussian-process posterior spread is the spread of a model that believes its own kernel, which for molecular fields it should not.

Conformal prediction supplies the missing guarantee and requires almost nothing in return. Take the absolute errors the model makes on molecules it did not fit, let $q$ be their $(1-\alpha)$ quantile — the $\lceil (n+1)(1-\alpha) \rceil$-th smallest, the +1 being what makes the result finite-sample rather than asymptotic — and report $\hat{y} \pm q$. Then, as long as the calibration molecules and the new one are exchangeable, the truth lies inside at least $(1-\alpha)$ of the time, **whatever the learner, whatever the kernel, however wrong the model is**. A badly specified model does not lose its coverage; it pays with a wider interval instead.

```
python cmf_cli.py build data.sdf --charges file --conformal 0.9 --save model.cmf
python cmf_cli.py predict model.cmf new.sdf --out predictions.csv
```

`--conformal` calibrates on the errors of the cross-validation the program is running anyway, so it costs nothing and adds no step, prints the half width, and — when there is a test set — the coverage it actually achieved on molecules that were never part of the calibration. The rule is stored in the `.cmf` file, so `predict` prints an interval beside every number and writes `lower` and `upper` columns into the CSV. `--conformal LEVEL` on `predict` re-reads the stored errors at a different level; the errors are kept, not just the width.

### What it costs, measured

Calibrating on out-of-fold errors and then predicting with the model refitted on everything is not exactly exchangeable — the errors come from models fitted on nine tenths of the training set and the predictions from a model fitted on all of it — so the guarantee becomes an approximation, and the size of the gap is an empirical question. `benchmarks/run_conformal.py` answers it on the eight Sutherland sets, ten random 70/30 splits each, with the width in units of the activity (whose standard deviation averages 1.36):

| nominal | out-of-fold (`cv`) | held-out third (`split`) | GP posterior |
|---|---|---|---|
| 80 % | **80.3 %**, width 2.28 | 80.1 %, width 2.59 | 77.7 %, width 3.00 |
| 90 % | **89.9 %**, width 3.10 | 90.1 %, width 3.61 | 83.1 %, width 3.85 |
| 95 % | **95.1 %**, width 4.09 | 94.9 %, width 4.04 | 86.9 %, width 4.59 |

The cheap variant holds its level to a few tenths of a per cent at every level, and is **narrower** than the exact split version, which loses accuracy because it fits on less data after holding a third out. The Gaussian-process interval, which is what one would otherwise reach for, misses its level at all three — and misses it unpredictably: across the eight sets its 90 % coverage ranges from 63 % to 99 %, while every conformal rule stays between 89 and 96 %.

**Normalised conformal** — dividing each error by the distance of its molecule from the training set, so that the interval widens where the model extrapolates — is available as `--conformal-normalise`, and on these sets it gains nothing: it over-covers (93.4 % at a nominal 90 %) and is a third wider. Softening it until it stops over-covering turns it back into the plain rule. The error on these series simply does not grow with distance from the training set strongly enough for the normalisation to be useful, which is itself worth knowing about one's data.

Two limits worth stating. The guarantee is **marginal**, not conditional: 90 % of molecules are covered, not 90 % of *every kind* of molecule, so a subclass the model handles badly can be covered less often than that while the overall figure holds. And a level needs molecules to support it — 19 calibration errors for 95 %, 9 for 90 % — below which the program says the level is not reachable rather than returning its widest error and calling it an interval.

Regression only: conformal classification produces prediction *sets* rather than intervals, and is not implemented.

## 7.18 Three estimates of the error, compared

The program now offers three answers to *should I believe this number*, and they come from three different places: the **Gaussian-process posterior variance**, which is the kernel's own statement

about how much of this molecule lies in the span of the training set; the **applicability domain**, which is a geometric statement about how far it is from its nearest training neighbours; and the **conformal interval**, whose normalised form widens where the model is extrapolating. They are in three different units — a variance, a distance in field space, a width in log units of activity — so they cannot be compared by their numbers. What can be compared is the order they put the molecules in, because that is what they are used for: which prediction do I act on, which do I measure first.

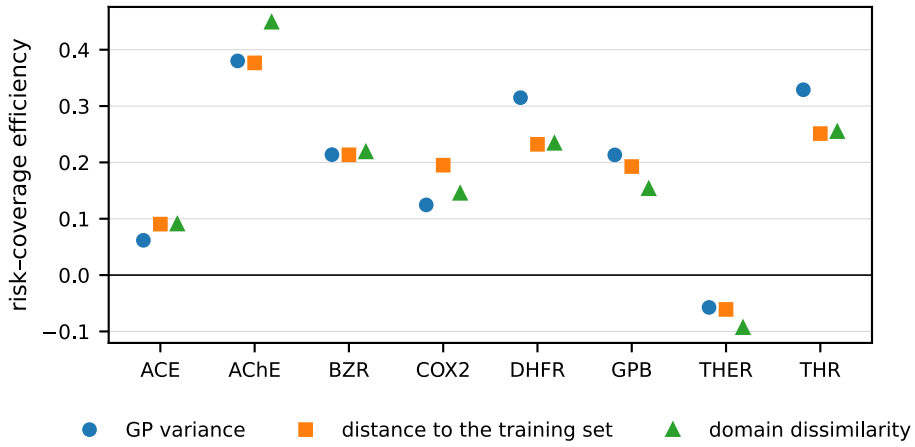
```
python cmf_cli.py build data.sdf --test-file test.sdf --domain \
       --conformal 0.8 --conformal-normalise --error-predictors
```

Each estimate is scored against the absolute error actually made on the test molecules, three ways. **Spearman's ρ** asks whether the order is right at all. The **risk–coverage efficiency** asks what is gained by acting on it: trust the most confident fraction, measure the RMSE of that fraction, and compare the area under that curve with two references — the *oracle*, which orders the molecules by the error they really made, and the flat curve of ignoring the estimate. One means the oracle, zero means no better than ignoring it, and a negative number means actively misleading. The **halves** are the crudest and most practical: the RMSE of the half it trusts against the half it doubts.

**Measured, the three are indistinguishable.** Eight Sutherland sets, five 70/30 splits each, α and λ chosen by the evidence on the training part, the conformal rule calibrated on that part's own out-of-fold errors:

| estimate | ρ | efficiency | RMSE trusted | RMSE doubted | ratio |
|---|---|---|---|---|---|
| GP variance | +0.142 | **+0.198** | 0.835 | 1.024 | 1.38 |
| distance to the training set | +0.125 | +0.186 | 0.853 | 1.009 | 1.31 |
| domain dissimilarity | +0.133 | +0.181 | 0.841 | 1.022 | 1.36 |
| normalised conformal width | +0.133 | +0.181 | 0.841 | 1.022 | 1.36 |

The GP variance is ahead by +0.011 in efficiency over the distance, with a 95 % interval of [−0.021, +0.044] — zero is inside it, and the choice between the three does not matter.




*The risk–coverage efficiency of the three estimates on each of the eight Sutherland sets, mean over five splits. One would be the oracle that knows the error of every molecule, zero an estimate that carries no information. The three move together from set to set, and on THER all three are below zero.*

Two things in that table deserve more attention than the ranking. The first is that the **normalised conformal width and the domain measure score identically, and must**: the normalisation divides the interval by the molecule's difficulty, which *is* its distance from the training set, so the width is a monotone function of the domain and cannot order the molecules differently. The conformal rule is justified by being calibrated — it covers what it says it covers, which neither of the others claims — and not by ordering. There are three estimates here, not four.

And the remaining two are nearly the same. Over the same splits the estimates agree with each other at a mean Spearman ρ of **+0.80 to +0.86** — the GP variance against the distance to the training set at +0.799, against the domain dissimilarity at +0.853 — while the conformal width against the domain measure is **+1.000**, exactly as the algebra says it must be. That is why the efficiencies in the table are indistinguishable: these are not three independent opinions about a molecule but one geometric fact, how far it sits from the training set, expressed in three sets of units.

The second is the size of all of them. The best captures about a fifth of what the oracle would, and the half it trusts has an RMSE of 0.84 against 1.02 for the half it doubts. That ratio of about 1.4 is real and useful — it is exactly the gain from measuring the confident half first — and it is nothing like a per-molecule error bar. On one of the eight sets (THER) every estimate comes out negative: an estimate of error is itself a model, and it can be wrong about a series. Use them to order molecules, and quote the conformal interval when a number has to be attached to one.

## 7.19 What the model thinks the groups are: clustering the kernel

Every grouped validation in this manual splits the molecules by their Bemis–Murcko scaffold, and the reasoning is sound: random folds put a molecule's own scaffold-mates in the training part and greatly overstate the quality of a mixed series. But it rests on an assumption nobody checks — that a scaffold is what makes two molecules alike *to the model*. The model does not see scaffolds. It sees fields, and its notion of similarity is the kernel.

*Spectral clustering* partitions that similarity directly. The kernel is already a similarity matrix, so there is nothing to build: form the normalised Laplacian, take the eigenvectors belonging to its smallest eigenvalues, put the rows on the unit sphere and run k-means there (Ng, Jordan & Weiss 2002). Groups that are not linearly separable in field space are separable in those coordinates, which is the point of the construction. How many groups is decided by the **eigengap** unless it is given, and the gap is reported beside the answer so that a weak one can be seen for what it is.

```
python cmf_cli.py cluster data.sdf --charges file
python cmf_cli.py cluster data.sdf --clusters 5 --out clusters.csv
```

The command prints the clusters, the spectrum they came from, the activity of each, and the **adjusted Rand index** between them and the scaffold groups — the number that says whether the usual grouping groups by what the model actually responds to.

### The two partitions do not agree, and it matters

Measured on the eight Sutherland sets, with both partitions cut into the same number of groups — holding out one of two clusters removes half the training set and holding out one of nine does not, so a difference in granularity would be read as a difference in difficulty:

| | $q^2$ |
|---|---|
| random 10-fold | +0.508 |
| leave one **scaffold group** out | +0.271 |
| leave one **field cluster** out | +0.057 |

and the adjusted Rand index between the two partitions averages **+0.140**, which is weak agreement: what the chemist calls a series and what the kernel finds similar are largely different things. On BZR they are different enough that holding out a field cluster takes $q^2$ to −0.740.

The consequence is uncomfortable and worth stating. This manual recommends the leave-one-chemotype-out figure as the more reliable one, and against random folds it is — 0.271 against 0.508. But it is not the hardest reliable test available: holding out a group of the model's own similarity costs another 0.214, because a scaffold group leaves molecules the kernel still considers close in the training part. Quote the scaffold figure, by all means; just do not read it as the worst case.

## 7.20 Where the validation appears

The three checks of *Building a model* — y-randomisation, the bootstrap intervals and the applicability domain — are set in one place and reported in another, which is worth stating plainly because there is no button and no separate window for them.

**Setting them**: the *Validation* row at the foot of the model panel's second group, directly under *External CV*. Three controls: **shuffles** (0 = off), **bootstrap** (0 = off), and the **applicability domain** tick. They are settings, not actions — choose them before pressing **Build model**, and they run as part of that build, after the fit and the test prediction.

**Reading them**: the *Statistics* tab of the results panel, where they are appended below the usual figures under a heading *Validation*. The names of the molecules that fell outside the applicability domain go to the *Log* tab instead, with their similarity values, so that the statistics stay a summary and the list of molecules outside the domain stays with the rest of the log.

Three cases produce no validation block at all, and each says why rather than leaving a blank:

- all three controls at their defaults — nothing was asked for;
- a classification or one-class model — y-randomisation and the bootstrap are implemented for regression only, and the block says so;
- the autocorrelation kernel source — the applicability domain is measured in the field kernel, which that source does not provide.

## 7.21 Statistics reported

Regression: $r^2$ and RMSE for the fit, $q^2$ and RMSEcv for the cross-validation, $R^2$pred and RMSEpred for the test set in both variance conventions, $q^2$ex and RMSEcvex for the external CV. Classification: accuracy, balanced accuracy, sensitivity, specificity, Matthews correlation, AUC and the ROC curve, each for the fit, the cross-validation and the test set. Screening: AUC, enrichment factors at 0.5, 1, 5 and 10 %, the best achievable enrichment at each of those fractions, BEDROC and RIE.

## 7.22 Saving and reusing

*File → Save model...* writes a `.cmf` file (joblib) holding the field set, the kernels' definition, the fitted coefficients, the training molecules, every setting needed to predict later, and the record of how the training set was prepared that *What has to match between fitting and predicting* describes. `cmf_cli.py predict model.cmf new.sdf --out predictions.csv` applies it, after checking the new molecules against that record; the GUI loads it with *File → Load model...* and writes the record's summary to the log.

## 7.23 A report to hand to a colleague

A `.cmf` file is for the program. The person who has to judge the model — a colleague, a reviewer, a supervisor — will never start the application, and what they need is one file they can open.

```
python cmf_cli.py report model.cmf --out model.html
```

That writes a single self-contained HTML page: every picture is inside it, so it survives being emailed, attached or put on a share, and it opens in any browser and prints to PDF. It has seven sections, in the order a sceptical reader asks for them.

- **What this model is** — fields, α, learner, kernel, how the charges were made.
- **How well it fits and predicts** — the statistics with the observed-against-predicted plots beside them, because a $q^2$ is a summary of that picture and not a substitute for it.
- **Whether to believe that** — the validation that was actually run: y-randomisation, the bootstrap intervals, the applicability domain, the conformal level, the comparison of error predictors. **A check that was not run is reported as not run**, never silently omitted, which is the difference between a report and an advertisement.
- **What the model rests on** — the mean absolute contribution of each field, which is what separates a model standing on its geometry from one standing on a single field.
- **The co-fields**, projected onto the three coordinate planes from fixed viewpoints: the strongest signed value along each viewing axis, with the training atoms as grey dots for scale. A projection rather than an isosurface, because a report has no camera and its reader has no mouse.
- **The extremes of the series** — the most and least active training molecules, each decomposed over the fields by exact Shapley values and over its own atoms by the contribution rule.
- **The pharmacophore** read out of the co-fields, and the provenance of the model file.

Everything comes out of the model file, so a report costs no refitting and **cannot disagree with the model it describes**. Missing parts are reported rather than causing a failure: a nonlinear kernel has no co-fields, no atom contributions and no pharmacophore, and the report says so in place of each and keeps the rest; a model fitted without a test set reports no $R^2$pred.

`--no-cofields` leaves out the pictures, which are the slow part — a report of a 50-molecule model takes about three seconds with them and under one without — and `--spacing` sets how finely they and the pharmacophore are computed. `--open` shows it in the browser straight away.

# 8 Reading the model

A CMF model is readable in a way a grid model is not, because the prediction can be taken apart exactly. The 2019 paper states three decomposition rules, and the program implements all three.

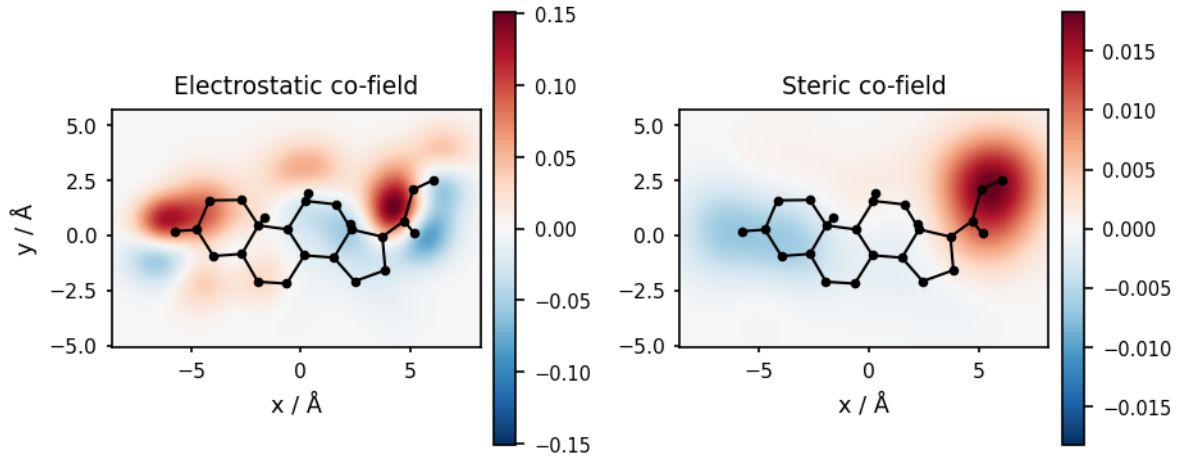
## 8.1 Co-fields

The co-field is the model as a function of space:

$$C_f(\mathbf{r}) = \sum_j a_j \, s_j \, X_{fj}(\mathbf{r})$$

a weighted sum of the training molecules' fields with the fitted dual coefficients as weights, each divided by the norm $s_j = K_f(j,j)^{-1/2}$ of its own field when the kernels are normalised, which they are by default. The prediction of any molecule is then the overlap of its own field with the co-field, field type by field type. Press **Compute co-field** and the program evaluates it on a grid around the training set; choose the field with the *Field* selector and set the *iso level* slider, which is a percentage of the largest absolute value rather than an absolute number, so the surfaces stay meaningful when you change field.

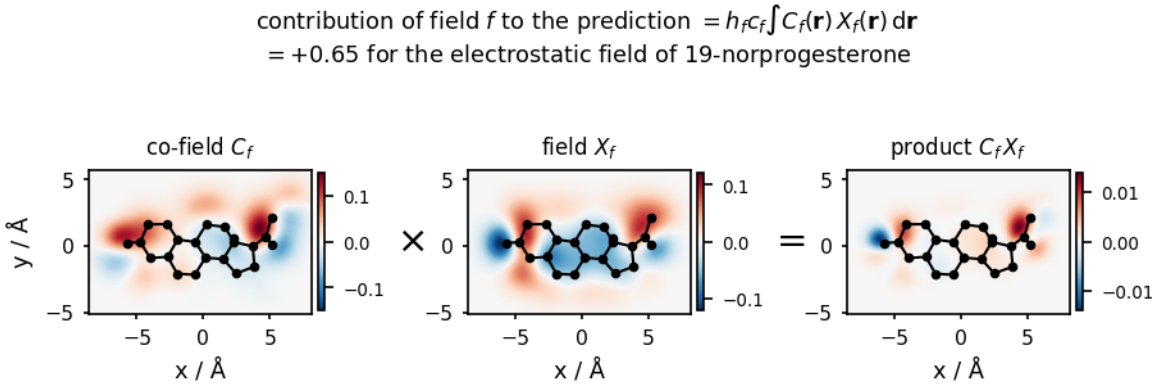
What the lobes mean: a positive co-field lobe is a region where *more* of that field raises the predicted activity. For the electrostatic field that means positive charge is favoured there, for the steric field bulk, for the hydrophobic field lipophilicity, and for an indicator field it means an atom of that type is favoured at that position.




*The electrostatic and the steric co-field of a model of the steroid training set (α = 1, λ = 1, q² = 0.79), cut in the mean plane of the series, with corticosterone, the most active training molecule, drawn over them. Red marks a region where more of that field raises the predicted affinity: positive charge for the electrostatic field, bulk for the steric one. The lobes sit at the two ends of the skeleton, which is where the series varies.*

Two further maps use the same grid, for the molecule currently shown:

- the **molecular field** $X_f(\mathbf{r})$ of that molecule, in the same colours. The two share a colour scale, so only the drawing style tells them apart: the style box is `auto` by default, which draws this one as a mesh while a co-field is on screen and solid when it is the only surface shown;
- the **overlap** $C_f(\mathbf{r})\,X_f(\mathbf{r})$, whose integral *is* that field's contribution to the prediction (Eq. 7), so the map shows where the prediction comes from.




*The prediction as an overlap, for the electrostatic field of one test molecule of the steroid set and the model of the previous figure: the co-field of the model, times the molecule's own field, gives the product whose integral over space, with the field's scale and mixing coefficient, is that field's contribution to the predicted affinity. The prediction is the sum of these contributions over the fields plus the bias. This is Fig. 1 of the 2013 paper drawn with the program's own maps; the application shows the same three maps in 3D for the molecule on screen.*

*File → Export current co-field (Gaussian cube)* writes a cube file readable by PyMOL, VMD or Jmol, which is the way to put a co-field into a publication figure.

## How much of it is noise

A co-field is read as a fact about the chemistry. It is a linear map of the measured activities, so part of its shape is a fact about the measurement error instead, and the program can say which part in closed form.

The dual coefficients are a linear map of the activities — with a bias, the bordered system of the 2013 paper, whose inverse's top-left block is the matrix below — so activities carrying independent noise of variance $\sigma^2$ give coefficients carrying covariance $\sigma^2\, G\, G^{\mathrm{T}}$, and the co-field, being linear in them, carries

$$\mathrm{Var}\,[C_f(\mathbf{r})] = \sigma^2 \left\| G^{\mathrm{T}} \mathbf{x}_f(\mathbf{r}) \right\|^2, \qquad x_{fj}(\mathbf{r}) = s_{fj} X_{fj}(\mathbf{r})$$

— one matrix product per block of grid points, no resampling and no refitting. The **hide noise** tick of the 3D view then draws only what lies more than two standard deviations from zero, and `pharmacophore --min-z 2` reads features off the same masked map, so that a lobe the noise could have produced never becomes a feature. The report (`report`) does both by default.

The noise level σ is the model's own out-of-fold error, RMSEcv, which is the assay's noise and the model's error together; `--sigma` sets it directly.

**What this is, precisely.** It is not the Bayesian posterior variance of the co-field, and the difference is worth stating. Under the Gaussian process whose kernel this is, the prior on the field weight function is white noise on $L^2$: its value at a point is not a random variable at all, and the pointwise posterior variance is infinite everywhere. What the prior does allow is the variance of $\langle w, g \rangle$ for a probe field $g$ of non-zero width — which is exactly the GP variance of a prediction for a one-atom molecule placed there, and the program already computes that. The quantity above is the other half of the question: the molecules and the kernel are held fixed, and the noise in the activities is propagated. It answers *how much of this lobe would survive the error in the measurements*, which is the question a co-field picture provokes.

**Measured, the mask keeps about a third of the picture, and that third is the part that is stable.** Eight Sutherland sets, co-fields on a 0.8 Å grid; the training set is then resampled with

replacement thirty times and refitted, and a voxel counts as stable when the sign of its co-field survives that — a harder perturbation than the formula models, since it moves the molecules too:

| | |
|---|---|
| kept by the mask, of what would have been drawn | **30 %** |
| of the kept part, holds its sign under resampling | **97.8 %** |
| of the cut part | **90.6 %** |
| difference | **+0.072**, 95 % interval [+0.064, +0.083], in the same direction on 8 of 8 sets |

One more figure, for completeness: the bootstrap moves the co-field only about 0.6 times as far as the closed form allows for, so cutting at two standard deviations is really cutting at about 3.2 of the bootstrap's — the threshold is conservative, which follows from taking σ to be the out-of-fold error. What it gets right is *where* the uncertainty is, and the two orderings agree across the grid at a rank correlation of +0.82. That is what a mask needs, since a mask is an ordering with a cut-off.

**A co-field at a point, and what an atom there would do.** A co-field is a value per unit of field, which is not a number anyone can check against anything: a field is not something one can put somewhere. An atom is. `cofields.atom_effect` gives the overlap of the co-field with one atom's own Gaussian, which is the same closed-form integral the kernel sums over atom pairs, so the answer is in the units of the activity -- ppm for a shift, log units for an affinity -- and says what adding that atom there would add to the prediction. For a continuous indicator field the atom of its type carries a weight of one, so the number is directly "what one oxygen here does"; for the electrostatic field the weight is a partial charge and has to be given. `cofield_at_points` reads the co-field itself at a list of places, `cofield_covariance_at` and `atom_effect_covariance` give the error that the noise in the activities allows, and `contrast_at` compares two places with the covariance between them taken into account, which matters because two points a few angstrom apart are built from the same training fields and their errors are far from independent. Plan item LP3 uses all of this to read a co-field at the ortho, meta and para positions of a ring; nothing in it is specific to that use.

## 8.2 Contributions

The *Contributions* tab answers two questions about the selected molecule:

- **Which training molecules** make its prediction, as $a_j K(X, X_j)$ — the similarity decomposition, rule 2. A prediction resting on one or two training molecules is a warning.
- **Which of its atoms**, as the atom contributions of Eq. 14 of the 2015 paper, drawn in the 3D view as red and blue spheres whose size is the magnitude. This is the map that is most useful for design: it says *this* substituent is helping and *that* one is not.

## 8.3 Shapley values

The *Shapley* tab decomposes a single prediction into additive contributions by the Shapley construction, over **atoms**, over **substituents of the common scaffold**, or over **field types**, with second-order interaction indices available. It serves two purposes. With the linear kernel it reproduces the atom contributions exactly, which is a useful check. With a nonlinear kernel layer, where co-fields do not exist, it is the only decomposition left — and it still works, because it needs nothing but the ability to evaluate the model on subsets.

## 8.4 A pharmacophore out of the co-field

*Tools → Extract pharmacophore from the co-field...* turns the co-field into a discrete, typed pharmacophore: the favourable lobes of each field become feature points of the matching kind (positive and negative centres, hydrophobe, donor, acceptor), each with a tolerance radius taken from the size of its lobe, and the unfavourable steric lobes — or the pocket, when a protein is loaded — become exclusion volumes. The result is a table and a PDB-style file of feature spheres that other software can read, and it draws in the 3D view; untick *pharmacophore* in the viewer controls to clear it.

This is the bridge from a CMF model to a screening query: the model is continuous and needs an alignment, the pharmacophore is discrete and does not.

### Taking the query out of the program

```
python cmf_cli.py pharmacophore model.cmf --min-z 2 --query query.json
python cmf_cli.py pharmacophore model.cmf --min-z 2 --query query.pml --enable-top 3
```

`.json` writes a **Pharmit** session and `.pml` the **LigandScout-style XML**; both are written to the documented shape of those formats, which is as far as this program can go — it cannot test any particular viewer. Three things about the file are worth knowing before it is opened somewhere else.

**A query is a conjunction**, so a dozen features at once matches nothing. The strongest `--enable-top` features are written switched on and the rest switched off, for the user to choose among in their own program; exclusion spheres are always on, since they only forbid.

**Hydrogen-bond features have no direction.** A co-field says that a donor *here* raises the activity; it says nothing about where the hydrogen points, so the features are written as isotropic points rather than as the vectors those formats usually carry.

**Three of this program's feature types cannot be expressed at all** and are reported rather than quietly dropped. `bulk` asks for atoms of any kind, which no query language expresses. The disfavoured `polar`, `acceptor_bad` and `donor_bad` say that one *particular* kind of group is unwelcome; the only negative a query has is an exclusion sphere, and that would forbid every atom, which is a much stronger claim than the co-field makes.

### Is the query any good, measured

Eight Sutherland sets, five 70/30 splits each. The pharmacophore is read out of the training model's co-fields — masked at two standard deviations first, so that a lobe the measurement error could have produced never becomes a feature — and its strongest four features are matched against the molecules held out, by the simple rule a screening program uses: an atom of the right kind inside the sphere. The actives are the most active quarter.

| set | n | features | AUC query | AUC random centres | AUC the model |
|---|---|---|---|---|---|
| DHFR | 361 | 7.4 | 0.573 | 0.498 | 0.901 |
| COX2 | 282 | 8.4 | 0.662 | 0.478 | 0.867 |
| BZR | 147 | 7.8 | 0.518 | 0.514 | 0.797 |
| ACE | 114 | 7.0 | 0.789 | 0.552 | 0.927 |
| AChE | 111 | 7.8 | 0.631 | 0.484 | 0.903 |

| set | n | features | AUC query | AUC random centres | AUC the model |
|---|---|---|---|---|---|
| THR | 88 | 8.4 | 0.595 | 0.468 | 0.780 |
| THER | 76 | 7.2 | 0.711 | 0.482 | 0.812 |
| GPB | 66 | 6.8 | 0.838 | 0.484 | 0.902 |
| **mean** | | | **0.665** | **0.495** | **0.861** |

The query separates the active quarter at an AUC of 0.665, against 0.495 for the same features moved to random places in the same box — a paired difference of +0.170 with a 95 % interval of [+0.102, +0.238], so the places the co-field picked are doing the work, and ahead on 36 of 40 splits. Over every split together the top tenth by the query holds 1.77 times as many actives as chance would put there.

**The model it was read from reaches 0.861 on the same molecules**, and that gap is the price of the summary: a pharmacophore is four spheres and a rule for counting them, while the model is every training molecule's field weighed by a fitted coefficient. What the query gains is that it can leave this program — it goes into a screening tool and runs over libraries no kernel method here could hold.

## 8.5 What a candidate is missing

*Tools → Field difference from a reference molecule...* subtracts two molecules' fields on the co-field grid and reports, per field type, the cosine between them, the distance, and how much of the reference's field the candidate **fails to reproduce** against how much it **adds**. The difference map draws as its own isosurface, and the table ranks the whole data set by field similarity to the reference. In a lead-optimisation reading: the "missing" column is what to put back, the "added" column is what may be causing trouble.

## 8.6 Several endpoints at once: the selectivity co-field

A separation problem, or any selectivity, is not one property but two: an extractant has an affinity for americium and an affinity for europium, and what matters is the difference. A model of the ratio works, but it throws information away. Each affinity has its own measurements, often on partly different compounds; a ligand that binds both metals strongly looks, to a model of the ratio, exactly like one that binds neither; and the ratio cannot say *which* of the two affinities a substituent moved.

*Tools → Model several endpoints (selectivity co-field)...* fits every endpoint you tick on the kernel of the model you have already built, with one bias per endpoint and a task covariance that lets them share information with each other:

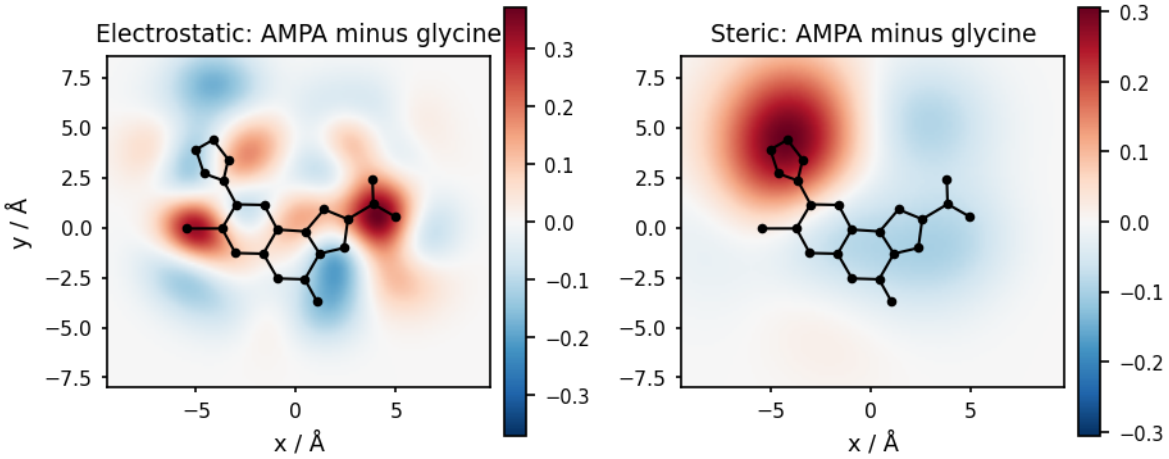
$$y_t(X) = \sum_j c_{jt}\, K(X, X_j) + b_t,$$

where the coefficients of the two endpoints are tied through a single correlation ρ. At ρ = 0 the endpoints are modelled separately and share only the kernel; as ρ rises they share information; at ρ = 1 the two sets of coefficients coincide and the model could only predict a *constant* selectivity, so 1 is excluded (and ρ may be negative, for endpoints that move in opposite directions). The dialog searches ρ and the ridge by cross-validation, and the cross-validation holds out whole **molecules**, never single measurements: with a molecule's first endpoint in the training set and its second in the test set, the correlation would reveal the answer.

Because each endpoint's prediction is linear in the field with its own coefficients, each endpoint has a co-field, and so does their difference — the **selectivity co-field**

$$C_A(\mathbf{r}) - C_B(\mathbf{r}) = \sum_j (c_{jA} - c_{jB})\, X_j(\mathbf{r}),$$

a map of where more of a field raises the first endpoint more than the second. Only a model linear in the field can produce it; it is available here for the same reason co-fields are available at all, and it is refused for a model with a nonlinear kernel layer.




*The selectivity co-field of the 57 AMPA / glycine-NMDA antagonists of* `examples/selectivity`*, AMPA minus glycine, in the plane of the tricyclic core, with the most AMPA-selective compound drawn over it. A red lobe is a place where more of that field shifts the preference towards the AMPA receptor, a blue one towards the glycine site. The two-endpoint fit has its correlation and ridge chosen by cross-validation.*

Once the fit is made, an **Endpoint** box appears in the *Maps* row of the control strip with one entry per endpoint and one for the difference of the pair you named — and every map, export and contribution table then refers to the chosen one; a cube exported while the selectivity is chosen carries it in its file name. The endpoints are numeric properties of the data set, so the ones beyond the first are imported with *File → Import activities from CSV...* or read from columns of the file. Molecules that carry only some of them are kept: they inform the endpoints they have, and through the covariance the others. Each endpoint needs at least five molecules.

What the chosen ρ means is itself a result. On the corrosion-inhibitor series of the benchmark (`benchmarks/results/multitask/table.md`) the cross-validation chose ρ = 0 on the two largest series and a non-zero ρ of either sign on the small ones, where it gained at most 0.04 in $q^2$: an endpoint that already has its own molecules gains little from another, and the ρ the dialog reports in the log is the measurement that says how much.

## The same model from the command line

`build --second-endpoint COLUMN` fits the pair without the application. The second endpoint is a numeric property of the same file, the correlation is chosen by the marginal likelihood unless `--task-rho` gives it, and the cross-validation that is reported holds out whole molecules. The fitted model then predicts the primary endpoint with the coefficients the two-task fit gives it, and carries the other endpoint with it:

```
cmf build guests.csv --second-endpoint alpha_logK --save model.cmf
cmf predict model.cmf new_guests.csv --out predictions.csv
cmf cofield model.cmf --endpoint selectivity --out-prefix maps
```

`predict` prints both endpoints and their difference and writes them as two more columns; `cofield --endpoint second|selectivity` draws the other endpoint's maps or the selectivity ones. A molecule with no primary activity is dropped before any model is built, as always, so the primary endpoint has to be the one every row carries; the second may be missing anywhere, which is the case the model exists for. `examples/make_two_endpoint_example.py` joins two of the cyclodextrin tables into such a file.

**Two cavities, measured.** `benchmarks/run_two_endpoints.py` takes the pairs of cyclodextrin hosts: the guests of two tables joined on the structure of the guest, one rotation-invariant kernel, the correlation chosen by the evidence, ten-fold cross-validation over whole molecules, and `alone` the same code at a correlation of zero, which is exactly the two separate fits.

| pair | guests (both) | correlation | first endpoint, shared - alone | second endpoint |
|---|---|---|---|---|
| beta, alpha | 1493 (496) | +0.788 | -0.004 [-0.011, +0.004] | +0.005 [-0.008, +0.019] |
| beta, gamma | 1253 (201) | +0.869 | -0.001 [-0.006, +0.004] | +0.032 [-0.011, +0.082] |
| alpha, gamma | 938 (181) | +0.706 | -0.002 [-0.009, +0.004] | -0.008 [-0.029, +0.016] |

**Sharing changes nothing here**, which is the same answer the corrosion series gave: every interval contains zero, the largest movement being +0.032 for the gamma endpoint when it borrows from beta's 1162 guests. An endpoint with a few hundred molecules of its own already has what it needs, and what the two endpoints have in common is carried by the kernel they share before any correlation is fitted.

**The selectivity is a different matter.** It is predictable -- q2 +0.273 for beta against alpha on the 496 guests measured against both, +0.223 for beta against gamma and +0.389 for alpha against gamma -- and **what it rests on is the size of the guest**, which is the known difference between these cavities. The correlation between the predicted selectivity and the number of heavy atoms follows the ordering of the three cavities exactly: beta over alpha rises with size (+0.468, the measured values +0.382), while beta over gamma falls (-0.739 against -0.480) and alpha over gamma falls further (-0.676 against -0.613). A big guest prefers the bigger cavity, a small one the narrower, and the model says so without being told that a cavity has a size.

## Beyond one correlation

One ρ says that every pair of endpoints is related to the same degree. For two endpoints that is no assumption at all; for seven it is a strong one, and wrong in the usual case where some endpoints are nearly the same and others are only weakly related. The dialog's **Task covariance** box therefore offers a free-form matrix,

$$\mathbf{T} = \mathbf{L}\mathbf{L}^{\mathrm{T}} + \mathrm{diag}(\mathbf{d}),$$

with **L** of shape (endpoints, rank) and **d** > 0. It is positive definite for every value of its parameters, which is why nothing has to be constrained; at rank 1 it contains the one-ρ model exactly, so the comparison is meaningful; and its **diagonal is free**, which matters whenever the endpoints are in different units — a potency and a free energy in kJ/mol need their own amplitudes, and one shared correlation cannot give them one.

Such a covariance has (rank + 1) × endpoints parameters, which is far too many for cross-validation: ten fits per candidate and several hundred candidates. It is chosen instead by the **marginal likelihood**, the multi-endpoint form of the criterion of *Choosing α and λ without holding anything out*: the mean is marginalised over the design columns under a flat prior — one constant per endpoint, or the task descriptors below — and the kernel amplitude is profiled out, so one factorisation scores a candidate and the value does not depend on the units of the activities. `cmfqsar/multitask.log_evidence` is that criterion and `fit_covariance` optimises it by Nelder–Mead from several starts, because the evidence over a covariance is not concave and a start at ρ = 0.8 and one at ρ = 0.3 do not always converge to the same answer.

## An endpoint nobody measured

A free-form row for an endpoint with no measurements is unidentifiable: there is nothing to fit it to. A row read off **task descriptors** is not. Give the endpoints their own descriptor vectors $\mathbf{z}_t$ — the medium, the metal, the temperature; in a host-guest problem the structure of the host itself — and set

$$T(t,s) = (1-\varepsilon)\exp\left(-\frac{|\mathbf{z}_t - \mathbf{z}_s|^2}{2\ell^2}\right) + \varepsilon\,\delta_{ts},$$

whose length scale ell and nugget varepsilon are chosen by the same evidence (`descriptor_covariance`). The nugget is not optional: without it two endpoints with identical descriptors would be perfectly correlated, the covariance singular, and their coefficients forced to coincide. A `linear` form is available too, and is the one to use when two endpoints are expected to be *anti*-correlated, since an RBF kernel cannot be negative.

The mean needs the same treatment, since an endpoint that was never measured has no bias to fit. Passing `task_features` to `Observations` replaces the indicator columns of the bias with the descriptors themselves, so that the mean is linear in them and extrapolates; with no features the design is the indicator matrix and the behaviour is exactly as before. Then

```
from cmfqsar.multitask import Observations, descriptor_covariance, fit
obs = Observations.from_columns(Y, names=hosts, task_features=Z_with_intercept)
dc = descriptor_covariance(K, obs, Z)
f = fit(K, obs, lam=dc.lam, T=dc.T)
y_new = f.predict_new_task(K_cross, dc.row_for(z_new), features=z_new_with_intercept)
```

predicts the unmeasured endpoint, and `f.new_task_coefficients(dc.row_for(z_new))` gives it a co-field like any other. What this cannot do is invent what is peculiar to that endpoint: it interpolates between the endpoints that were measured and inherits only what they have in common.

## What it measures

Measured on `examples/cyclodextrin/` (`benchmarks/run_tasks.py`), which has seven real endpoints and descriptors for them: 1669 distinct guest molecules, 2671 association constants against the seven cyclodextrin hosts (beta 1162, alpha 827, gamma 292, hp_beta 135, sbe_beta 135, methyl_beta 77, dimethyl_beta 43), and the hosts are molecules too, so the task descriptors are a structure rather than a label — cavity size, lipophilicity, charge, hydroxyl count, heavy atoms. One kernel for every row, the rotation-invariant field autocorrelation at α = 0.1 (there is no frame available here, and *Working without an alignment* found that kernel the best of the three on this data), and 10-fold cross-validation over **guests**, so that no guest has one host in the training set and another in the test set.

| task covariance | mean $q^2$ | beta | alpha | gamma | hp_beta | sbe_beta | methyl_beta | dimethyl_beta | what it is |
|---|---|---|---|---|---|---|---|---|---|
| independent | **+0.412** | +0.448 | +0.338 | +0.473 | +0.545 | +0.331 | +0.377 | +0.374 | rho = 0 |
| one rho, CV | **+0.454** | +0.448 | +0.338 | +0.524 | +0.586 | +0.389 | +0.441 | +0.454 | rho = +0.90 |
| one rho, evidence | **+0.454** | +0.449 | +0.341 | +0.523 | +0.592 | +0.391 | +0.435 | +0.451 | rho = +0.857 over 25 values |
| free rank 1 | **+0.475** | +0.447 | +0.339 | +0.525 | +0.604 | +0.444 | +0.504 | +0.465 | 2 x 1 starts, 1603 evaluations of the evidence |
| free rank 2 | **+0.488** | +0.447 | +0.340 | +0.527 | +0.624 | +0.452 | +0.536 | +0.490 | 2 x 1 starts, 2274 evaluations of the evidence |
| descriptors | **+0.448** | +0.446 | +0.342 | +0.505 | +0.608 | +0.390 | +0.430 | +0.413 | rbf, length scale 4, nugget 0.1 |

**Sharing is worth +0.076 mean $q^2$** — +0.412 fitting the seven hosts independently on the same kernel against +0.488 for the free rank 2 covariance — and the gain is largest on the endpoints with fewest measurements, which is what sharing information between endpoints is supposed to do (the gain and the number of measurements correlate at -0.89, and the biggest gain goes to methyl_beta, while the smallest set is dimethyl_beta). A free-form covariance beats one shared correlation by +0.034, so with seven endpoints the single number misses part of the gain.

The evidence was the objective for every row but the cross-validated one, and it is worth checking whether it chose what cross-validation would have:

| task covariance | log evidence | mean $q^2$ |
|---|---|---|
| free rank 2 | -2605.3 | +0.488 |
| free rank 1 | -2608.8 | +0.475 |
| one rho, evidence | -2635.3 | +0.454 |
| one rho, CV | -2635.8 | +0.454 |
| descriptors | -2647.6 | +0.448 |
| independent | -2713.1 | +0.412 |

It did: the covariance with the highest evidence is also the one with the highest out-of-fold $q^2$, which is the agreement that is assumed without a check elsewhere in this manual.

## An endpoint nobody measured, measured

Each host was then removed **entirely** — the whole endpoint, not a fold of its guests — and predicted from the other six through the descriptor covariance, with the mean linear in the descriptors so that there is something to extrapolate. Spearman ρ is the column to read: the absolute scale of an endpoint nobody has titrated is not recoverable, and a screen ranks.

| host removed | n | nearest host | ρ, descriptor covariance | ρ, nearest host alone | ρ, mean of the measured hosts | $q^2$, descriptor |
|---|---|---|---|---|---|---|
| beta | 1162 | sbe_beta | +0.577 | +0.320 | +0.623 (615) | +0.231 |
| alpha | 827 | beta | +0.431 | +0.433 | +0.470 (522) | +0.066 |
| gamma | 292 | sbe_beta | -0.284 | +0.145 | +0.533 (225) | -5.499 |
| hp_beta | 135 | sbe_beta | +0.664 | +0.294 | +0.806 (105) | +0.483 |
| sbe_beta | 135 | beta | +0.516 | +0.515 | +0.498 (74) | +0.200 |
| methyl_beta | 77 | dimethyl_beta | +0.595 | +0.529 | +0.894 (73) | +0.416 |
| dimethyl_beta | 43 | methyl_beta | +0.703 | +0.708 | +0.863 (38) | +0.044 |

Mean ρ **+0.457** through the descriptor covariance, against **+0.421** for the single-task model of the nearest host by descriptors and **+0.670** for simply averaging the other hosts' measured values of the same guest. So the covariance does carry real information about an endpoint it has never seen — it beats the nearest host's own model 4 times out of 7 — and it does not beat the cheapest baseline of all. That baseline is not a model, which is its only disadvantage: it can only say something about a guest some other host was measured against, and says nothing about a new molecule. The $q^2$ column is the necessary warning: it is negative on 1 of the 7 hosts, so the numbers this route produces are a ranking and not a binding constant.

What governs the spread is whether the removed host is interpolated between the others or extrapolated to. Cavity size is the first descriptor, and of the seven hosts five have seven glucose units — β-cyclodextrin and its four derivatives — while **α** has six and **γ** has eight. Remove one of those five and the covariance interpolates; remove α or γ and it has to extrapolate outside the range it was built on. The five interpolated hosts average ρ **+0.611** against **+0.473** for the nearest host's own model, a margin of +0.138; the two extrapolated ones give +0.431 (α) and **−0.284** (γ, with $q^2$ −5.50, a prediction that is worse than useless).

So the claim this route supports is *interpolation between endpoints that were measured*. γ is the warning: it is not merely the largest cavity but the only one of its size, the task kernel has no neighbour to take its shape from, and the mean — linear in the descriptors — is extrapolated past every host it was fitted on. The length scale the evidence chose is itself the symptom: 4 standardised units, which is the *top of the grid* on every one of the seven fits, so the criterion would

prefer an even wider task kernel. With seven endpoints there is not enough to localise it, every endpoint is treated as a neighbour of every other, and that is generous towards γ and wrong.

## 8.7 Comparing two models by their co-fields

The selectivity co-field above is the *difference* of two co-fields. Their *inner product* is the other half of the same algebra, and it answers a different question: how far do two models ask the same thing of the chemistry?

```
python cmf_cli.py transfer targetA.cmf targetB.cmf
python cmf_cli.py transfer round1.cmf round2.cmf round3.cmf --fields --out map.csv
```

It needs no grid. With $C_f^A(\mathbf{r}) = \Sigma_j\, a_j\, s_{fj}\, X_{fj}(\mathbf{r})$ and the same for B, the overlap is a sum of overlap integrals of atom-centred Gaussians — which is the field kernel between the two training sets:

$$\langle C_f^A, C_f^B \rangle = \sum_{j,k} a_j b_k s_{fj} s_{fk} K_f^{\mathrm{raw}}(j,k) = \mathbf{a}^{\mathrm{T}} \mathbf{K}_f(A,B)\, \mathbf{b}$$

one matrix product on a block the program computes for a prediction anyway. The number to read is the **cosine**, because it does not move when either activity is rescaled, when the mixing coefficients change, or when the field scaling does: it compares the *shapes* of two demands rather than their sizes. A unit test checks it against the same integral taken on a 175 000-point grid; they agree to four decimals.

What it is for, in the order the question arises. **Two targets**: a positive cosine says both want the same thing in the same place, which is why selectivity is hard, and a negative one says they want opposite things, which is where selectivity comes from. **Two series on one target**: agreement means the SAR transfers, disagreement means two chemotypes are being fitted by two different stories and a model pooled over both will be an average of them. **One series across rounds**: the cosine with the previous round says how much the new compounds changed the model, and the per-field cosines say in which field.

### Is it a map of SAR transfer, measured

Every Sutherland set split into its chemotypes, a model on each group of fifteen molecules or more, the attenuation shared across the set, and every ordered pair scored twice: by the co-field cosine, and by how well one model orders the other group's molecules (Spearman's ρ — not $R^2$, since two chemotypes often sit at different activity levels and an offset is not a failure of the SAR).

| | mean cosine | mean transfer ρ |
|---|---|---|
| two halves of one chemotype (the control, 16 pairs) | **+0.401** | **+0.673** |
| two different chemotypes of one target (64 pairs) | +0.178 | +0.334 |

**The cosine separates the two cases by 0.22**, which is what it is for: two models of one SAR look alike and two models of two SARs do not. And within the chemotype pairs the cosine and the transfer **correlate at ρ +0.667** over the 64 pairs, which is the claim of this feature, measured — a table of co-field overlaps is a map of SAR transfer.

**Everything depends on one assumption.** Two co-fields are functions on one space only if their molecules were posed in one frame. The program checks what it can from the model files — the

same fields, the same attenuations, training boxes that overlap — and refuses with a reason when they fail. What it cannot check is whether two alignments *mean* the same thing, and nothing in the cosine would reveal that they did not. A cosine is also not a prediction: that two models want the same thing does not mean either is right.

## 8.8 Chemical space maps

*Tools → Chemical space map...* builds a two-dimensional map of the set from the CMF kernels. Fields are not vectors, so Generative Topographic Mapping cannot be applied to them directly; the program performs kernel PCA on the CMF kernel — which is functional PCA, and reproduces feature-space distances exactly — and runs the ordinary EM fit of GTM in those coordinates. The out-of-sample projection stays a single kernel row, so new molecules can be placed on an existing map.

**Two projections, and they answer different questions.** The *Projection* box chooses between the GTM landscape and the **kernel PCA score plot**, which draws the functional principal components themselves — by default the first two, with any other pair available. The GTM spreads the set over a square grid, which is what makes a landscape possible, but it is a nonlinear mapping: a tight cluster is pushed apart to fill the grid and no distance on the map can be read as a distance between molecules. The score plot is the linear projection the GTM is fitted on, so its distances are the kernel's own as far as the two eigenvalues reach, and each axis is labelled with the share of the kernel variance its component explains — on the steroid set, 32.9 % and 15.1 %. Ask the map *where the activity is*; ask the score plot *whether this series is one cluster or three*, and how much of the kernel a flat picture of it can genuinely show. The score plot needs no EM fit, so it also costs a fraction of the time.

From the responsibilities follow the **activity landscape** (responsibility-weighted mean activity per node), the **class landscape** (the fraction of actives) and the **data density**, which is a picture of the applicability domain; nodes with no support are left blank. The landscape is a model in its own right: on the thrombin training set it reproduces the activities with $R^2$ 0.92.

Two cautions. Use cosine-normalised kernels, or the leading components follow molecular size. And a CMF kernel between molecules that share no frame is meaningless, so a map spanning several series would partly be showing alignment quality — unless you use the pose-free count kernel of *Working without an alignment*, which separates three targets' ligands cleanly with no alignment at all.

# 9 Classification and virtual screening

## 9.1 Setting up a classification task

Choose *Binary classification* as the task. The classes come either from a threshold on the activity (with *active below* for data where small is potent, such as an IC50 in molar units) or from a class property in the file. The learners are C-SVC, ν-SVC and the Laplace Gaussian-process classifier; the *ROC* and *Scores* tabs replace the scatter plot.

```
python cmf_cli.py build examples/sutherland/AChE.sdf --charges file \
       --task classification --threshold 6.84 --method gpc --Cs 1,10,100
```

## 9.2 One-class models: screening with actives alone

Often there are no reliable inactives, only a set of known actives, and the question is whether a new molecule belongs to the same class. That is one-class classification, applied to CMF by Karpov et al. (2011). Five descriptions of the active class are implemented:

| Method | What it measures |
|---|---|
| `ocsvm` | the 1-SVM: a hyperplane in feature space separating the actives from the origin |
| `ocmean` | distance to the mean embedding of the actives, the centroid of the fields |
| `ocknn` | distance to the k-th nearest active |
| `ocgp` | the one-class Gaussian process, optionally scored by mean over standard deviation |
| `ockpca` | reconstruction error against the principal subspace of the actives (kernel PCA) |

On the sets measured here the 1-SVM is the **most fragile** of the five: the centroid, k-NN and kernel-PCA descriptions are as good or better and have one parameter fewer to go wrong. Try more than one; they disagree in interesting places.

## 9.3 Decoys and the figures that matter

A screening model is only as meaningful as the background it is tested against. *File → Add decoys (background set)…* or `--decoys` loads presumed-inactive molecules used **only** for evaluation, never for fitting. `examples/make_decoy_sets.py` builds property-matched decoy sets, which is the fair kind: decoys matched to the actives in molecular weight, logP, rotatable bonds and formal charge, so that the model cannot win by noticing size.

Two problems the program handles explicitly:

- **A decoy must not be rejected for its pose.** `--align-decoys` places every decoy in the frame of the training set by maximising its field similarity, with several random restarts, and `--decoy-conformers N` generates N conformations of each decoy and keeps the one that poses best (the file's own conformation is always among the candidates).

- **AUC saturates.** With cross-target decoys almost any method reaches 0.95 and above, which tells you nothing. Read the **enrichment factors** at 0.5, 1, 5 and 10 % instead, which the program reports beside the maximum achievable value at each fraction, and **BEDROC/RIE** (Truchon & Bayly), which weight the top of the list.

A measured caution from the benchmarks in this repository: field similarity on these tasks performs about as well as ECFP4 fingerprint similarity, and cross-target decoys are trivially separable by either. If a screening result looks spectacular, check what the decoys were.

## 9.4 Measured inactives, and learning from positives alone

Every background in the previous section is *presumed* inactive, and a presumed inactive is an easier to reject than a measured one. A compound that was tested and did nothing is the kind of molecule a screen actually has to reject, and there is a public collection of them: **LIT-PCBA** (Tran-Nguyen, Jacquemard & Rognan 2020), assembled from high-confidence PubChem dose-response assays — 15 targets, 7761 actives and 382674 inactives, every one with an experimental verdict. The AVE-unbiased subsets are the form the authors recommend, since the asymmetric validation embedding of Wallach & Heifets removes the analogue bias that otherwise lets a screen score well by recognising that actives resemble actives.

What LIT-PCBA does not have is the thing a field method needs: a conformation and a pose. `examples/make_litpcba_examples.py` supplies them. For each target it embeds the actives and a seeded sample of the measured inactives with ETKDG and MMFF, then poses **both classes by one rule** — maximum field similarity to the target's crystal ligand, which the archive ships, so the frame is the one the receptor defines and the placement knows nothing about which molecule is active. The 57 MB archive is downloaded on demand and is not redistributed here; what is kept are the derived structures, `examples/litpcba/<target>.sdf` and `<target>_decoys.sdf`.

### Why the posing rule is half the measurement

`--pose-against actives` poses the inactives to maximise their similarity to the posed actives instead, which is what `cmfqsar/decoys.py` does for the background sets of the previous section: no decoy is then rejected for having been posed badly. That sounds like the fair choice and is the generous one, for a reason that is not the obvious one. Posing a decoy where it most resembles the actives does make it harder to reject — but the pose search also *fails* on a molecule whose field cannot be made to match, and the similarity it settles at is then a shape-matching score in its own right. Optimising every candidate's pose against the actives silently makes the alignment part of the scoring function, which is what shape-based screening does on purpose and what a field model measured this way gets for free. The same molecules posed both ways say what it is worth, and the figure below says it is most of the result.

### Unlabelled is not inactive

The compounds that came back negative in one assay at one concentration are **unlabelled** rather than inactive: some are inert, some were mis-measured, some would have worked at another concentration, and a few are actives nobody has recognised. Training a classifier on that mixture as a negative class teaches it to reject the very molecules a screen is looking for, and the damage is worst at the top of the ranking, which is the only part anyone reads.

`cmfqsar/pu.py` implements **bagging PU** (Mordelet & Vert 2014), the method that needs neither an estimate of the class prior nor an assumption about how the positives were sampled:

1 draw a small random sample of the unlabelled set and call it negative;

2 fit on the positives against that sample;

3 score the unlabelled molecules that were *not* in the sample — out of bag, so a molecule's score never comes from a model that was told it was negative;

4 repeat, and average each molecule's out-of-bag scores.

An unlabelled molecule that looks like the positives is called negative in only a few bags and scores high in the ones that leave it out; one that is genuinely unlike them scores low wherever it appears. Everything works on a precomputed kernel, so the CMF kernel enters unchanged:

```
from cmfqsar.pu import pu_scores
scores = pu_scores(K, positives=train_actives, unlabelled=library, test=held_out, bags=30)
```

## What it measures

Measured on two LIT-PCBA targets (ESR1_ant, TP53), 152 measured actives against 2999 measured inactives, 70/30 on the actives and half the inactives offered to the methods that can use them. AUC on the held-out actives against the held-out inactives:

| method | ESR1_ant | TP53 | what it sees |
|---|---|---|---|
| distance to the centre of the actives | 0.536 | 0.365 | the training actives |
| distance to the k-th active | 0.515 | 0.382 | the training actives |
| one-class Gaussian process | 0.542 | 0.546 | the training actives |
| distance from the kernel-PCA subspace | 0.537 | 0.444 | the training actives |
| ECFP4 Tanimoto to the actives | 0.668 | 0.701 | the training actives |
| classifier, inactives = negative | 0.544 | 0.613 | + half the inactives, as negative |
| classifier, inactives = unlabelled (bagging PU) | 0.662 | 0.715 | + half the inactives, as unlabelled |

Three things in that table are worth saying plainly.

**The one-class field descriptions do not work here.** The best of them reaches 0.544 against 0.685 for the maximum ECFP4 Tanimoto to the same actives, and some are below chance (distance to the centre of the actives at 0.365 on TP53: the held-out actives look *less* typical of the training actives than the measured inactives do). Nor is that a badly chosen setting. Every parameter that can reorder the ranking was swept — the field attenuation over 0.3, 0.5, 1 and each description's own parameter, 45 fits per set — and keeping whichever scored best *on the test set*, a bound no legitimate protocol could reach, lifts the best description only to 0.574. ($\nu$ is not in that sweep, and cannot be: it places the threshold so that a fraction $\nu$ of the training actives falls outside the description, which shifts every decision value by the same constant and leaves every ranking metric unchanged.) What the gap reflects is the benchmark's design: the AVE debiasing removes the analogue bias on purpose, and a method whose entire content is "this molecule resembles the actives" then has nothing left to say.

**The kernel is not what fails.** The same CMF kernel used discriminatively, with the inactives treated as unlabelled, reaches 0.689 — level with the fingerprint, and +0.110 above the same learner told that the inactives are negatives. For screening against measured inactives, use the

kernel discriminatively; the one-class descriptions are for the case the earlier sections describe, a congeneric series against a structurally different background.

**The posing is half of any screening number.** Posed instead to maximise their similarity to the actives, the same inactives move the one-class AUCs by +0.141 on average while the fingerprint shifts by +0.000, because it never looks at a coordinate. That is the generous placement the background sets of the previous section are given, and the difference is a measurement of how much a field-based screening figure depends on how its decoys were placed.

## 9.5 Screening a library that does not fit in memory

Everything above loads a data set, builds every molecule's fields and keeps them. That is right for a training set and impossible for a library: a hundred thousand molecules will not fit, and nothing about them needs to be kept once their score is written.

Predicting with a kernel method needs only the rows `K(X, X_train)` — one row per new molecule, as wide as the training set. So the `screen` subcommand reads the library in blocks and never holds more than one:

```
python cmf_cli.py screen model.cmf library.sdf --out scores.csv --beta 1.0 --top 25
```

It reads `--chunk` molecules (500 by default) with a forward supplier, so the file is streamed and never indexed; builds their fields with the model's own field set; computes the block's kernel rows; writes the scores to the CSV as it goes, so a run stopped after an hour keeps everything it had scored; and keeps only the best `--top` of them in memory. A SMILES file is accepted too and its molecules are embedded as they are read.

Measured against the same library loaded whole, scored by a 200-molecule model:

| library | how it was read | seconds | molecules per second | peak process memory |
|---|---|---|---|---|
| 4166 molecules (22 MB) | streamed, 500 per block | 48 | 86 | 295 MB |
| 4166 molecules (22 MB) | streamed, 100 per block | 59 | 71 | 232 MB |
| 4166 molecules (22 MB) | loaded whole | 46 | 90 | 571 MB |
| 16664 molecules (87 MB) | streamed, 500 per block | 201 | 83 | 288 MB |
| 16664 molecules (87 MB) | streamed, 100 per block | 237 | 70 | 237 MB |
| 16664 molecules (87 MB) | loaded whole | 178 | 94 | 1706 MB |

The memory column is what matters. Over a four-fold library the whole-file path grows 571 to 1706 MB, a factor of 3.0, while the streamed path at 100 molecules per block stays at 232 and 237 MB. Extrapolating the first to 100 000 molecules asks for about 10 GB; the second would still be using 237 MB. The price is 33 % of wall clock, from parsing the file in blocks rather than once.

**What the score is.** For a regression model it is the prediction, or with `--beta B` the lower confidence bound mean $-$ $B$ sd, which is what a library should normally be ranked by, and the reason the standard deviation is always in the CSV: a library is mostly molecules unlike the training set, a kernel method answers the training mean for those, and a point estimate cannot be told from

a real prediction. Screen the thrombin series against a cyclooxygenase model and every molecule comes back at the training mean with a spread four times the residual error — which is the model saying, correctly, that it has nothing to contribute.

**What it does not do.** It does not invent poses. A field kernel compares molecules in one frame, so the library has to be in the training set's frame already, or `--pose` has to be given, which places each molecule by maximising its field similarity to the training molecules. That is the placement the background sets of the benchmarks are given, it costs about half a second per molecule — far more than the scoring — and it is optimistic, because a real library has to choose a pose without knowing the actives. `--pose-conformers N` generates N conformations of each molecule and poses every one, which multiplies the cost by N.

## 9.6 Which molecule to make next

The model predicts an activity and, through the Gaussian-process view, a standard deviation beside it. Those two numbers answer a question the 2019 paper poses and leaves as a statement — that maximising the overlap tells one how to modify the molecule — as an operation with a number:

```
python cmf_cli.py suggest model.cmf candidates.sdf --acquire ei --top 10
```

The candidates have to be posed in the training set's frame, as for `predict`. What comes back is the ranked list with the prediction, the standard deviation, and the distance to the nearest molecule already made.

**The rule depends on what the asking is for**, and the program keeps the two apart rather than pretending one serves both.

| `--acquire` | **what it maximises** | **what it is for** |
|---|---|---|
| `ei` | expected improvement over the best made so far | finding an active |
| `ucb` | prediction + β standard deviations | the same, more crudely |
| `greedy` | the prediction | the same, with no allowance for being wrong |
| `variance` | the posterior standard deviation | improving the model |
| `maxmin` | distance in field space to everything made | the same, with no model at all |
| `random` | nothing | the baseline the others have to beat |

Expected improvement counts only the upside, which is why it is the usual choice:

$$\mathrm{EI}(x) = (\mu - y^{*} - \xi)\,\Phi(z) + \sigma\,\varphi(z), \qquad z = \frac{\mu - y^{*} - \xi}{\sigma}$$

Note that this is the **opposite** of how the same two numbers are used to rank a screening library, where `rank_score` subtracts the standard deviation instead of adding it. The difference is not a matter of taste. A screening library is dominated by molecules unlike anything in the training set, a kernel method returns the training mean presented as a prediction for exactly those, and a molecule chosen on that basis is wasted. A molecule *made* to answer a question is different: being wrong about it costs little, and being wrong is informative. Optimism is right for one and wrong for

the other.

## What it gains, measured

Eight Sutherland sets, ten runs each, starting from twenty molecules drawn at random that hold no hit and acquiring a quarter of what is left one at a time. α and λ are reselected by the evidence at every step from what has been acquired and nothing else — which is what the evidence was built for. A **hit** is a molecule in the top 5 % of its set.

**Finding an active:**

| rule | best found, of the best in the set | runs that found a hit | median acquisitions to the first hit |
|---|---|---|---|
| random | 92.2 % | 79 % | 6 |
| maxmin | 90.9 % | 64 % | 9 |
| variance | 90.6 % | 71 % | 9 |
| greedy | 95.0 % | 100 % | 4 |
| **ucb** | **96.6 %** | 95 % | 4 |
| ei | 95.2 % | 89 % | **3** |

**Improving the model**, as the RMSE on everything not yet acquired (against the variance of the whole set, so that a rule which acquires the extremes is not rewarded for making the rest look alike):

| rule | mean RMSE over the run | RMSE at the end | $q^2$ at the end | against random, paired |
|---|---|---|---|---|
| random | 1.122 | 1.064 | +0.366 | — |
| maxmin | **1.108** | 1.047 | +0.384 | −0.014 [−0.034, +0.006] |
| variance | **1.108** | 1.042 | +0.386 | −0.014 [−0.034, +0.006] |
| greedy | 1.179 | 1.195 | +0.158 | **+0.057 [+0.038, +0.078]** |
| ucb | 1.144 | 1.127 | +0.279 | +0.022 [+0.001, +0.044] |
| ei | 1.121 | 1.086 | +0.331 | −0.001 [−0.022, +0.021] |

**The optimistic rules work and the exploring rules do not.** For finding actives, `ucb` reaches 96.6 % of the best activity in the set against 92.2 % for picking at random, hits in a median of four acquisitions against six, and the paired gain in hit rate is +0.163 [+0.062, +0.275]; `ei` hits soonest of all, in three. For improving the model, the best exploration rule gains −0.014 RMSE over random with an interval of [−0.034, +0.006] — zero is inside it, so on these sets **exploring gains nothing measurable**, while `greedy` is clearly worse, by +0.057 [+0.038, +0.078].

The two tables are kept apart because an acquisition rule cannot be judged without saying what the acquiring is for, and three things in the numbers are worth more than the ranking.

**The uncertainty rule and the diversity rule are the same rule.** `variance` and `maxmin` give the same result, which is the error-prediction finding seen from a different direction: the posterior standard deviation of a CMF model is very nearly the distance to the training set in other units. Acquiring by what the model does not know and acquiring by what the series does not cover are the same experiment with two names.

**Exploring gains nothing measurable.** The molecule the model is least sure of is the molecule least like the series, and what is learned from an outlier does not transfer to the molecules anyone will ask about next. Uncertainty sampling is a good rule for a space one intends to cover and a poor one for a series one intends to understand.

`greedy` **does better here than it would in general.** It ignores the posterior entirely, so a candidate unlike anything made is predicted at the training mean and cannot be told from a real answer. These sets are series of analogues, where nothing is far enough away for that failure to matter; on a library of diverse candidates the protection is gone, and `ei`, which at least tempers its optimism with the posterior, is the rule to use.

## 9.7 Designing the next compound

`suggest` ranks candidates somebody has already drawn. The design loop makes the suggestion itself: take a parent compound of a series, take one of its attachment points, put every fragment of a library there in turn, and rank what comes out.

```
python cmf_cli.py design series.sdf --parent 45 --position 2 --acquire ei --top 15
```

The command does the whole thing: it finds the scaffold, builds a **topomer model** of the series, chooses α and λ by the evidence, reports the $q^2$ that the suggestions rest on, enumerates, scores and ranks. There is no model file to pass, because the model has to be a topomer model and the core and the attachment points come from the series anyway.

**Why a topomer model and not an aligned one.** A substituent nobody has made has no crystal structure and no alignment, and superimposing it on anything would be a guess. The topomer rule builds a fragment into a pose that is a deterministic function of its own topology, so a candidate is posed by exactly the rule that posed the molecules the model was fitted on, and the comparison is like with like at one attachment point. A model fitted on superimposed whole molecules has no way to place a group that was never synthesised, which is why the command builds the topomer model itself instead of taking a model file: there is no aligned model that could answer the question.

Three numbers come back for each candidate, and they answer different questions: the **prediction**, the **posterior standard deviation** (a fragment unlike anything in the series gets the training mean and a wide spread, which is what stops the loop from proposing molecules it knows nothing about), and the **acquisition score** that combines them — the same rules as `suggest`, expected improvement by default. Then the annotations: whether the series already uses that group at that position, whether the whole molecule has been made and what it measured — the only free check there is on a suggestion — and, for the top few, what the candidate's fields have that the parent's do not.

The library is `--fragments common` (a built-in list of about forty ordinary medicinal-chemistry substituents), `series` (what this series already puts there), `both`, or a file of SMILES with one attachment point each.

### Would it have suggested the right group, measured

Leave one substituent out: every molecule carrying a given R-group at the varied position is removed, the model is refitted on what is left, and the whole library is enumerated and ranked from one fixed parent. The group's measured activity is the mean over the molecules that carry it.

| set | groups held out | position | $q^2$ of the models | ρ of the ranking with the measurements | the group it puts first | library mean | library best |
|---|---|---|---|---|---|---|---|
| THER | 46 of 46 | 0 | +0.251 | **+0.279** | 6.53 | 5.24 | 10.17 |
| GPB | 41 of 47 | 2 | +0.566 | **+0.516** | 4.40 | 2.94 | 6.80 |

Over 87 held-out groups the ranking agrees with the measurements at a mean Spearman ρ of **+0.397**, and the group the loop puts first is worth **+1.37** log units more than the library average — it lands in the best 20 % of the library **48 %** of the time. The sharper question is where the groups that really are good end up: of the five best held out on each set, the median rank is 7 of 42 on GPB and 12 of 46 on THER, and the single best lands at 13 and at 32. A loop run twice over would have found most of them and missed one.

Only 2 of the eight sets can be reached at all: ACE (the core comes off cleanly in only 44% of the series), AChE (the core comes off cleanly in only 26% of the series), BZR (the core comes off cleanly in only 42% of the series), COX2 (the core comes off cleanly in only 22% of the series), DHFR (the core comes off cleanly in only 12% of the series), THR (the core comes off cleanly in only 25% of the series).

## What it cannot do

The loop varies **one position of one parent**, and it states plainly the three things that this limits.

It needs a series with a scaffold that comes off at acyclic single bonds. Of the eight Sutherland sets only two are that; on the rest the Bemis–Murcko framework is fused into the molecules and what comes off is not a substituent. The command says so instead of ranking the pieces anyway.

It ranks groups, not conformations. The topomer pose is a convention, and a suggestion is a statement about which group belongs at a position, not about how it will sit when it is there.

And it cannot tell you about a position the series never varied. A model learns from contrast; a site that is the same in every training molecule carries no information about what else might go there, and the posterior standard deviation is the only thing that will say so.

## A series that has no scaffold: the invariant route

The three limits above belong to the topomer route. `--kernel autocorrelation` takes the other one (plan item SC7): the rotation-invariant kernel needs no frame, so **a candidate is built as a whole molecule and embedded once**, and nothing is superimposed on anything. The series then needs no common scaffold at all -- the core is only what names the position being varied on the parent -- which is what makes a design loop possible on a table of guests, where the molecules share nothing.

```
cmf design examples/cyclodextrin/beta_cyclodextrin.csv --kernel autocorrelation \
    --core "C(=O)O" --parent "CC(=O)O" --acquire greedy
```

Two things change with it. There is no field difference to report, because there is no frame to draw one in; in its place each candidate carries **how much of its prediction comes from the group**

**that was put in**, which the invariant model's atom contributions give exactly. And the acquisition score needs watching: on a table that is not a congeneric series the best molecule is far above anything the loop can reach, expected improvement is then the same for every candidate, and the command says so and asks for `--acquire greedy` rather than presenting an order that is no order.

**Does the ranking mean anything?** `benchmarks/run_guest_design.py` answers with a leave-one-group-out test on the 1162 guests of the beta-cyclodextrin table. For each group of an eleven-group library of R-COOH, **every guest carrying that group is removed**, a model is fitted on what is left, and the library is ranked by it; a group's prediction therefore comes from a model that never saw a molecule containing it. A group carried by more than 30% of the table cannot have all of its guests removed -- a methyl is in every one of them -- and those rows remove the molecule alone and say so.

| group | guests removed | predicted | rank | measured |
|---|---|---|---|---|
| methyl | 1 (the molecule only) | +1.545 | 11/11 | +1.301 |
| propyl | 1 (the molecule only) | +1.864 | 10/11 | +1.505 |
| butyl | 1 (the molecule only) | +2.022 | 9/11 | +1.954 |
| pentyl | 1 (the molecule only) | +2.397 | 6/11 | +2.467 |
| hexyl | 261 | +2.483 | 3/11 | +2.041 |
| phenyl | 1 (the molecule only) | +2.081 | 8/11 | +2.127 |
| 4-methylphenyl | 1 (the molecule only) | +2.435 | 5/11 | +3.225 |
| cyclopentyl | 83 | +2.134 | 7/11 | +3.204 |
| cyclohexyl | 122 | +2.243 | 6/11 | +3.620 |
| cyclohexylmethyl | 111 | +2.613 | 2/11 | +3.519 |
| adamantyl | 6 | +3.297 | 1/11 | - |

Over the 10 groups whose acid is a measured guest the ranking follows the measurements: Spearman +0.709. **The two kinds of row are not equally hard**, and the mean over both would hide it: where only the molecule was removed the model had seen hundreds of its relatives and the mean absolute error is 0.26 over 6 groups, while where every guest carrying the group was removed (adamantyl, cyclohexylmethyl, hexyl, cyclohexyl, cyclopentyl) it is 0.95 over 4, with a bias of -0.73. A group the table no longer contains is pulled towards the mean, which is what a model should do with a group it has never seen.

**What survives that is the order at the top.** Adamantyl -- whose six guests in the table measure 2.9 to 4.3 against a median near 2, and whose own acid is not in the table at all -- comes **first of the 11** with every one of those six removed, and cyclohexylmethyl, the strongest of the measured groups that were fully held out, comes second. Taken over every row the 4 groups measuring 3.0 or more have mean rank 5.0, so the list is worth reading at the top and not in the middle. That is the claim a design loop has to make, and it is the one this measurement supports: it will name the group worth trying, not tell you what it will measure.

## 9.8 A design loop with more than one objective

The suggestion rules of the earlier section close the loop on potency alone. Lead optimisation has at least two objectives, and since the two-task model they live on one kernel: one model predicts the potency and the anti-target activity of the same candidate, each with a posterior. Plan item MC11 is what goes between that model and a chemist's list, and it is `cmfqsar/multiobjective.py`.

**The Pareto front** is the candidates no other candidate beats on every objective at once. A weighted sum needs weights, and nobody knows them before the compounds are made; a front does not. `pareto_layers` gives the front and the fronts behind it, so a list can be as long as the chemist wants.

**The acquisition** is the expected improvement of each objective against the best already made, added with weights. Each term is already in the units of its own objective's improvement, so the weights say how much a log unit of one is worth against a log unit of another and nothing is hidden in them.

**The batch** is the rule of the previous section with more than one task: after a pick, every objective's posterior covariance is conditioned on it and the acquisition is computed again. What that takes away is the uncertainty and not the promise -- two compounds the model is sure are a log unit better are both still worth making -- so the spreading acts through what the model does not yet know, which is the only honest place for it to act.

**Every suggestion comes with the trade it makes**: `multiobjective.trade` reports, against the best compound already made, which objectives a candidate gains and which it loses, with a change smaller than the model's own error reported as no change. A loop that gives only a score hides the trade it is making.

### What happened when it was run backwards

The series of the previous section comes in two parts, which makes a retrospective test possible. The loop is trained on the 78 compounds of part 1 with both their endpoints, and then scores 425 candidates: every other histamine H3 ligand in ChEMBL that fits the same frame. 16 of them are the compounds **the same authors went on to make** in part 2 of the paper. The loop sees no activity of any candidate.

| the list the loop reports | candidates | of them made next | share |
|---|---|---|---|
| the Pareto front | 5 | 0 | 0.0% |
| the first 10 by the acquisition | 10 | 0 | 0.0% |
| the first 50 by the acquisition | 50 | 0 | 0.0% |

**The gate is not met.** The front holds none of them, and the compounds the authors made sit at Pareto layer 23 of 47 against 19 for the rest of the pool, at rank 343 of 425 against 208: the loop puts them in the worse half. The best of them reaches layer 2 and rank 79, so they are not invisible to it.

Three things explain the miss, and they are worth more than the verdict. The model does see that these compounds are good: their measured H3 affinity averages 9.15 against 7.98 for the rest of the pool, and the model predicted 8.68 against 8.59 -- the order is right and the size is not, which is what a q2 of +0.28 on a narrow series buys. What sinks them in the ranking is the other objective: the model predicts their hERG activity at 6.29 against 5.89 for the pool, and an acquisition that adds improvements prefers a candidate extreme in one objective to one that is good in both. And

no hERG number was ever measured for them, so whether the model is right about that cannot be checked.

The last point is the real limit of this test. The compounds the authors made next were chosen with knowledge the loop does not have -- synthetic access, a chemical idea, the rest of a programme -- so the ones they did not make are not known failures, and only one of the two objectives can be checked against what happened. What the item asked for and what this provides is the machinery: a front over several objectives on one kernel, a batch taken from it by the rule of the previous section, and for every suggestion the trade it makes, written out.

## 9.9 An anti-target beside the potency

Where may field be added for the target that is **not** also added for the off-target? The two-task model answers it as a difference of co-fields, and the section on two endpoints above measured that on the Am/Eu extractants. Plan item MC4 asks the same question in its medicinal-chemistry form, and what it needed was data: one series with a potency and an anti-target activity measured on the same compounds.

**The series** is `examples/antitarget/h3_herg.sdf`, built by `examples/make_antitarget_examples.py` from one ChEMBL document, which is one paper: Lazewska and co-workers, *Novel and highly potent histamine H3 receptor ligands. Part 1: withdrawing of hERG activity*, Bioorg. Med. Chem. Lett. 21 (2011) 5378. The paper's own subject is this question. 78 compounds carry both numbers: the H3 affinity as a displacement of [125I]iodoproxyfan, and the hERG activity as a displacement of [3H]dofetilide, **one assay each, in one laboratory**. That matters, and the assay descriptions travel with the file as columns: an anti-target number from another paper is a different measurement. The two endpoints have the same spread, 0.68 log units each, and correlate only -0.22, so there is a selectivity to find rather than a single axis seen twice.

**The first part of the gate: the potency must not suffer.** The task correlation chosen by the evidence is -0.431 -- in this series a compound good for H3 tends to be worse at hERG -- and with it:

| endpoint | q2 with the other endpoint | q2 on its own |
|---|---|---|
| pKi_H3 | +0.341 | +0.319 |
| pIC50_hERG | +0.357 | +0.338 |

On the potency the two-task fit is +0.022 [-0.005, +0.048] against the single fit, paired over the molecules: not worse, so the selectivity co-field is not bought with a worse model of the thing that matters. The selectivity itself is predicted at q2 +0.471, better than either endpoint alone, which is the 2003 paper's claim met again on a different kind of series.

**The second part: where the difference co-field puts its weight.** `cofields.atom_effect` says what one more atom of a given field type, at a given place, would do to the predicted selectivity, in log units, with the error the activities allow. Read at the basic nitrogen of every molecule, at the carbons within 5 A of it, and at every other heavy atom:

| place | field | effect on the selectivity |
|---|---|---|
| the basic nitrogen | steric | +12.84 [+9.68, +15.99] |
| the basic nitrogen | hydrophobic | +6.96 [+5.46, +8.47] |
| the carbons beside it | steric | +11.56 [+8.86, +14.26] |
| the carbons beside it | hydrophobic | +7.35 [+5.92, +8.78] |

| place | field | effect on the selectivity |
|---|---|---|
| every other heavy atom | steric | +0.03 [-2.39, +2.45] |
| every other heavy atom | hydrophobic | +0.70 [-0.77, +2.17] |

The weight is of one sign and it is concentrated exactly where the item says hERG is driven from: at the amine and the carbons beside it two fields stand clear of the noise and both are positive, while over the rest of the molecule neither does. A co-field that said the same thing everywhere would be saying nothing about the basic nitrogen in particular.

**The sign is the opposite of what the item expected, and that is the interesting part.** Lipophilicity does drive hERG here as the textbook says: cLogP correlates +0.38 with the hERG activity and -0.31 with the selectivity over a range of 3.0 to 7.0. But the co-field is about a **place** and cLogP is about a molecule, and what this series says is that the lipophilicity costing selectivity is the lipophilicity spread over the rest of the molecule, while the region around the basic nitrogen is where the H3 affinity is made. That is also what the paper did: it kept that region and changed what is far from it. A whole-molecule number cannot make that distinction, and it is the only thing a co-field adds.

```
        cmf build examples/antitarget/h3_herg.sdf --activity pKi_H3 --second-endpoint pIC50_hERG \
            --save h3.cmf
        cmf cofield h3.cmf --endpoint selectivity --out-prefix selectivity
```

`benchmarks/run_antitarget.py` is the measurement above. It is one series from one paper, with hERG measured by displacement rather than by patch-clamp and an alignment by common substructure rather than by a crystal structure; a second series, measured another way, is what would turn this reading into a rule.

## 9.10 Choosing ten compounds at once

A laboratory makes ten or twenty compounds at a time, not one. The acquisition rules of the previous section pick one molecule and refit; a whole plate has to be chosen before any of it is measured, so something has to keep the plate from being one molecule ten times over.

**The cheap rule** has been in `cmfqsar/active.py` from the start: take the best scores in order, and skip a candidate whose cosine with a pick already made is above a threshold. It works, and it has a threshold in it that nothing fixes.

**The rule the posterior already implies** is what plan item MC2 asks for. After a pick, the posterior covariance conditioned on that pick says how much uncertainty is left everywhere else:

cov' = cov - cov[:, j] cov[j, :] / cov[j, j],

which is one step of a Cholesky factorisation with `j` as the pivot. The acquisition is then evaluated again on the conditional standard deviations, so a candidate whose uncertainty the first pick has already removed scores low by itself, with no threshold and no refitting (`active.condition_on`, `active.batch_by_covariance`). With the **variance** acquisition the score is the conditional standard deviation, so the rule is the greedy maximisation of the determinant of the batch's own covariance block -- the classical D-optimal design -- and a test checks the greedy batch against the exhaustive best on a small matrix.

`model.predict_with_covariance` returns the whole posterior covariance of a block of candidates, which is what the rule needs and what `predict_with_variance` only gives the diagonal of.

**Measured** by `benchmarks/run_batch.py` with the protocol of the previous section -- the eight Sutherland sets, twenty molecules to start, the attenuation and the ridge reselected by the evidence at every fit, ten repeats, the same starting set for every rule -- in batches of ten:

| how the batch is chosen | fits | best acquired | error on what is left |
|---|---|---|---|
| one at a time | 35 | 8.878 | 1.034 |
| ei, similarity filter | 3 | 8.902 | 1.041 |
| ei, covariance | 3 | 8.939 | 1.025 |
| variance, similarity filter | 3 | 8.540 | 1.029 |
| variance, covariance (largest determinant) | 3 | 8.403 | 1.043 |
| random | 3 | 8.709 | 1.011 |

**The gate is not met, and the item says that is an answer.** Paired run by run, the covariance rule minus the filter is +0.037 [-0.015, +0.110] in the best acquired and -0.016 [-0.035, +0.002] in the error for expected improvement, and -0.137 [-0.267, -0.013] and +0.014 [-0.008, +0.037] for the variance acquisition. None of the four favours the covariance rule beyond its interval and one favours the filter, so the cheap filter is enough. That is the outcome plan item E2 predicted from the posterior standard deviation being nearly flat over these pools: if conditioning on a pick barely moves the ordering, the two rules choose almost the same plate by different arithmetic.

**What a batch costs is nothing measurable here**, which is the more useful finding. Against the one-at-a-time loop, which sees every measurement before choosing the next molecule and pays thirty-five fits for it, expected improvement with the filter differs by +0.025 [-0.025, +0.079] in the best acquired and +0.008 [-0.016, +0.033] in the error. A plate of ten chosen at once is worth as much as ten chosen one after another, at a tenth of the fitting.

Both rules are on the command line:

```
cmf suggest model.cmf --candidates library.sdf --kind ei -n 10 --batch-rule covariance
```

## 9.11 Growing a bound fragment into the gap

A fragment that binds fills part of a site and leaves the rest empty. The question a fragment campaign asks next is what to put there. `cmfqsar/fraggrow.py` answers it with the fields alone, which is what plan item MC7 asks for, and the answer it gives is a measured no.

**The construction.** The pocket's offer at a point is its own field with the sign convention of the complementarity section: the electrostatic field with a minus, because a positive ligand region should meet a negative protein region; the hydrophobic field with a plus; the pocket's donors paired with the ligand's acceptors and the other way round. The **gap** is that offer in the space that is free -- free of the protein, and free of the fragment, which is how "minus what the fragment already places" is taken here. An offer is open only where a new atom could sit. (The other reading, subtracting the fragment's own field from the pocket's offer, compares two different quantities and erases exactly the contact region a new group would use; it was tried and it does not work.)

**The candidates** are built from their SMILES with a distance-geometry embedding and a force field, never from a crystal structure, and placed on the growth vector by rule: the origin at the atom that grows, the x axis along the direction that atom's own bonds leave empty, the bond length the sum of two covalent radii, and the turn about the new bond scanned. The true answer is built and placed

exactly like every other candidate, so nothing about it enters the geometry. A placement whose atoms come closer than 2.5 A to the protein is refused.

**Measured** by `benchmarks/run_fraggrow.py` on 25 growing cases from the fragment sets of the Schrodinger free-energy collection -- the only public sets here with a pocket and with the fragment and the compound it grew into in one crystal frame -- against a library of 201 groups taken from the ligands of the same collection:

| ranking | median rank of the true group | among the first ten | first |
|---|---|---|---|
| at random | 82 | 0% | 0% |
| by whether it fits, at random inside that | 15 | 40% | 0% |
| by whether it fits, then by the size of the group | 28 | 12% | 0% |
| by whether it fits, then by the field score | 25 | 28% | 0% |

**The gate is not met, and the field score is worse than not scoring at all.** Refusing the placements that clash and shuffling the rest puts the true group at a median rank of 15 of 201, and finds it among the first ten in 40 per cent of the cases. Sorting those same candidates by the field score moves it to 25 and 28 per cent. The score is not neutral: it rewards whatever fills most of the gap, so it prefers the largest group that still fits, while the additions in these cases are a fluorine, a methyl, a chlorine. What carries the ranking is the geometry -- whether the group fits at all -- and the field adds nothing to it.

Under rule 5 that makes the gap a display tool: a picture of where there is room next to a bound fragment and what the pocket would reward there, which is worth drawing before a chemist chooses a group, and not a way of choosing one. `fraggrow.gap` returns it on a grid, with one array per field type, ready for the same cube export as any other field of this program.

**What the measurement cannot settle.** The library holds 201 groups and the true one is always in it, which is the easy part of a real enumeration. The cases are few, because a public set with a pocket, a fragment and the compound it grew into is rare. And every candidate gets one embedded conformation and a turn about the new bond, where a real growing tool would minimise the group inside the site. A negative result here is evidence against the one-body field score, not against fragment growing.

## 9.12 A matched pair as a field placed in space

A matched molecular pair is usually a row in a table: a transformation, written as two group names, and the activity change it produced. In a series whose molecules are in one frame it is also something else -- **a difference of two fields**, a local object sitting where the two groups differ, with its shape and its charge kept. Plan item MC6 asks whether that object carries the effect of the transformation with it: can a change measured in other series predict the change in a series the model has never seen?

Plan item D5 is the warning and the reference. Increments fitted on one series did not transfer, and the vectors fitted on two sets agreed to +0.89 at best. An increment is a number attached to an atom type; the field difference is the same transformation with its geometry kept, so it can tell a methyl that points into a pocket from one that points at the solvent. Whether that is enough is a measurement.

**The construction** (`cmfqsar/transformfield.py`). The two members of a pair are already in one frame, since the series is aligned, so their fields are comparable with each other. What they do not share with another series is a frame, so the difference field is placed in a frame of its own: the origin at the core atom that carries the group, the x axis along the bond to the group, and the plane fixed by one more neighbour of that atom, chosen by atom type so that both members pick the same one. The difference field is the second group minus the first, each restricted to its own atoms by the mask of plan item MC1, and it is an ordinary field object: every kernel and every display of this program applies to it. The overlap kernel is bilinear, so reversing a pair reverses its kernel row, and a ridge with no intercept therefore predicts exactly the opposite change for the reverse of a pair -- the one property such a model must have.

**The data** are the only sets here with a crystal pose for every ligand: the ten Merck KGaA FEP series of Schindler and co-workers and the eight Schrodinger JACS series, 3814 matched pairs in 17 congeneric series. The folds hold out a whole series, the ridge is chosen inside the training series the same way, and q2 is taken against each series' own mean change, which is the strongest form of the reference the item names.

| what predicts the change | q2 against each series' own mean |
|---|---|
| no change at all | -0.025 |
| the transformation as a field, inside one series (no transfer) | +0.465 |
| the transformation as a number (plan item D5) | -0.073 |
| the transformation as a field, without a frame | -0.025 |
| the transformation as a field | -0.025 |

**The gate is not passed, and the reason is not the construction.** Across series the model reaches q2 -0.025, with an interval over series of -0.118 to +0.035: it predicts nothing, and the ridge chosen in every fold is at the strong end of the grid, which is the model saying so. The same pairs described without a frame, as the rotation-invariant autocorrelation of the difference field, give the same -0.025, so a frame convention is not what went wrong. Trained **inside** each series, where nothing has to transfer, the same difference fields reach q2 +0.465 with whole molecules held out. The descriptor works; the transfer does not.

**The data say it before any model is fitted.** Of the twelve transformations that occur in most series, eleven raise the activity in one series and lower it in another:

| transformation | series | pairs | mean change | lowest | highest |
|---|---|---|---|---|---|
| methyl to chloro | 10 | 23 | +0.15 | -0.93 | +1.07 |
| methyl to ethyl | 8 | 13 | -0.17 | -1.15 | +1.01 |
| fluoro to methyl | 7 | 14 | +0.12 | -0.53 | +1.37 |
| chloro to cyano | 5 | 12 | -0.48 | -1.05 | +0.09 |
| isopropyl to ethyl | 5 | 8 | +0.30 | -0.64 | +1.62 |
| methyl to methoxy | 4 | 13 | -0.14 | -1.49 | +0.40 |

Replacing a methyl by a chlorine is worth about a tenth of a log unit on average and anything between -0.9 and +1.1 depending on the protein. There is no single effect for a field or for a number to carry, and that is the answer to the item rather than a failure of the machinery.

**What is worth keeping** is the object itself. `transformfield.average_field` builds the mean difference field of all the pairs of one transformation, which is a picture of what that transformation does to the field, and the within-series row above says that inside one series the difference field predicts a change well. The place for it is inside a project, on one target, where plan item MC1's table already uses the same masking.

## 9.13 Which groups are interchangeable: a bioisostere table

`cmf design` ranks candidate substituents by the predicted activity of the whole molecule. The table a medicinal chemist asks for answers a different question: **which groups can replace each other at this position**, so that one of them can be chosen for a property the model knows nothing about -- solubility, cost, a patent, a metabolic liability. Plan item MC1 builds that table (`cmfqsar/bioisostere.py`).

**It needs no new model.** The prediction of the overlap kernel is a sum over the atoms of the molecule (decomposition rule 3), so the part of a prediction that belongs to a substituent is the sum over that substituent's own atoms, and the predicted effect of exchanging group A for group B at one position is the difference of those two sums. That is one integral per pair of groups instead of one model fit per group. The masking is exact rather than approximate: the kernel is a sum over pairs of atoms, so restricting one side of it to a subset of the atoms is the same thing as building the fields of that subset and nothing else, and `masked_field` does that. A test checks the split: for any division of a molecule in two, K(whole, whole) is K(a,a) + 2K(a,b) + K(b,b).

**Three ways to say how far apart two groups are**, which is the ladder of rule 1:

1. the **fingerprint distance** between the two groups, one minus their Morgan Tanimoto: two dimensions, no geometry, no model;
2. the **field distance** between the two posed groups, restricted to their own atoms: geometry, but no model. This is not a weak reference -- it already knows the shape and the charge each group puts where;
3. the **predicted change**, the difference of the two atom-contribution sums: geometry and the co-field, so it also knows whether the activity depends on the place where the two groups differ.

**Measured** by `benchmarks/run_bioisostere.py` on matched molecular pairs found by RDKit's `rdMMPA` -- one single bond cut at a time, both pieces tried as the core, two molecules with the same core and different groups, nothing chosen by eye. Each measure is compared with the size of the measured activity difference by rank correlation. For every pair the dual coefficients are refitted **without both of its members**, so the predicted change has not seen the two activities it is judged against.

| set | pairs | fingerprint | field | co-field (pair left out) | co-field (whole model) |
|---|---|---|---|---|---|
| GPB, 66 molecules | 730 | -0.025 | +0.005 | +0.561 | +0.631 |
| THER, 76 molecules | 142 | +0.084 | +0.162 | +0.277 | +0.641 |
| THR, 88 molecules | 751 | +0.144 | +0.242 | +0.335 | +0.711 |

**The co-field weighting earns its place.** On GPB and THR the predicted change ranks the measured activity difference better than both references by more than the interval of rule 4: against the fingerprint distance +0.586 [+0.505, +0.669] and +0.191 [+0.106, +0.279], against the field distance +0.556 [+0.479, +0.632] and +0.093 [+0.008, +0.181]. On THER it is ahead of both as well, +0.193 [-0.059, +0.437] and +0.116 [-0.089, +0.313], but the interval contains zero; that is the set with the fewest pairs. On no set is it worse than either reference.

**How much of it is memory** is the gap between the last two columns, and the gap is large: the whole model, which has seen both activities of every pair it judges, reaches +0.631, +0.641 and +0.711, while refitting without the two members leaves +0.561, +0.277 and +0.335. The honest column is the one above. Two things still come from the whole set and are worth naming: the field types, the attenuation and the ridge were chosen on all the molecules, and the alignment is the published one.

**What the table is read for is the size, not the ranking.** A predicted change smaller than twice the conformal half width is a change the model cannot see, and the groups inside that bound are the ones the table calls interchangeable. That part is the model's own error bar rather than a claim about the chemistry. On a model of GPB whose half width is 1.5 log units the bound covers every pair of the 35 groups at that position, which is the table saying honestly that this model cannot separate them; what it still gives is the order, and the groups it puts closest to the carboxamide are the methyl carbamate (+0.02), butyramide (+0.04) and acetamide (+0.05).

```
cmf isosteres model.cmf --molecules series.sdf --activity pKi --csv table.csv
```

The command finds the position carrying the most different groups, prints for each group the partners the model puts closest with the predicted change of the exchange, marks those the model can actually tell apart, and writes the whole square table.

## 9.14 The gradient of a prediction

Everything above ranks candidates that somebody enumerated: a library of fragments, a table of guests, a set of poses. A prediction is also a differentiable function, and the derivative answers a different question -- not *which of these* but *which way*.

The kernel is a sum of Gaussian overlaps over pairs of atoms, and a Gaussian overlap is an analytic function of the distance between the two atoms. So the prediction of a molecule is an analytic function of the positions of its atoms and of the per-atom weights that carry its fields, and both derivatives are available in closed form:

```
from cmfqsar.gradients import prediction_gradient

g = prediction_gradient(model, mol)
print(g.value)                      # the prediction, the same number predict() gives
print(g.coords.shape)               # (N, 3): d(prediction) / d(atom position)
print(g.weights.shape)              # (N, F): d(prediction) / d(per-atom field weight)
print(g.largest_weight_terms(3))    # the three atoms and fields that matter most
```

Nothing has to be enumerated first, and nothing is approximated: the normalisation, the field scaling and mixing, the nonlinear kernel layer and the interpolation between attenuation grid points are all differentiated with the kernel. The non-field descriptors contribute nothing, because they are computed from the molecular graph and do not change when an atom moves.

**The co-field is the potential the atoms move in.** For an unnormalised linear model the two readings of the model are the same statement. The derivative of the prediction with respect to the position of one atom is the atom's weight, times the gradient of the co-field at the atom's own position, smoothed by the atom's own Gaussian, summed over the fields. `cofield_gradient` evaluates that gradient anywhere, with or without the smoothing, and needs no molecule -- only a place to stand. A design rule that was a picture (*put something electronegative in the red lobe*) becomes a force.

**Three uses.** A three-dimensional generative model proposes atom positions, and a gradient lets a fitted CMF model steer that proposal instead of only judging the result. A pose or a conformation can be refined by predicted activity, the way `varalign` and `flexalign` refine it by confidence. And the derivative with respect to one atom's weight is a local reading of the model: how much the prediction would change if that atom carried a little more charge, or were a little more hydrophobic -- the question a co-field picture is usually asked, answered as a number per atom.

## Is it the derivative, and what does it cost

The only independent check is a finite difference of `predict` itself. A correct gradient disagrees with it by the square of the step until round-off in the predictions takes over, so a scan over the step shows a minimum, and the depth of that minimum is the accuracy of the comparison rather than of the gradient (`benchmarks/run_gradient.py`).

| set | molecule | atoms | step 1e-2 | 1e-3 | 1e-4 | 1e-5 | 1e-6 |
|---|---|---|---|---|---|---|---|
| steroids | prednisolone | 54 | 3.4e-04 | 3.4e-06 | 3.4e-08 | **6.2e-09** | 4.1e-08 |
| THR | 01 | 75 | 5.4e-05 | 5.4e-07 | **5.4e-09** | 2.3e-08 | 2.1e-07 |

The alternative to the analytic form is that finite difference, which costs two predictions for every degree of freedom -- 3N for the positions and N per field for the weights.

| set | atoms | degrees of freedom | analytic | finite differences | gain |
|---|---|---|---|---|---|
| steroids | 54 | 432 | 5.6 ms | 1.07 s | 190 times |
| THR | 75 | 600 | 29 ms | 7.4 s | 258 times |

## What happens if you follow it

Steepest ascent on the atom positions of one held-out steroid, with every step capped so that no atom moves more than 0.05 A. The last two columns are the important ones.

| step | prediction | largest displacement | rmsd | posterior sd | worst bond length change |
|---|---|---|---|---|---|
| 0 | +7.934 | 0.00 A | 0.00 A | 0.234 | 0.00 A |
| 1 | +8.149 | 0.05 A | 0.01 A | 0.247 | 0.06 A |
| 5 | +8.875 | 0.25 A | 0.07 A | 0.356 | 0.29 A |
| 10 | +9.534 | 0.50 A | 0.12 A | 0.508 | 0.54 A |
| 20 | +10.329 | 1.00 A | 0.22 A | 0.701 | 0.80 A |

The prediction rises by 2.4 log units, which is more than the whole range of some series. Two things rise with it. The posterior standard deviation triples, so the molecule is leaving the training set at the same rate as the number is improving. And the bond lengths are pulled apart by up to 0.8 A, because nothing in the gradient knows that a molecule has bonds. The same walk on a thrombin ligand gains 2.0 log units and distorts a bond by 0.44 A.

## What it cannot do

**A weight is not a molecule.** The weight of an atom in the electrostatic field is its partial charge, and the charges of a real molecule are not free: they sum to its total charge and they follow from its structure. A positive derivative says the model would like more of that quantity there. It does not say that a molecule exists in which only that one atom has changed. The weight gradient is a direction in the field representation, to be realised by choosing another atom or another substituent.

**Maximising an overlap is not designing a molecule.** The table above is the demonstration: the overlap with a positive lobe of a co-field grows without limit as atoms are piled into that lobe. The gradient is a term to add to a generative model or a force field that enforces geometry, and not an optimiser to run on its own.

**The model still has a domain.** A step along the gradient goes where the training set is not. The posterior standard deviation, the conformal interval and the two-sample test are what say how far it can be trusted, and the lower confidence bound of *Which molecule to make next* is the objective to follow when the aim is a molecule to make rather than a number to maximise.

**It needs a frame.** The gradient is defined for the overlap-integral kernel, which is the one that needs the molecules superimposed. The rotation-invariant, surface and periodic kernels are built from the distances inside each molecule, and the program refuses them with that reason instead of returning a number.

# 10 Conformations: the fourth dimension

A 3D-QSAR model represents each molecule by one conformation, and the answer depends on which one. 4D-QSAR (Hopfinger et al. 1997) replaces the choice by an ensemble and averages over it. *Model → Conformations* switches between the two; **single is the default and is unchanged**, and the tests assert that by comparing its fields against those built before ensembles existed.

In CMF the averaging needs no new mathematics. A field is a sum of atom-centred Gaussians and the kernel is bilinear in the fields, so the mean field of an ensemble is itself a field — its atoms are the atoms of every conformation with their weights multiplied by that conformation's weight — and the kernel of two ensembles comes out as

$$K(A,B) = \sum_k \sum_l w_k^A w_l^B \, K\left(X_A^{(k)}, X_B^{(l)}\right)$$

the mean-map kernel of the two conformational distributions, positive definite by construction. A test checks that identity against the pairwise kernels of the conformations. The model stays linear in that mean field, so **co-fields, the decomposition rules, Shapley values and the complementarity scores all keep working**; contributions arrive per atom per conformation and fold back onto the atoms by summation.

*Tools → Generate conformations...* builds the ensembles with ETKDG and MMFF94s, keeping the pose a molecule already has as its first conformation, so an earlier alignment survives. A multi-conformer file can be read instead, with `--ensemble-by name` grouping the records of one compound. Weights are uniform or Boltzmann from the stored energies, with kT and a cap on the number kept.

**Where the other conformations go** is a choice of its own, and not a small one: what matters is not where a conformation sits relative to its own molecule but where it sits relative to the rest of the series.

| `--conformer-align` | **Where each generated conformation is put** |
|---|---|
| `self` | on the molecule's own pose, over all heavy atoms — the default, so nothing changes silently |
| `core` | on its own pose, over the common substructure with a reference only, so the scaffold stays where the published alignment put it and only the flexible part moves |
| `reference` | on a reference molecule of the series, through that same common substructure, which puts every conformation of the series in one frame |
| `field` | as `reference`, then refined by maximising the normalised CMF kernel — the 2013 paper's criterion, applied per conformation |

**Multi-instance learning** (`cmf_cli.py mil`) goes further: instead of averaging, the model chooses which conformation of each molecule to use, alternating between fitting and selecting. Use it when you believe one conformation is the bioactive one and the others are noise; use the ensemble when you believe the molecule is genuinely flexible.

## 10.1 The variance of an ensemble: a flexibility map

The ensemble above represents a molecule by the **mean** of the fields of its conformations. A mean throws away how much the field moved between them, and that is a flexibility map: where a molecule's field is the same in every conformation, and where it is not. Plan item D10 asks whether that is worth carrying.

**It is a field of the same class, which is what makes it cheap to ask.** With the Gaussians written as exp(-b/2 |r - c|^2), the variance across conformations is

V(r) = (1/k) sum_m X_m(r)^2 - ( (1/k) sum_m X_m(r) )^2,

and every term is a product of two Gaussians. A product of two Gaussians is a Gaussian, so V is again a sum of Gaussians: at the **midpoints of atom pairs**, at **twice the width**, with a coefficient that falls off as the two atoms move apart. `cmfqsar/fluctuation.py` builds it, and `variance_fieldset` gives the field set it must be used with -- the widths are doubled, so a variance field and an ordinary one must not go into the same kernel call. The construction is exact and the tests check it against the variance evaluated directly on a grid.

**What it costs.** The pairs are what costs: k conformations and n atoms give k^2 n^2 pseudo-atoms before anything is dropped, so the field grows with the square of the molecule. A floor on the pair coefficient keeps that in hand -- pairs whose two atoms are far apart contribute almost nothing -- and `kept_share` says what the floor dropped. On the sets measured here that is 1065 to 2119 pseudo-atoms per molecule at an attenuation of 2 and a floor of 0.01, keeping 99% of the pair weight, and a kernel over a set takes 286 seconds at most. Hydrogens are dropped before the construction for the same reason: their fields are small and they double the atom count.

**It does not help.**

| set | molecules | the mean alone | the variance alone | the best mixture | difference |
|---|---|---|---|---|---|
| the 31 CBG steroids, the classic CoMFA benchmark | 31 | +0.794 | +0.425 | +0.785 at 0.10 | -0.009 [-0.079, +0.062] |
| glycogen phosphorylase b, 66 molecules | 66 | +0.460 | +0.180 | +0.462 at 0.10 | +0.002 [-0.015, +0.020] |
| thrombin, 88 molecules | 88 | +0.308 | -0.013 | +0.302 at 0.10 | -0.006 [-0.032, +0.023] |

Every mixture that includes the variance is at or below the mean alone, and every interval contains zero or lies below it, so under rule 1 the block adds nothing. The best mixture is the smallest one offered on all three sets, which is what a block that is not helping looks like when a search is made to choose it.

**What the variance carries by itself runs out as the series grows.** Alone it reaches q2 +0.425 on the 31 steroids, +0.180 on the 66 of GPB and -0.013 on the 88 of thrombin: something on a small set of one rigid scaffold, nothing on a larger and more varied one. And on none of them is it something the mean does not already have. That is the reading the plan expected, and it leaves the variance field where it is useful: as a map. Where a series is rigid and where it moves is a different question from where it is active, and the two pictures are worth putting side by side -- `cmfqsar/cofields.py` draws a variance field like any other, with the field set that `variance_fieldset` returns.

## 10.2 Scoring a conformational restriction

Tying two parts of a molecule together into a ring, so that a bond can no longer rotate, is one of the few moves in medicinal chemistry whose effect is meant to be purely conformational. The variance field of the previous section is a map of where a molecule places field only part of the time, which is exactly what such a move is supposed to change. Plan item MC9 asks whether it can be turned into a number: when a chemist closes a ring, does the field that matters stop moving, and does the potency follow?

**Two amounts per molecule** (`cmfqsar/restriction.py`). The co-field C_f(r) says how much the field of type f at point r matters for the activity, so its **square** is a weight: positive everywhere, large where the model reads the field, near zero where it does not. With that weight a molecule has

- **relevant**, the integral of C_f(r)^2 Xbar_f(r)^2 -- how much activity-relevant field the molecule places, averaged over its conformations;
- **intermittent**, the integral of C_f(r)^2 V_f(r) -- how much of that field is there only part of the time.

Both are summed over the field types and taken on a grid, the same grid machinery that draws a co-field. A pair of a flexible molecule A and its restricted analogue B is then two shares: **kept** = relevant(B)/relevant(A), and **removed** = 1 - intermittent(B)/intermittent(A).

**The pairs are found by a rule and not by eye**, so that the measurement cannot be steered. The restricted member has exactly one ring more, at least one rotatable bond fewer, at most two heavy atoms more, an atom composition differing by at most two atoms, and a maximum common substructure covering at least 80 per cent of the smaller member. A fingerprint is deliberately **not** used for the similarity: closing a ring changes the environment of every atom near the new bond, so butylphenol and 2-methylchroman -- the same eleven heavy atoms, one bond more -- are only 0.15 alike as Morgan fingerprints, and a threshold high enough to be useful would throw away exactly the pairs this item is about. Both members get the same conformer generator with the same settings, since the comparison is between two ensembles and not between two molecules.

**Measured** by `benchmarks/run_restriction.py` on four aligned Sutherland sets, 439 pairs in all, with three scores on the same pairs:

| score | pairs | correlation with the potency gain |
|---|---|---|
| rotatable bonds removed | 439 | +0.236 [+0.151, +0.317] |
| variance removed, unweighted | 439 | -0.034 [-0.140, +0.062] |
| variance removed, weighted by the squared co-field | 439 | +0.043 [-0.030, +0.111] |
| the same, with the pair left out of the co-field | 439 | +0.020 [-0.061, +0.093] |
| relevant field kept, weighted | 439 | +0.220 [+0.120, +0.315] |

**The score does not pass the gate of the item.** How much intermittent field a ring closure removes is not related to whether the closure gained potency: +0.043 [-0.030, +0.111], an interval that contains zero. Counting the rotatable bonds the closure removed reaches +0.236 [+0.151, +0.317] on the same pairs and needs no geometry, no conformations and no model. The co-field is also rebuilt without both members of every pair, so the weight cannot have learned the answer from the two potencies it is compared with; that changes the score to +0.020, which says there was

nothing to learn.

**What does track the potency gain is the other half of the score.** How much of the activity-relevant field the restricted analogue still places reaches +0.220 [+0.120, +0.315]. That is the **mean** field and not the variance, and it says the expected thing: a ring closure helps when the molecule still puts its field where the model reads it, and hurts when the ring holds it somewhere else.

This is the third place in this manual where the conformational ensemble carries less than it promises. Averaging a field over conformations does not predict better than one geometry; the variance adds nothing as a block of the kernel; and it does not explain what a restriction did either. The variance field stays a map, which is where the previous section left it.

**From the command line.** The score is a subcommand of its own, because it needs a model (for the co-field), a set of molecules in that model's frame, and conformations of each:

```
cmf restriction model.cmf --molecules series.sdf --activity pIC50 --conformers 3 \
    --csv pairs.csv
```

It finds the pairs by the rule above, prints the two shares of each, correlates **removed** with the potency gain when the file carries potencies, and writes one row per pair. It is the right tool for asking the question of a series of your own; the answer on the four public sets above is that the question has a negative answer there.

**Two limits of the measurement.** The ensembles are three conformations from a distance-geometry generator superimposed on the pose the series was aligned to, which is a narrow view of what a molecule does in water. And a ring closure is never purely conformational in practice: it changes the shape, and often an atom or two as well. A set of pairs that differ in nothing but a single bond would be a better test, and no public set of that kind with measured potencies was found.

## 10.3 Protonation and tautomers: ensembles of different molecules

A series measured in acid is modelled in one protonation state, chosen when the data was prepared. That choice is a guess wherever a pKa falls near the pH of the experiment: an azole with a pKa of 1 in 1 M HCl is about half protonated, and neither the neutral form nor the cation is the molecule that was measured. The same applies to a tautomer: an azole can be written with its proton on either nitrogen, and the two forms have different fields.

**The averaging needs no new mathematics, and the members need not be one molecule.** The argument of the previous section used only bilinearity: the mean field of an ensemble is itself a field whose atoms are the atoms of every member with their weights multiplied by that member's weight. Nothing in it requires the members to be conformations. They can be protonation states, or tautomers, which are different molecules with different numbers of atoms (`cmfqsar/protonation.py`, `cmfqsar/tautomers.py`). The kernel is still the mean-map kernel, still positive definite, and the model is still linear in the mean field, so the co-fields and the decomposition rules keep working. Tests check that identity against the pairwise kernels of the members for both kinds of ensemble, and that a conformational ensemble built through the general builder reproduces the tiled one atom for atom.

| Option of `build` | What it does |
|---|---|
| `--ph 0` | represent every molecule by the pKa-weighted mean field of its protonation states at this pH |
| `--ph-states N`, `--ph-min-population F` | how many states to keep, and how light a state may be |
| `--tautomers score` | the mean field over the tautomers of the state the file gives |
| `--tautomers uniform`, `--tautomers given` | keep every enumerated form, or weigh the forms by energies in the molecule property `tautomer_energies` |
| `--tautomer-forms N` | forms kept at most |

**The weights of the protonation states are populations.** Each ionisable site is found by Dimorphite-DL's patterns, given its regression pKa, and weighted by the Henderson-Hasselbalch fraction

$$f = 1/(1 + 10^{\mathrm{pH} - \mathrm{p}K_\mathrm{a}})$$

for the protonated form and its complement for the deprotonated one; a state's weight is the product over the sites. **The weights of the tautomers are not populations.** A tautomer has one only if the free energies are known, and none are computed here, so `score` keeps the forms RDKit scores highest and counts them the same. Energies from a semiempirical calculation would replace that assumption with a Boltzmann weight.

The states are built on the geometry the molecule arrives with: the heavy atoms keep their coordinates and only hydrogens are added, removed or moved, so **an aligned series stays aligned** and the ensemble costs no further superposition. Atom contributions come back on the atoms of the molecule as usual; a proton that only one state carries is reported on the heavy atom it sits on, which is the atom a chemist would read it on.

## What it is worth, measured

`benchmarks/run_protonation.py` measures four models on the 23 corrosion-inhibitor series in 1 M HCl (997 molecules, `examples/corrosion_hcl`), with the same five fields, the same attenuation and ridge grids, the same ten-fold cross-validation and the same scaffold alignment as `run_corrosion.py`: `file`, the state the data file gives; `dominant`, the likeliest state by pKa; `ensemble`, the pKa-weighted mean over the states; and `tautomer`, the mean over the tautomers of the file's state.

| Model at pH 0 | mean $q^2$ | best on |
|---|---|---|
| `file` (the state the data file gives) | **+0.324** | 13 of 23 series |
| `dominant` (the likeliest state by pKa) | +0.260 | 3 of 23 |
| `ensemble` (the pKa-weighted mean) | +0.261 | 1 of 23 |
| `tautomer` (the mean over the forms) | +0.314 | 6 of 23 |

**At pH 0 the ensemble has almost nothing to average over.** Only 1% of the molecules are a real mixture in 1 M HCl, because a pKa two units away from the pH already settles the question, so

`ensemble` differs from `dominant` by +0.001. Where the two differ is in which state is chosen, and the pKa choice is the worse one here: -0.064 against the state the data file gives, with `file` best on 13 of the 23 series. The likeliest state by pKa carries the charge the file gives it for only 52% of the molecules, and its mean charge is +1.57 against +0.77 in the file: treating the sites as independent protonates a second and a third basic nitrogen that would not both be protonated in reality. **So the limit is the pKa model, not the ensemble.** Averaging the tautomers moves the result by -0.010, on the 5% of molecules that have more than one form worth keeping.

Where a mixture does exist, the ensemble has something to do: the share of molecules that are a real mixture is 1% at pH 0, 18% at pH 2, 83% at pH 3 and 27% at pH 7.4.

**And it is not the lack of mixtures that holds the ensemble back.** Repeating the measurement at pH 3, where 83% of the molecules are mixtures and a molecule has 3.0 states on average, gives the same order and no gain from the averaging: `file` +0.324, `dominant` +0.255, `ensemble` +0.246, which is -0.009 for the averaging itself. The endpoint was measured in 1 M HCl, so a pH-3 model is not the chemistry of the experiment; it is there to show what the averaging does when there is something to average, and the answer is nothing measurable.

**When to use it.** On this evidence, not as a default: the state prepared by hand gave the higher mean q2 at both pH values measured and was the best of the four models on 13 of the 23 series. Use it as a check -- if the ensemble and your chosen state disagree, the pKa of that molecule is near the pH and the state is worth looking at -- and use it as the model only where no one has chosen the states by hand. Read the mean charge the program prints beside the ensemble: where it is far above the charge of your file, the independent-site assumption is adding protons that a real molecule would not carry, and the ensemble is then worse than the single state, not better. The pKa values are Dimorphite-DL's and cannot be replaced from the command line yet; tautomer energies can be, through the molecule property `tautomer_energies`. Use the tautomer ensemble where the series contains azoles or other groups whose form the file decided arbitrarily, and read its numbers as the weaker claim they are. A conformational ensemble and a protonation or tautomer ensemble cannot be combined yet: the program refuses rather than build a product of members it cannot describe.

# 11 Working without an alignment

This chapter is the one to read if your molecules came from SMILES, if your series has several scaffolds, or if you simply do not trust the superposition you have. The program offers three frame-free routes, and the scripts in `benchmarks/` measure all of them.

## 11.1 Why the frame is the problem

An alignment is an assumption, and a wrong one is expensive. Measured on the nine targets that occur both in the curated Sutherland collection and in the automatically aligned Mittal collection — same compounds, same activities, same fields and grids — the curated superposition gives a mean $q^2$ of **+0.597** and the automatic one **+0.295**. Worse, the gain over a baseline with no geometry at all (atom-type counts) is +0.25 on the curated files and **−0.06** on the automatic ones: on five of the nine targets, fields in an automatic frame predict *worse* than counts. An alignment that is not trustworthy is not merely imperfect, it is actively harmful.

## 11.2 The pose-free count kernel

As $\alpha \to 0$ an indicator field tends to the count of atoms of that type, so the kernel tends to the dot product of the atom-type count vectors and nothing of the geometry survives. The program implements that limit in closed form (`chemotypes.count_kernel`), which is exact rather than approached, costs one pass over the atoms, and is 10 to 34 times faster than the Gaussian route. It is the **baseline** any aligned model should be made to beat, and it is what the leave-one-chemotype-out workflow reports beside the aligned model.

One subtlety, because it is easy to get wrong: the normalisation must be applied to the joint kernel, not field by field. A field normalised on its own tends to 1 whenever both molecules contain the type at all, so the counts cancel and the baseline loses the very descriptors it is built from.

## 11.3 The rotation-invariant autocorrelation kernel

Geometry without a frame. For each pair of field types the program computes the smoothed field autocorrelation

$$\Phi_{fg}(d) = \sum_{l}\sum_{m} w_{lf} w_{mg} \exp\left(-\frac{(|\mathbf{r}_l - \mathbf{r}_m| - d)^2}{2\sigma^2}\right)$$

as a function of interatomic distance $d$, which is invariant under rotation and translation by construction.

**Where the two molecules are.** That equation describes *one* molecule: both sums run over its own atoms, which is what makes it an autocorrelation. It is a descriptor, not a kernel — one number per channel $c$, a channel being one pair of field types at one distance — and the kernel is the cosine-normalised dot product of two such vectors:

$$K(A,B) = \frac{\sum_c s_c^2 \, \Phi_c(A)\,\Phi_c(B)}{\|s\,\Phi(A)\|\;\|s\,\Phi(B)\|}$$

with $s_c$ the scale channel $c$ inherits from its block: the reciprocal root mean square of that block over the training molecules, times the block's mixing weight. This is the opposite construction to

the overlap kernel of *How a prediction is made*, which is defined on two molecules from the start and has no finite feature vector at all — its feature map is the field itself. That reversal is also what decides how each model can be read afterwards: a dual expansion over a feature map that is a function of space is itself a function of space, which is the co-field, while a dual expansion over channels is a vector of channel weights, which is the co-pattern.

The $d = 0$ channel reproduces the CMF self-overlap exactly when $\sigma^2 = 2/\alpha$, so the descriptor contains the ordinary kernel's diagonal information and adds the distance profile. Select *Autocorrelation* as the kernel source in the model panel (`--kernel autocorrelation`, with `--ac-sigma` and `--ac-dmax`), and the model fits, cross-validates, saves and predicts exactly as any other — only the co-field, which needs a frame, is unavailable.

On the chemotype benchmarks the invariant kernel recovers most of the value of the geometry without a frame (0.172 against 0.189 for the aligned model and 0.072 for counts), and on automatically aligned files it frequently beats the aligned model outright — on six of the nine shared targets, including thrombin at +0.452 against +0.021.

**Two points, and what that misses.** The descriptor counts pairs of atoms at a distance, and a mirror image has every pair at exactly the same distance, so this kernel cannot tell a molecule from its enantiomer. It is blind for the same reason to the orientation of one field with respect to another — whether the donor sits clockwise or anticlockwise from the bulk, going round the ring. The quantity that is not blind is the signed volume of three displacement vectors, and `--ac-chirality` adds it as a second block of channels, the signed triple products of three field types resolved over `--ac-chirality-shells` radial shells (four by default). The block is appended rather than mixed in, so the two-point part is unchanged to the last bit, and it is scaled separately because a volume carries the cube of a length.

**Measured, it does not justify its cost.** `benchmarks/run_chirality.py` asks the question in the regime the alignment-free kernel exists for — leave-one-chemotype-out on the five chemotype-split sets, with three regularisations by three distance grids chosen inside the training part of every fold, and the number of shells kept out of the search so the signed channels never get more freedom than the kernel they extend:

| | two-point | signed, 1 shell | signed, 2 shells | signed, 4 shells |
|---|---|---|---|---|
| mean $q^2$ over the five sets | +0.063 | +0.082 | +0.095 | +0.108 |
| paired difference | — | +0.019 | +0.033 | +0.045 |
| 95 % interval over the sets | — | [−0.051, +0.113] | [−0.031, +0.121] | [−0.028, +0.121] |
| better on | — | 2 of 5 | 3 of 5 | 3 of 5 |

The mean moves in the right direction and the interval contains zero at every setting. More to the point, **one set accounts for all of it**: at four shells COX2 gains +0.187 and the other four sets average +0.009 between them, with AChE and BZR made worse. On five sets that is not a result, and the channels cost 16 % more columns.

**The experiment those sets cannot do.** None of the five contains an enantiomeric pair, so the kernel's blindness costs it nothing there and the table above is about something else. The same script therefore builds a set where handedness *is* the activity: twenty molecules, each with its exact mirror image — one conformation reflected, not a second embedding, so a pair differs in handedness and in nothing else and their two-point channels agree to one part in $10^{15}$ — with the

activity `(1 − w)·composition + w·handedness`, the handedness read as the CIP descriptor rather than the SMILES tag, which is not the same label from one scaffold to the next.

At `w = 0` the model works: q² +0.97. At `w = 1`, where the answer is the handedness and nothing else, **no variant learns it** — two-point −0.391, and the signed channels −0.34 to −0.37 however many shells they are given and even when scaled up to carry half the kernel.

**And the reason is measurable.** A kernel method carries what it learnt from one molecule to another along a direction in feature space. The signed channels of a molecule and its own mirror image are exact negatives — that is what the block is for, and a test asserts it — but between *different* molecules they share nothing: the cosine of the signed block is **−0.016 for two molecules of the same handedness and +0.016 for two of opposite handedness**, a difference of 0.03 against a spread of 0.55. Even within one scaffold family, amino acids differing only in the side chain, the difference is 0.10 against a spread of 0.48. Handedness is not a direction in this space, so there is nothing for a linear model to transfer, and the two negative results above are one result seen twice.

That is the conclusion: the block is correct and does what it says about a molecule and its mirror image; it does not put the handedness of two different molecules on a common axis, which is what a QSAR model needs. The channels stay available and off by default.

**The construction that does work reads one centre at a time.** `cmfqsar/stereo.py` takes every atom with at least three neighbours, orders those neighbours by their weight in a field — the field weights playing the part the CIP rules play for a chemist, so that the descriptor stays a property of the fields — and takes the signed volume of the ordered frame. It negates under reflection and survives any rotation, like the three-point block, and unlike it two *different* molecules of the same configuration agree: the cosine between same-handedness molecules is **+0.468 against −0.468 for opposite**, a difference of 0.935, where the three-point block manages 0.03.

That difference explains everything, and the synthetic set shows what it gains. Where the activity is handedness and nothing else, the stereo centres reach **q² +0.721** while every variant of the three-point block sits at −0.35; at every mixture from a quarter handedness upward they are ahead of the two-point kernel, and where there is no handedness in the activity at all they cost little (+0.924 against +0.970).

On the five real sets they add nothing — mean +0.039 against the two-point kernel's +0.063 — which is the expected answer rather than a disappointment: those sets contain no enantiomeric pair, so the channels carry no signal and 25 extra columns dilute the kernel. **Turn them on for a series whose stereochemistry varies, and leave them off otherwise.** An order that rests on two nearly equal weights is not an order, so a centre whose top three priorities are within `min_separation` of each other contributes nothing rather than a guessed sign; that guard is what takes the same-handedness separation from 0.18 to 0.86.

**Mixing coefficients over the channel blocks.** The channels come in blocks, one per pair of field types, each scaled to unit root mean square before all of them enter one cosine-normalised dot product. The overlap kernel weights its fields with $h_f$; the three invariant sources (autocorrelation, surface, periodic) weight their blocks with `--block-weights` (library: `ac_block_weights` on `CMFModel`), one coefficient per block applied to the scaled features before the normalisation, so that a block's share of the kernel goes as the square of its weight and a weight of zero removes the block exactly. Three ways to set them: `balanced` gives every *kind* of block — the pairs within a group of fields, the pairs across two groups — the same share of the kernel however many blocks each kind has; `KIND=W,...` names the kinds' weights (`indicator=0.3,indicator*physchem=0.5`; unnamed kinds weigh 1); `align` or `alignf` derive one weight per block by centred kernel alignment of the per-block kernels

(`align_block_weights`, the `align_mixing` of the fields carried over, with `--block-rho` the shrinkage of the joint solution), re-derived from the training part of every cross-validation fold, so the reported $q^2$ never sees the left-out activities. The groups default to the continuous indicator fields (`indicator`) against the rest (`physchem`); `--block-groups 'GROUP=field,field;GROUP=cif:*'` names others. The fitted weights are printed by kind and stored with the model (`block_weight_report`).

## What the model says about one molecule

The invariant kernel has no co-field, because it has no frame, but **it does have a decomposition over atoms**. Every channel is a double sum over atom pairs, and the prediction is a fixed linear functional of the channels, so a pair of atoms is one term of the prediction and the only choice is how to divide a pair between its two atoms. Dividing it equally gives one number per atom that adds up to the prediction exactly:

$$y(x) - b = \sum_a q_a, \qquad q_a = \frac{1}{\|s\,\Phi(x)\|} \sum_c s_c\,\gamma_c\,\Phi_c^{(a)}(x)$$

with γ the co-pattern -- the training molecules' channel rows summed with their dual coefficients -- and $\Phi_c^{(a)}$ that atom's half of the channel. `CMFModel.atom_contributions` returns them for an invariant model as it does for an aligned one, the report lists the largest, and the viewer colours the atoms by them. `invariant_pair_contributions` splits the same total over the pairs of field types instead, which says which kind of channel the prediction came from. The signed three-point channels are a sum over *triples*, so a model built with `--ac-chirality` has no split over two atoms and the program says so rather than dropping them.

**Does it name the right atoms?** `benchmarks/run_guest_atoms.py` asks the question on the cyclodextrin guest tables, where the chemistry is known: binding is driven by a hydrophobic group entering the cavity, and polar groups stay in the water. For every guest the mean share per heavy atom inside a group is compared with the mean outside it, with the contributions taken from models fitted without that guest, so a guest's own constant never enters its own explanation.

| Group on beta-cyclodextrin | guests | share per atom, inside minus outside |
|---|---|---|
| adamantyl | 6 | +0.0646 ± 0.0368 |
| cyclohexyl | 122 | +0.0378 ± 0.0202 |
| phenyl | 817 | +0.0158 ± 0.0041 |
| carboxyl | 347 | -0.0366 ± 0.0046 |
| hydroxyl | 221 | -0.0916 ± 0.0126 |
| amine | 217 | -0.0886 ± 0.0270 |

**On beta-cyclodextrin it does.** The three groups that are known to enter the cavity take more of the prediction than the rest of their own molecule and the three polar groups take less, and the order among the three -- adamantyl, then cyclohexyl, then phenyl -- is the order of the affinity they are known to give. The largest sets carry the claim: phenyl on 817 guests and the three polar groups on 347, 221 and 217, every interval away from zero. Adamantyl is the strongest effect and the weakest evidence, 6 guests.

**On the other two hosts it does not.** On alpha-cyclodextrin, whose cavity is too narrow for a benzene ring, phenyl comes out at -0.0158 and the carboxyl at +0.0703; on gamma-cyclodextrin, whose cavity is wide enough that none of these groups fills it, every one of the three is negative

(-0.0333 for phenyl, -0.0558 for cyclohexyl). Read that as the limit of the reading rather than a fault in the arithmetic: the split is exact for all three, and what differs is whether the group a chemist would name is the group this host's data makes the model lean on.

**It is a statement about a series, not about one molecule.** Shuffling an atom's number among the atoms of the same molecule gives a null, and the group's z-score against it is +0.35 for phenyl and +1.38 for adamantyl: within one guest the group does not stand out far above the spread of its own atoms. What is solid is the mean over hundreds of guests. Use the atom numbers to see what a model has learned from a table, not to decide where to put the next substituent on one compound.

## 11.4 Molecules on a surface: the in-plane invariant kernel

A molecule adsorbed on a surface has one direction that is not like the others. Its height above the metal and its tilt are physics — they are what the relaxation computed, and they are most of what decides the adsorption energy. Where it sits in the plane and how it is turned about the normal are not: the slab is periodic and its origin is wherever the cell was cut. The overlap kernel cannot tell the two apart, so it demands a frame for a degree of freedom that has no preferred value, and the invariant kernel of the last section throws the height away along with the rest.

This kernel keeps exactly what is physical. It is the two-point distribution of the field weights over the **in-plane** distance ρ, resolved by the heights of both atoms:

$$\Phi_{fg}(\rho_k, z_i, z_j) = \sum_l \sum_m w_{lf} w_{mg}\, G_\sigma(\rho_{lm} - \rho_k)\, G_{\sigma_z}(z_l - z_i)\, G_{\sigma_z}(z_m - z_j)$$

$$G_s(u) = \exp\!\left(-\frac{u^2}{2s^2}\right), \qquad \rho_{lm} = |\mathbf{r}_l^{xy} - \mathbf{r}_m^{xy}|$$

It is therefore invariant to translation in the plane and to rotation about the normal, and to nothing else: tilt, height and the vertical order of the atoms all survive. Averaged over the direction of ρ it is, by the Wiener–Khinchin theorem, the in-plane power spectrum of the fields resolved by height — the molecule described in reciprocal space across the surface and in real space along the normal. The surface is at $z = 0$ with the normal along $+z$, which is how the adsorption scripts of `examples/corrosion_fe110` write their poses; nothing checks that convention, so a set of poses in some other frame will be described accurately but uselessly.

The channels run over in-plane distances to 12 Å in steps of 0.75 Å (`--ac-dmax`, `--ac-sigma`) and heights from 1 to 6 Å in steps of 1 Å (`--surface-z ZMIN ZMAX ZSTEP`), the spacing of a channel being also its width on both axes. That is 17 × 6 × 6 = 612 channels for each pair of field types, 9,180 features on the five physico-chemical fields, which is why the blocks are scaled and may be weighted exactly as in the preceding section. Pairs of different field types are summed both ways round, as the three-dimensional descriptor counts them, and the $l = m$ terms are kept. Those channels are again a descriptor of one molecule, and the kernel of two of them is the dot product of the preceding section, over this section's channels:

$$K(A,B) = \frac{\sum_c s_c^2\, \Phi_c(A)\, \Phi_c(B)}{\|s\,\Phi(A)\|\;\|s\,\Phi(B)\|}, \qquad c = (f, g, \rho_k, z_i, z_j)$$

The command is `--kernel surface`; the model panel does not offer it.

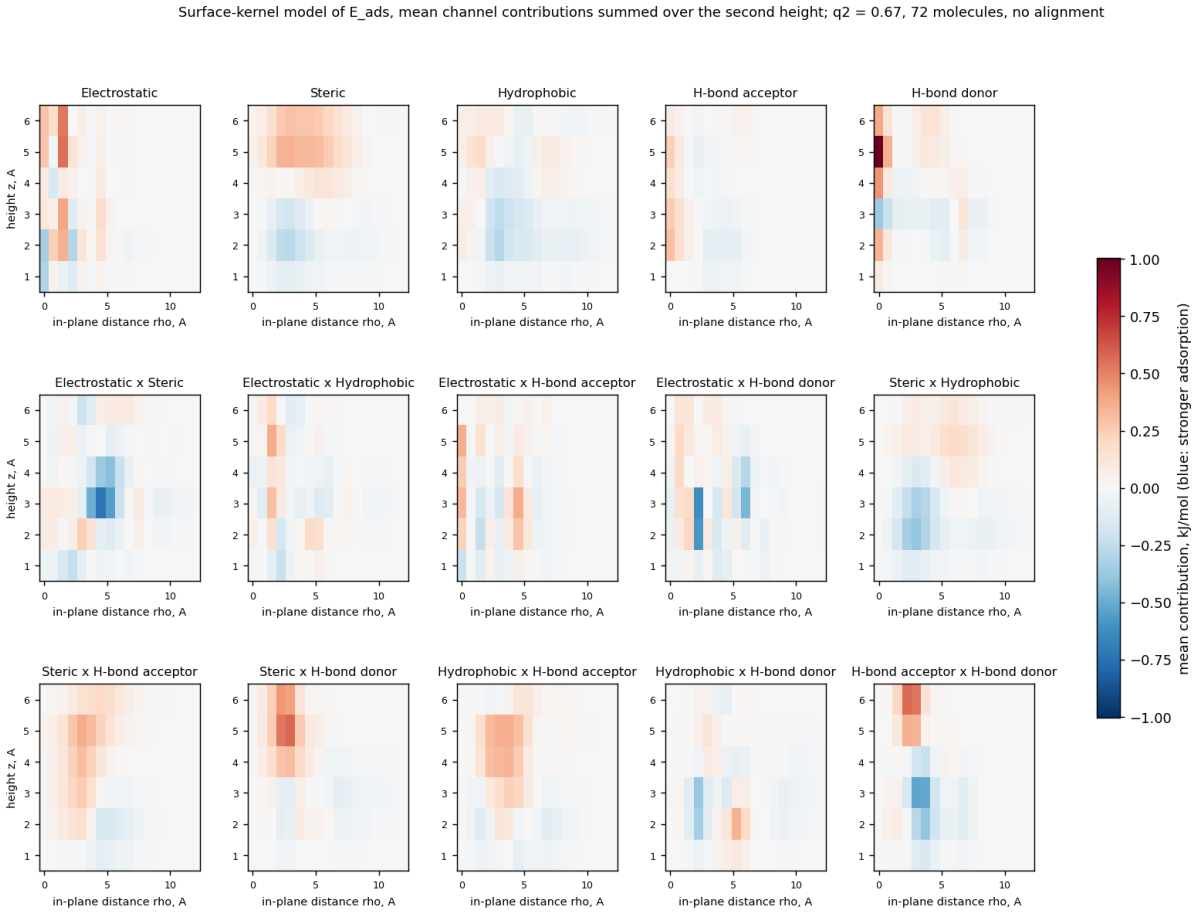
### What it is worth

On the benzimidazole corrosion node — 72 molecules relaxed on Fe(110) with the UMA machine-learned potential, the target their adsorption energy, five physico-chemical fields, KRR with α and λ chosen in a 10-fold inner cross-validation, 5 outer folds, 10 repeats:

| model | $q^2$ | RMSE (kJ/mol) |
|---|---|---|
| **surface kernel**, poses as the relaxation wrote them | **0.655** | 31.2 |
| **surface kernel**, the same poses shuffled in the plane | **0.655** | 31.2 |
| overlap fields, poses as written | 0.505 | 37.4 |
| overlap fields, poses registered in the plane by their common substructure | 0.517 | 36.9 |
| overlap fields, poses shuffled in the plane | 0.480 | 38.3 |
| rotation-invariant autocorrelation, which discards the height | 0.109 | 50.2 |

Three things follow. The **invariance is demonstrated rather than asserted**: turning every pose by a random angle about the normal and sliding it up to 6 Å in the plane changes the model by nothing at all, the second row reproducing the first to three decimals. **The height is what produces the score** — the fully invariant kernel differs from this one only in discarding *z*, and it falls from 0.655 to 0.109. And **the frame the overlap kernel wants is not worth building**: registering the poses laterally on the common substructure moves it from 0.505 to 0.517, a negligible difference, while the kernel that was never given a frame is 0.14 above the better of them.

There is no co-field, for the reason every invariant source has none: the kernel is not an integral of a product of fields, so the prediction cannot be turned back into a function of space. What it offers instead is the co-pattern — the weight of every channel of the fitted linear model, which names the pair of field types, the in-plane distance and the two heights that carry the signal, which `top_channels` ranks against the labels of `surface_channel_labels`. The measurements above come from `surface_kernel_eads.py`, in `examples/corrosion_fe110`, which writes its table next to the script.




*The co-pattern of the surface model of the adsorption energies of 72 molecules on Fe(110), $q^2 = 0.67$: the mean contribution of every channel to the prediction, in kJ/mol, summed over the second height, by pair of field types, in-plane distance and height. Blue means stronger adsorption. The signal sits in a few channels, among them the electrostatic and the hydrogen-bond donor field at zero in-plane distance. Drawn by* `surface_channels_eads.py` *in* `examples/corrosion_fe110`*.*

## 11.5 Posed by rule: the surface frame without the slab

The surface kernel needs only the normal, but it cannot draw a co-field; the overlap kernel can, if the molecules are in the surface frame, and so far that frame came from relaxing every molecule on a slab. `cmfqsar/planealign.py` puts a molecule there by rule — flat, with the plane of its two longest principal axes parallel to the surface and the lowest heavy atom at a chosen height; upright, with its most basic nitrogen pointing a lone pair at the surface; or whichever of those is most similar in field to one trusted pose — and `cmf align --by plane` applies it to a set:

```
cmf align poses.sdf --by plane --pose flat --height 1.9 --out onsurface.sdf
cmf align poses.sdf --by plane --pose auto --refine plane --reference 0 --out onsurface.sdf
```

`--pose flat` turns the flattest face down and `flat-up` the other; `--pose auto` tries both faces and the upright pose and keeps the most similar to the reference, after `--refine plane` (moved in the plane and about the normal only) or `--refine plane+tilt` (also tilted by up to 30° and lifted by up to 1 Å). Nothing else moves: the conformation is never touched.

What the rule is worth was measured on the Fe(110) sets (`benchmarks/run_fe110.py`, the README section of the same name). A chemisorbed ring that the relaxation left flat is reproduced to 0.4 Å

*given its face*; the face of a planar molecule is a mirror in the plane that no rule and no field similarity to one reference decides, and the molecules the relaxation left upright or oblique are 3 Å from any rule pose. The overlap kernel in the rule frame recovers of the adsorption energy exactly what it recovers in the relaxed frame once that frame's height — the outcome of the relaxation, which alone predicts E_ads with q² 0.4–0.5 — is removed, and the surface kernel on the same rule poses does as well. The aligner therefore does not replace the slab; it is the way to put a set in a surface frame and draw a co-field in it when the frame is all that is wanted, and `--pose flat-up` is there because the user may know the face that the rule cannot.

## 11.6 Crystals: the periodic autocorrelation kernel

The same descriptor for a periodic structure, summed over the neighbour list of one cell with its periodic images out to a cutoff $r_{\max}$, per atom of the cell, with the uniform background of each field pair removed and the cutoff tapered:

$$\Phi_{fg}(d) = \frac{1}{N}\Big[\, P_{fg}(d) \;-\; \frac{S_f S_g}{V}\, B(d) \,\Big]$$

The first term is the tapered pair histogram of the two fields, over every atom $l$ of the cell and every neighbour $m$ inside the cutoff, periodic images included:

$$P_{fg}(d) = \sum_{l \,\in\, \text{cell}} \; \sum_{m \,:\, r_{lm} < r_{\max}} w_{lf} w_{mg}\, \omega(r_{lm}) \exp\!\left(-\frac{(r_{lm}-d)^2}{2\sigma^2}\right)$$

The second is what that histogram would be if the same total weight of each field were spread uniformly through the cell, which is the part of $P_{fg}$ that reports only how dense the crystal is:

$$B(d) = \int_0^{r_{\max}} 4\pi r^2\, \omega(r) \exp\!\left(-\frac{(r-d)^2}{2\sigma^2}\right) \mathrm{d}r$$

Here $N$ is the number of atoms in the cell, $r_{lm}$ the distance from atom $l$ to image $m$, $S_f = \Sigma_l\, w_{lf}$ the total weight of field $f$ over the cell, $V$ its volume, σ the width of the distance smearing (`--sigma`) and ω a half-cosine taper over the last `--taper` ångström. Both sums run over ordered pairs, so a pair of two different field types is counted both ways round, as the molecular descriptor counts it, and its background with it; the $l = m$ terms are kept, which makes the $d = 0$ channel the composition of the cell. As in the two preceding sections Φ describes one structure, and the kernel of two of them is the same cosine-normalised dot product over the channels:

$$K(A,B) = \frac{\sum_c s_c^2\, \Phi_c(A)\, \Phi_c(B)}{\|s\,\Phi(A)\|\;\|s\,\Phi(B)\|}, \qquad c = (f, g, d_k)$$

It is invariant to the choice of cell, origin and orientation, so a supercell is the same material. Averaged over orientations it is the sine transform of the product of the typed structure factors, which makes the pair products of the element indicator fields the partial structure factors a diffraction experiment measures.

The fields of a solid are built by `inorganic_fieldset`: electrostatic (oxidation states guessed from the composition, or the `oxi_states/charges` array of the file), steric (atomic volume), electronegativity, one indicator field per element and one for the rest, and any per-atom array as `prop:<name>`. The command is `build-crystals` (an ASE-readable structure file, `--target KEY` or `--target-file CSV`, `--cutoff`, `--sigma`, `--taper`, `--no-background`, `--elements`, `--charges`, `--external-cv`, `--save`); in the library, `kernel_source="periodic"` on a `CMFModel` with a table from

`crystal_table`. There is no co-field, but there is a co-pattern: the weight of every channel in the fitted linear model, which `top_channels` ranks by weight times spread, with the sign that says whether the pair at that distance raises or lowers the property.

On the Matbench `phonons` task (1,265 crystals, the top of the phonon density of states) the linear kernel on four fields -- electrostatic, steric, electronegativity and the inverse square root of the mass -- reaches a mean absolute error of 117 $cm^{-1}$ on the Matbench folds, 106 with a Gaussian layer, against 324 for the dummy predictor and 29 for the best graph network; keeping the uniform background improves the 58-field model from 181 to 119 on this task, so the switch is worth trying on a property that depends on composition and density (the four-field model kept it too reaches 92). The element indicator fields dilute the joint kernel — seventeen hundred equally scaled element-pair blocks against the ten physico-chemical ones — and `--block-weights` is the remedy: with the background kept, the aligned weights take the 59-field model from 119 to 97; they do not make it better than the four physico-chemical fields alone. The full table and the co-patterns are in `benchmarks/results/matbench_phonons/table.md`.

## 11.7 An atom instead of a molecule: the local invariant kernel

Every model so far predicts one number per molecule. A carbon shift, an acidity, a local reactivity or a site's energy belongs to an **atom in a molecule**, so the sample has to be the atom and its surroundings (plan item LP2). The section on per-atom targets above holds the targets of that shape (`cmfqsar/atomtargets.py`); `cmfqsar/localenv.py` builds the samples that go with them.

**The sample is the field under an envelope.** The environment of atom *a* of molecule *A* is the molecule's own field, for every field type *f*, multiplied by a Gaussian envelope centred on that atom:

$$X_{fa}(\mathbf{r}) = X_{fA}(\mathbf{r}) \exp\left(-\rho\,|\mathbf{r}-\mathbf{r}_a|^2\right)$$

Here $\mathbf{r}_a$ is the position of the atom and ρ sets the width of the envelope, in $Å^{-2}$. An atom near the centre keeps its weight and one far away loses it. A CMF field is a sum of atom-centred Gaussians and the envelope keeps that form, which is the point: an environment **is** a field of the same class as a molecule's, so the kernels, the regression, the variance and the atom decomposition all apply to it unchanged. The centre is the first atom of the environment, and `member_atom_map` records which atom of the molecule each atom of the environment came from, so a per-atom quantity can be put back where it belongs.

**ρ says how far the model looks**, and it is a hyper-parameter like α (the option is `rho`): the envelope falls to 1/*e* of its height at a distance 1/√(ρ) from the centre, which `envelope_width` reports. The default ρ = 0.5 $Å^{-2}$ is a width of 1.4 Å, which is a first-shell view; ρ = 0.1 $Å^{-2}$ is 3.2 Å and takes in most of a small molecule. It has to be chosen for the property, and the measurement below shows that it decides everything. Atoms whose envelope factor falls below 1e-4 are left out, which costs nothing measurable and saves carrying them; `kept_share` says what share of the enveloped weight that dropped.

**The kernel needs no alignment.** The kernel to use on an environment first is the rotation-invariant field autocorrelation of the sections above, taken from the centre instead of over a whole molecule: two environments of two different molecules need not be superimposed for their autocorrelations to be compared. That matters here more than anywhere else, because a common frame for carbons of unrelated molecules does not exist. On an environment the autocorrelation becomes a double sum over atom pairs in which each atom carries its envelope factor, and the distance axis is shorter -- `LOCAL_SPEC` runs to 8 Å in steps of 0.5 Å, because the envelope has already removed everything beyond a few Å.

**An acidity is the field of a change.** A pKa is the free energy of losing a proton, so the sample for it is not the acid but the **difference field** of the acid and its anion, which `difference_field` builds: a kernel on CMF fields is bilinear, so a difference of two fields is again a field, whose atoms are the atoms of both molecules with the second one's weights negated. That is the signed counterpart of the ensemble builder of the protonation section, which averages members and therefore needs non-negative weights. The two molecules have to be in one frame for the difference to mean anything, which is why the anion is built on the geometry of the acid.

**The reference is an atom-centred count.** What such a model has to beat is the simplest thing that could work: the type of the centre itself plus the counts of atom types in distance shells around it, which is what a per-atom increment scheme is. `count_features` builds it from the same molecules, the same samples and the same folds, so the only difference between the two numbers is the descriptor.

**The folds hold out whole molecules.** `AtomTargets.folds` splits molecules, never atoms. Two carbons of one molecule, one in the training set and one in the test set, would let the model see the molecule it is being tested on. For the same reason the bootstrap interval resamples molecules, with every measured atom of a drawn molecule going in with it.

```
cmf atoms examples/local/shifts_cdcl3.sdf --list        # what could be a target here
cmf local examples/local/shifts_cdcl3.sdf --target shift --reference
cmf local examples/local/carboxylic_pka.sdf --target pka --rho 0.1 0.5 --reference \
    --out with_predictions.sdf
```

`cmf local` fits the local kernel on the per-atom target, searches the ridge on the folds that hold out whole molecules, and with `--reference` fits the count model on the same folds and bootstraps the difference over molecules. `--out` writes an SDF carrying the measured value and the out-of-fold prediction on the atom each belongs to.

## What it is worth, measured

`benchmarks/run_local.py` measures the two properties the plan names, against the count reference on the same folds. Both sets are prepared by `examples/make_local_examples.py` from free downloads and are in `examples/local/`, with their origin, their licences and every filter in `examples/local/README.md`. The last column but one is the mean kernel value between two different samples: a cosine-normalised kernel of 1 says the descriptor cannot tell two samples apart, whatever the regression does with it afterwards.

**13C shifts in chloroform, 5096 assigned carbons in 400 molecules.**

| descriptor | envelope width rho | q2 | error (ppm) | mean kernel between two samples | difference in q2 against the counts |
|---|---|---|---|---|---|
| atom-centred counts (18 atom types in 6 shells) | -- | +0.853 | 20.464 | 0.566 | -- |
| local field | 0.25 | +0.902 | 16.731 | 0.462 | +0.049 [+0.030, +0.067] |
| local field | 0.5 | +0.962 | 10.404 | 0.486 | +0.109 [+0.097, +0.123] |

| descriptor | envelope width rho | q2 | error (ppm) | mean kernel between two samples | difference in q2 against the counts |
|---|---|---|---|---|---|
| local field | 1 | +0.973 | 8.824 | 0.537 | +0.120 [+0.107, +0.133] |
| local field | 2 | +0.969 | 9.460 | 0.562 | +0.116 [+0.103, +0.130] |

**The field around a carbon predicts its shift better than a count around it does, and the envelope width decides how much.** At a width of 1.0 Å the error is 8.8 ppm against 20.5 ppm for the counts, a difference in q2 of +0.120 [+0.107, +0.133], which is far outside the interval. At the module's default width of 1.4 Å, which was fixed before either set here was measured, it is +0.109 [+0.097, +0.123]. A wider envelope is worse (+0.902 at a width of 2.0 Å) and so is a narrower one (+0.969 at 0.7 Å), so the best view of a carbon is about one angstrom across, which is what a shift should ask for: it is a property of the atom, its bonds and the next shell.

What these numbers are **not** is a shift predictor. Programs built for that task reach one to three ppm on 13C by using tens of thousands of spectra and often a calculated geometry; this model has 400 molecules and one force-field conformation each. The question the item asks is narrower, and the measurement answers it.

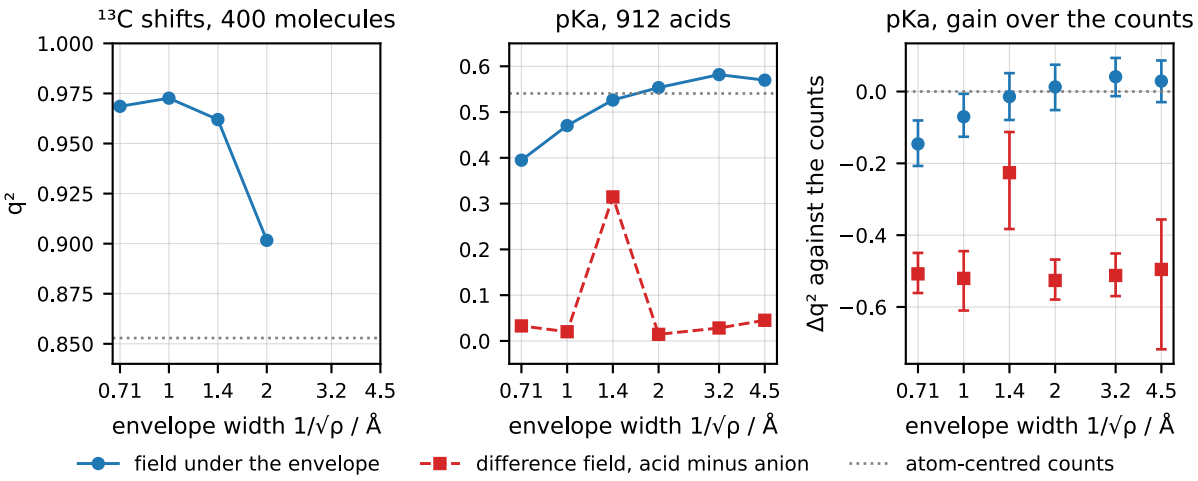
**Carboxylic acid pKa, 912 acids with the value on the carboxyl carbon.**

| descriptor | envelope width rho | q2 | error (pKa units) | mean kernel between two samples | difference in q2 against the counts |
|---|---|---|---|---|---|
| atom-centred counts (20 atom types in 6 shells) | -- | +0.541 | 0.563 | 0.577 | -- |
| local field | 0.05 | +0.570 | 0.545 | 0.716 | +0.029 [-0.030, +0.086] |
| local difference field | 0.05 | +0.045 | 0.811 | 1.000 | -0.495 [-0.718, -0.356] |
| local field | 0.1 | +0.582 | 0.537 | 0.775 | +0.041 [-0.013, +0.094] |
| local difference field | 0.1 | +0.028 | 0.818 | 1.000 | -0.512 [-0.570, -0.451] |
| local field | 0.25 | +0.553 | 0.555 | 0.877 | +0.013 [-0.052, +0.075] |
| local difference field | 0.25 | +0.014 | 0.824 | 1.000 | -0.526 [-0.579, -0.468] |

| descriptor | envelope width rho | q2 | error (pKa units) | mean kernel between two samples | difference in q2 against the counts |
|---|---|---|---|---|---|
| local field | 0.5 | +0.526 | 0.571 | 0.957 | -0.014 [-0.079, +0.051] |
| local difference field | 0.5 | +0.315 | 0.687 | 1.000 | -0.226 [-0.383, -0.113] |
| local field | 1 | +0.470 | 0.604 | 0.990 | -0.070 [-0.126, -0.006] |
| local difference field | 1 | +0.020 | 0.822 | 0.999 | -0.520 [-0.610, -0.445] |
| local field | 2 | +0.395 | 0.646 | 0.997 | -0.146 [-0.207, -0.081] |
| local difference field | 2 | +0.033 | 0.816 | 0.998 | -0.508 [-0.561, -0.449] |

**Here the local kernel is no better than the count, and the difference field, which the plan proposed as the sample, is far worse.** The best of the plain environments is a width of 3.2 Å, +0.582 against +0.541 for the counts, a difference of +0.041 [-0.013, +0.094] -- inside the interval, so by rule 4 it is not a result. The direction the plan expected is right, though: an acidity wants a **wider** envelope than a shift, and a tight one fails: +0.526 at a width of 1.4 Å and +0.395 at 0.7 Å, both below the count.

The mean-kernel column says why, and it is the useful part of this measurement. For the difference field it is 0.998 or more at every width, and 0.9999 at the three widest: **every carboxylic acid loses its proton in nearly the same way**, so the difference of the two fields is nearly the same pattern for all 912 of them, and after the cosine normalisation the samples are indistinguishable. The free energy of deprotonation is indeed a response to that change, but what distinguishes one acid from another is not in the change itself -- it is in what surrounds it, which is exactly what the difference removes. The same column explains the plain environments: at a width of 1.4 Å the mean kernel is already 0.957, because the first shell of a carboxyl group is the same group in every molecule, and the model only starts to see the substituent as the envelope widens.




*q² against the envelope width 1/√ρ for the two per-atom targets, with the atom-centred counts as the dotted line. Left: a ¹³C shift wants a narrow envelope of about 1 Å. Middle: an acidity wants a wide one, and the difference field fails at every width. Right: the gain of the pKa models over the counts with its 95 % bootstrap interval; only an interval that excludes zero is a result.*

**The conformations the molecule visits.** A measured shift is an average over the conformations a molecule visits, and so is an acidity; the single geometry that distance geometry returns is a stand-in for that average, and both measurements above named it as their weakest part. Plan item LP4 is the test of what it costs. It needs no new kernel: an environment is a field, so the mean of the per-conformation environments is a field too and the kernel on it is the mean-map kernel (`localenv.ensemble_environment`). What it cannot be is the environment of the mean field of the molecule: the atom sits in a different place in every conformation, so each conformation has its own environment, centred on its own copy of the atom, and it is those that are averaged. With a frame the same order applies -- pose each conformation, then average (`localalign.posed_ensemble`) -- because the site moves with the conformation.

`benchmarks/run_local.py` asks for 5 conformations per molecule from the distance-geometry generator with a force-field refinement, keeps the prepared geometry as the first of them, and measures four samples: that geometry alone, the lowest in energy of the pool alone, and the ensemble weighed equally and by the force field's energies.

**On the shifts, averaging costs, and the control says it is the averaging.**

| conformations | q2 | error (ppm) | atoms per sample | difference in q2 against the prepared one |
|---|---|---|---|---|
| one, as prepared | +0.973 | 8.824 | 13 | -- |
| one, lowest in energy | +0.972 | 8.880 | 13 | -0.000 [-0.001, +0.000] |
| the ensemble, weighed equally | +0.952 | 11.632 | 71 | -0.020 [-0.024, -0.017] |
| the ensemble, weighed by energy | +0.956 | 11.149 | 71 | -0.016 [-0.019, -0.014] |

The prepared geometry gives q2 +0.973 and 8.82 ppm. Averaging over the ensemble gives 11.63 ppm weighed equally and 11.15 ppm weighed by energy, which is -0.020 [-0.024, -0.017] and -0.016 [-0.019, -0.014] in q2 -- both outside their intervals, so both are real losses. The second row

is the control: a different single conformation, the lowest in energy of the same pool, which differs from the prepared one for 285 of the 400 molecules and costs -0.000 [-0.001, +0.000] -- nothing. So the loss is not that the generated geometries are worse than the prepared one. It is the averaging.

The reading is that the mean field smears every atom over the positions it takes in the conformations. At an envelope width of 1.0 Å what is left inside the envelope is the first shells, and those are the same in every conformation: the averaging adds blur and no information. The atom count per sample says the same thing from the other side -- 13 atoms become 71 and the model gets worse.

**On the acidities, it neither helps nor costs.**

| conformations | q2 | error (pKa units) | atoms per sample | difference in q2 against the prepared one |
|---|---|---|---|---|
| one, as prepared | +0.582 | 0.537 | 25 | -- |
| one, lowest in energy | +0.576 | 0.540 | 25 | -0.005 [-0.043, +0.035] |
| the ensemble, weighed equally | +0.562 | 0.550 | 127 | -0.020 [-0.061, +0.025] |
| the ensemble, weighed by energy | +0.549 | 0.557 | 127 | -0.032 [-0.070, +0.013] |

The prepared geometry gives q2 +0.582, the ensemble +0.562 weighed equally and +0.549 weighed by energy, which is -0.020 [-0.061, +0.025] and -0.032 [-0.070, +0.013]. The control costs -0.005 [-0.043, +0.035]. Both ensemble intervals include zero, so by rule 4 this is not a measurable loss; what it is not is a gain. The envelope here is wider, 3.2 Å, and does reach the parts of a molecule that move between conformations, which is the case where averaging had the best chance of paying. It did not take it.

```
cmf local examples/local/shifts_cdcl3.sdf --target shift --conformers 5 \
    --conformer-weighting boltzmann
```

The option is there because the question is a fair one to ask of another set, not because it helps on these two.

**Many atoms at once.** A per-atom target is a large problem where a per-molecule one is a small one: the 400 molecules in `examples/local/` are 5096 carbons, and the free NMR databases hold hundreds of thousands. An exact kernel over samples costs the square of their number in memory and the cube in time, so plan item LP5 asks for a Nyström or inducing-point approximation. For this kernel none is needed, and that is the first thing `benchmarks/run_local_scale.py` measures.

The rotation-invariant local kernel is a **linear kernel on a finite feature map**. The field autocorrelation of an environment is a fixed number of channels -- 255 for the five physico-chemical fields -- scaled per block and cosine-normalised, and the kernel is the inner product of those rows. So the ridge regression can be solved in the space of the 255 coefficients instead of the space of the samples (`cmfqsar/locallarge.py`), which is the same model and not an approximation of it:

| route | q2 | error (ppm) | memory | the folds |
|---|---|---|---|---|
| dual, an n by n kernel | +0.972645 | 8.8240 | 0.19 GB | 13 s |

| route | q2 | error (ppm) | memory | the folds |
|---|---|---|---|---|
| primal, n by d rows | +0.972645 | 8.8240 | 10 MB | 0.1 s |

The predictions of the two differ by 2e-06 ppm, which is round-off. What differs is the cost, and it is the memory that decides: the kernel grows with the square of the samples and the rows grow with the samples. At twenty times these 5096 carbons the kernel would be 77 GB and the rows 198 MB.

**More data does not help.** With the primal route a much larger set becomes easy, so the question the item really asks can be put: is the error of the sections above the method's or the data set's? The set is 58705 carbons in 6000 molecules, built from the whole nmrshiftdb2 dump rather than from its chloroform spectra alone, so it is noisier by about a ppm and far larger. An exact kernel over it would be 25.7 GB; the primal route cross-validates it in 7 seconds. Each size is a subset of the next.

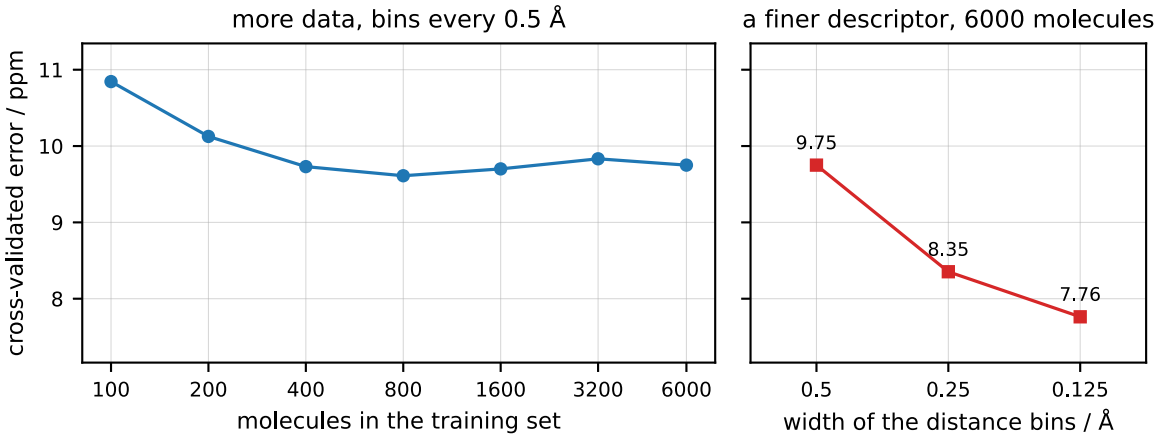
| molecules | carbons | q2 | error (ppm) | the folds |
|---|---|---|---|---|
| 100 | 1105 | +0.953 | 10.84 | 0.2 s |
| 200 | 2131 | +0.961 | 10.13 | 0.3 s |
| 400 | 4113 | +0.964 | 9.73 | 0.4 s |
| 800 | 8079 | +0.965 | 9.61 | 1.0 s |
| 1600 | 15780 | +0.965 | 9.70 | 1.7 s |
| 3200 | 31221 | +0.964 | 9.83 | 3.2 s |
| 6000 | 58705 | +0.965 | 9.75 | 6.8 s |

From 100 molecules to 6000 the error goes from 10.84 to 9.75 ppm, and from eight hundred molecules onwards it does not move at all. Sixty times the data is worth 1.23 ppm, nearly all of it spent before the first thousand molecules.

**A finer descriptor helps more than sixty times the data.** A flat curve says the error is not the data's. What is left is the descriptor, and the cheapest test of that is a finer one on the same samples: the distance axis of the autocorrelation at 0.5 A per bin against two finer spacings, everything else unchanged.

| distance bins | channels | q2 | error (ppm) |
|---|---|---|---|
| every 0.5 A | 255 | +0.965 | 9.75 |
| every 0.25 A | 495 | +0.974 | 8.35 |
| every 0.125 A | 975 | +0.978 | 7.76 |

Halving the bin width gains 1.40 ppm and halving again gains 1.99 ppm in all -- against the 1.23 ppm that sixty times the data gave. **The error of this model is set by what the descriptor can tell apart, not by how many carbons it has seen.** The route to the shift predictors this item names runs through a richer descriptor, which is plan item C5, rather than through a larger training set. What the scale work gives is that the question could be asked at all: the whole curve above, every fold and every ridge, is a few minutes of arithmetic.




*The learning curve of the local shift model on the 6000-molecule nmrshiftdb2 set, cross-validated error against the number of training molecules, each size a subset of the next, and the same error at the full size when the distance bins of the descriptor are halved twice. Sixty times the data gains about one ppm and stops gaining after eight hundred molecules; two halvings of the bin width gain two. Drawn from benchmarks/results/local_scale/curve.json.*

**The kernel that does need landmarks.** The posed overlap kernel of the next section has no finite feature map -- its features are the field itself, a function on space -- so there the dual form is the only exact one and the Nyström approximation is the route the plan names. `locallarge.nystrom_kernel` builds it from landmark environments, and on the 1225 aromatic carbons of that section it works very well: 200 landmarks of 1225 samples, 16 per cent of the rank, reproduce the exact q2 of +0.899, and 100 already give a relative kernel error of 0.001, while 25 give 0.072. The environments of one site are alike, which is why so few span them.

| landmarks | relative error of the kernel | q2 | error (ppm) |
|---|---|---|---|
| all 1225 (exact) | 0.000 | +0.899 | 4.217 |
| 25 | 0.072 | +0.801 | 5.908 |
| 50 | 0.031 | +0.859 | 4.970 |
| 100 | 0.001 | +0.883 | 4.539 |
| 200 | 0.000 | +0.899 | 4.203 |
| 400 | 0.000 | +0.898 | 4.231 |
| 800 | 0.000 | +0.899 | 4.217 |

## 11.8 A co-field around one atom: the local alignment

The local kernel of the previous section needs no frame, and that is also its limit: with no frame there is no co-field, so it cannot say **where** around the atom, and in which field, a feature raises or lowers the property. For one site a frame costs nothing, and plan item LP3 is that frame (`cmfqsar/localalign.py`).

**The rule.** The centre is already the origin and the first atom of the environment, so what is left is the rotation, and two of the centre's bonded neighbours fix it: the first gives the x axis, the second the xy plane. Which two is decided by the **atom type and nothing else**. A bond length would make a poor rule -- the two ring bonds of an aromatic carbon differ by hundredths of an angstrom, and by which of them is longer in one conformation -- so neighbours of the same type are treated

as interchangeable. When that leaves the choice open, as it does for the two ring neighbours of an aromatic carbon or the three hydrogens of a methyl group, **every pose is kept and averaged**: the environment becomes the mean field of its poses, which is a mean-map kernel over the frames the site allows. That is the device the symmetrised Hammett set uses for the four images of a monosubstituted benzene, applied to one atom instead of one molecule.

**The ring site.** `ring_frames` is the site the plan names: an aromatic carbon of a six-membered ring, with the bond that leaves the ring on +x, the ring in the xy plane, the ring normal on z, and the two mirror images kept. In that frame the positions a substituent occupies on the ring are fixed numbers, which `RING_POSITIONS` holds: ipso at 1.4 A on +x, then ortho at 2.4 A, meta at 3.7 A and para at 4.2 A from the centre. A carbon where two rings meet has three aromatic neighbours and no such frame, and the rule refuses it rather than posing it anyway.

**Then everything applies.** The posed environments are in one frame, so the ordinary overlap kernel of this program works on them exactly as it works on molecules, and so does the co-field: `overlap_kernel` builds the `AtomTable` and the kernel, and `CMFModel.cofield_coefficients` gives the co-field around the atom. `cofields.cofield_at_points` reads it at places a chemist can name, `cofield_covariance_at` says how far the noise in the measured values moves it, and `contrast_at` compares two places with the right error bar: the co-field at two positions a few angstrom apart is built from the same training fields, so the two values are correlated and the error of their difference is not the two error bars added.

```
        cmf local examples/local/shifts_cdcl3.sdf --target shift --align --site ring --rho 0.25 \
            --cofield cofield.cube
```

`--align` replaces the rotation-invariant kernel by the posed overlap kernel, `--site ring` uses the ring rule and leaves out every atom it does not fit, and `--cofield` writes the co-field around the centre as a cube file in the frame of the pose, which any molecular viewer draws.

## What it is worth, measured

`benchmarks/run_local_cofield.py` measures two things on the aromatic carbons of the shift set: 1225 carbons of six-membered aromatic rings in 170 molecules, with folds that hold out whole molecules, and the atom-centred count of the previous section as the reference.

**The frame gains where the envelope is wide, and nothing where it is narrow.**

| descriptor | envelope width rho | width (A) | q2 | error (ppm) | ridge | difference in q2 against the counts |
|---|---|---|---|---|---|---|
| atom-centred counts (18 atom types in 6 shells) | -- | -- | +0.768 | 6.383 | 0.01 | -- |
| posed overlap | 0.05 | 4.5 | +0.872 | 4.745 | 0.1 | +0.104 [+0.070, +0.144] |
| invariant local | 0.05 | 4.5 | +0.116 | 12.461 | 0.1 | -0.652 [-0.695, -0.594] |

| descriptor | envelope width rho | width (A) | q2 | error (ppm) | ridge | difference in q2 against the counts |
|---|---|---|---|---|---|---|
| posed overlap | 0.1 | 3.2 | +0.887 | 4.448 | 0.01 | +0.119 [+0.086, +0.158] |
| invariant local | 0.1 | 3.2 | +0.366 | 10.554 | 0.1 | -0.402 [-0.457, -0.344] |
| posed overlap | 0.25 | 2.0 | +0.899 | 4.217 | 0.01 | +0.131 [+0.098, +0.170] |
| invariant local | 0.25 | 2.0 | +0.803 | 5.882 | 0.001 | +0.035 [+0.003, +0.068] |
| posed overlap | 0.5 | 1.4 | +0.889 | 4.425 | 0.0001 | +0.121 [+0.083, +0.162] |
| invariant local | 0.5 | 1.4 | +0.847 | 5.187 | 0.01 | +0.079 [+0.048, +0.116] |
| posed overlap | 1 | 1.0 | +0.861 | 4.935 | 1e-06 | +0.093 [+0.046, +0.143] |
| invariant local | 1 | 1.0 | +0.836 | 5.361 | 0.01 | +0.068 [+0.036, +0.105] |

At the same envelope width the posed kernel is ahead of the invariant one by +0.756 [+0.718, +0.791] at a width of 4.5 A; +0.522 [+0.477, +0.572] at a width of 3.2 A; +0.096 [+0.069, +0.124] at a width of 2.0 A; +0.042 [+0.022, +0.061] at a width of 1.4 A; +0.025 [-0.015, +0.060] at a width of 1.0 A. The last of those is inside its interval, so at a width of 1 A the frame adds nothing that can be measured. Each at its own best width -- posed at rho 0.25 (+0.899) against invariant at rho 0.5 (+0.847) -- the difference is +0.052 [+0.033, +0.071].

That is what a frame is and is not for. An invariant descriptor of a wide environment has only distances, and at a width of 4.5 A the surroundings of an aromatic carbon are nearly the same distances in every molecule; the pose keeps the directions, so the ortho position and the meta position are two places rather than two atoms at similar distances. Squeeze the envelope down to 1 A and there is little left but the first shell, which distances alone describe perfectly well, and the frame adds nothing that can be measured.

**Where the co-field sits, and what it does not say.**

| field | ipso | ortho | meta | para | \ | para\ | - \ | meta\ | |
|---|---|---|---|---|---|---|---|---|---|
| electrostatic | -16.7 +- 20.3 | +33.0 +- 15.4 | -28.5 +- 15.0 | -10.0 +- 20.0 | -18.5 +- 24.7 | | | | |
| steric | +15.5 +- 1.3 | -2.6 +- 1.0 | -2.7 +- 0.9 | -4.2 +- 1.2 | +1.5 +- 1.5 | | | | |

| field | ipso | ortho | meta | para | \ | para\ | - \ | meta\ | |
|---|---|---|---|---|---|---|---|---|---|
| hydrophobic | +1.4 +- 4.5 | -2.0 +- 3.4 | +6.6 +- 3.3 | +2.2 +- 4.7 | -4.4 +- 5.9 | | | | |
| hba | -57.4 +- 21.4 | +17.8 +- 15.0 | +29.6 +- 13.4 | -41.1 +- 18.8 | +11.5 +- 25.2 | | | | |
| hbd | +7.1 +- 33.6 | +58.4 +- 25.4 | -0.2 +- 23.7 | +62.6 +- 30.4 | +62.4 +- 39.2 | | | | |

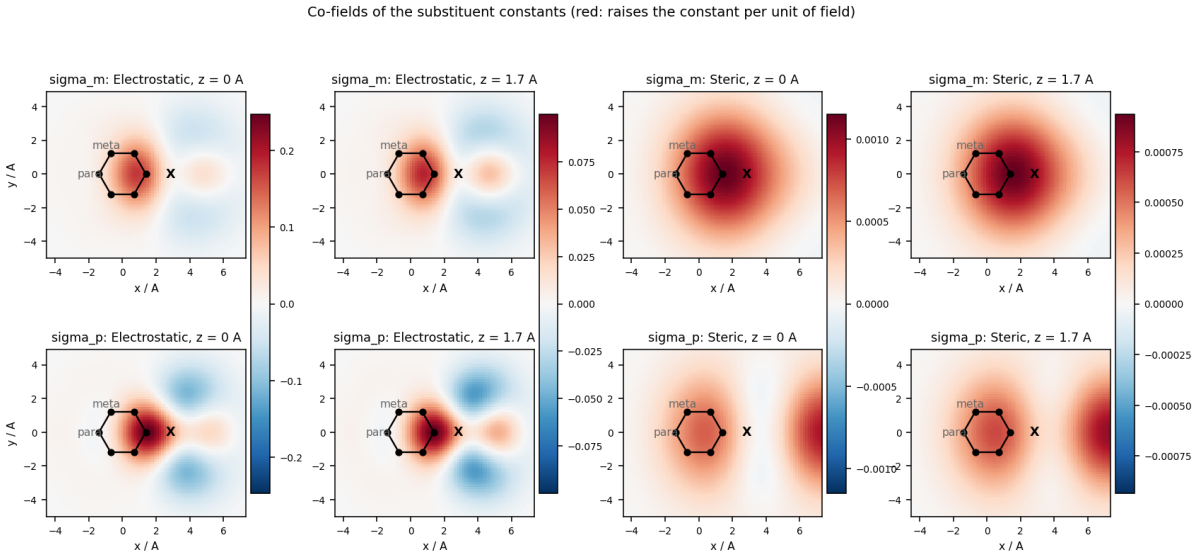
**The co-field says the large thing and not the subtle one.** The numbers above are the co-field at the four positions with the error the noise in the shifts allows, and one feature stands far above everything else: the **steric field at the ipso position**, +15.5 +- 1.3, 12 standard deviations from zero where nothing else reaches four. It says that putting anything bulkier than hydrogen on the measured carbon raises its shift, which is the dominant term of the 13C substituent effect and true of nearly every group: methyl +9 ppm, chlorine +6, nitro +20, methoxy +31.

**What it does not reproduce is the alternation around the ring, and that is the half of this item that fails.** The question the plan asks is whether the co-field puts more at the para position than at the meta one. The para position is further from the measured carbon (4.2 A) than the meta one (3.7 A), so the envelope and the overlap both favour meta, and a co-field that still came out larger at para would be saying what the geometry does not. None does. For the electrostatic field the difference is -18.5 +- 24.7, the wrong sign and inside the error. The closest any field comes is the hydrogen-bond-donor one, +62.4 +- 39.2 (z +1.6), which is the right sign and the right field -- a hydroxyl or an amino group is where the ortho and para effects on an aromatic shift are largest -- and still short of two standard deviations.

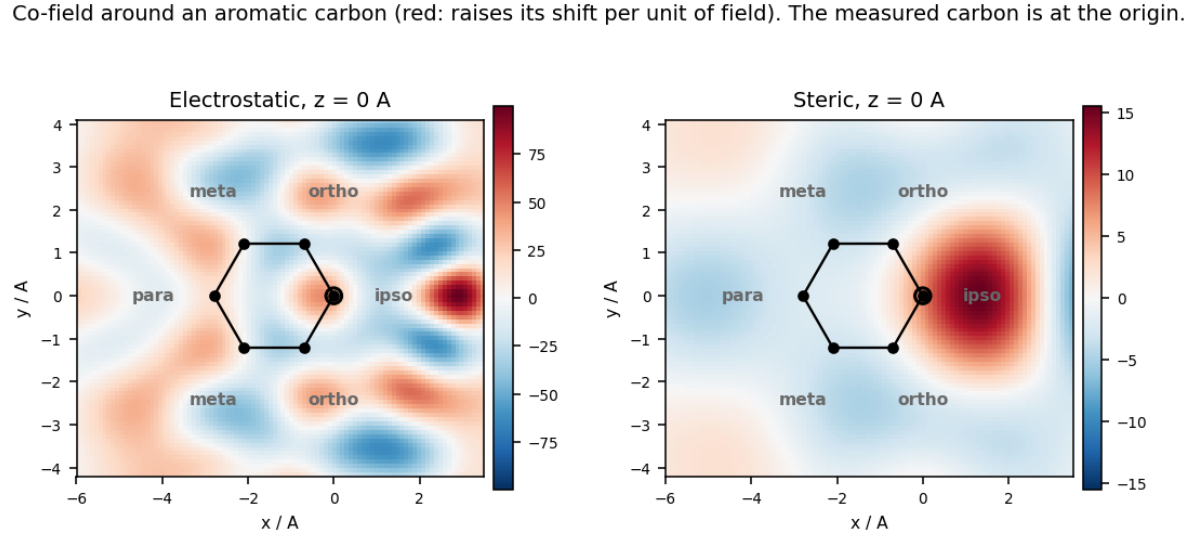
Read the five co-fields together rather than one at a time. The fields overlap in what they describe -- an oxygen is a heavy atom, an acceptor and a negative charge at once -- so a co-field per field is a decomposition of one effect over five bases, not five independent statements, and the hydrogen-bond-acceptor co-field correcting the steric one at the ipso position is that decomposition at work rather than a claim about acceptors.

The signs are the more interesting half, and they are not nothing. The electrostatic co-field comes out -16.7 at ipso, +33.0 at ortho, -28.5 at meta and -10.0 at para, and a negative value means that negative charge at that place raises the shift of the carbon at the origin. A methoxy group raises the shift of the carbon it sits on by 31 ppm and lowers the ortho and para carbons by 14 and 8 ppm, and its oxygen is negatively charged, so a co-field that had learned that effect would be negative at ipso and positive at ortho and para. This one agrees at 3 of the four positions, with a correlation of +0.55 over them, but the y-scramble null for that correlation is wide (z +1.2): a correlation over four points cannot be sharp, and this one is not.

There is a reason to expect no better from these fields. A nitro group, whose nitrogen is **positively** charged, also raises the ipso carbon, by 20 ppm, and raises the para carbon where methoxy lowers it. No model linear in fragment charges can follow both groups, because what separates them is resonance and fragment charges do not carry it. That is the same wall the Hammett co-fields of this program met for sigma_p, and the remedy there was quantum charges computed on the whole molecule, which carry the polarisation of the ring by the substituent (`examples/orca_atom_properties.py` and the `prop:` fields). The shift set here has no such charges. The other way round the wall is to stop asking the charge and ask the atom type, which is what the next part does and what settles the question.




*The co-fields of the Hammett constants $\sigma_m$ and $\sigma_p$, from the Hammett benchmark: the electrostatic and the steric co-field in the ring plane and 1.7 Å above it, with the ring and the substituent position X marked. Red raises the constant per unit of field. The electrostatic map of $\sigma_p$ has lobes of both signs around the substituent position, a pattern that fragment charges cannot deliver for a nitro group and a methoxy group at once. Drawn by* `benchmarks/run_hammett.py`.




*The electrostatic and the steric co-field of the shift model around the measured carbon, at the origin, in the ring plane, with the ring and its four positions marked. Red raises the shift of that carbon per unit of field. The steric lobe at the ipso position is the feature twelve standard deviations from zero; the electrostatic map alternates in sign around the ring. Drawn by* `benchmarks/run_local_cofield.py`.

**The same co-field, read from the indicator fields.** A partial charge cannot tell a pi-donor from a pi-acceptor, which is why the electrostatic co-field above could not reproduce the ring picture. A continuous indicator field carries the atom **type** instead: there is one field per Sybyl type, so an ether oxygen at the ortho position and a carbonyl carbon at the ortho position are different features with co-fields of their own. That is how a table of substituent effects is written, and the second part of `benchmarks/run_local_cofield.py` asks whether the model writes the same table. It uses 17 types beside the five physico-chemical fields, at an envelope width of 4.5 A -- wider than the co-field above, because an envelope that damps the para position to nothing cannot be asked what sits there -- and predicts about as well (q2 +0.882, 4.55 ppm out of fold).

Two things make this a comparison of numbers rather than of pictures. First, `cofields.atom_effect` turns a co-field into **what one atom placed at a point would add to the prediction**, in ppm: the co-field is a sum of Gaussians and so is the atom, the overlap of two Gaussians is the closed-form integral the kernel itself sums over atom pairs, and the test suite checks the result against what the model's own prediction does when that atom is added. Second, the atom enters with the weight the envelope gives it at that distance (ipso 0.91, ortho 0.75, meta 0.50, para 0.41), so the number is what a real substituent at that ring position contributes rather than what an undamped one would.

| type | ipso | ortho | meta | para | measured, for the group named |
|---|---|---|---|---|---|
| H | -1.6 +- 0.6 (n 774) | +0.2 +- 0.4 (n 1483) | +0.5 +- 0.3 (n 1448) | -0.1 +- 0.3 (n 733) | -- |
| O.3 | +5.4 +- 1.6 (n 170) | -6.4 +- 1.1 (n 342) | -1.3 +- 0.8 (n 334) | -1.5 +- 0.9 (n 157) | +31.4, -14.4, +1.0, -7.7 (methoxy (anisole)) |
| C.3 | +13.6 +- 0.8 (n 126) | +0.5 +- 0.5 (n 242) | -1.1 +- 0.4 (n 239) | -0.5 +- 0.5 (n 120) | +9.3, +0.7, -0.1, -2.9 (methyl (toluene)) |
| C.2 | +9.0 +- 1.1 (n 87) | -0.2 +- 0.7 (n 172) | -0.7 +- 0.5 (n 168) | +0.7 +- 0.7 (n 84) | +9.3, +0.2, +0.2, +4.2 (acetyl (aceto phenone)) |
| C.ar | +1.0 +- 0.2 (n 0) | -0.3 +- 0.2 (n 68) | -0.1 +- 0.1 (n 108) | -0.1 +- 0.2 (n 56) | -- |
| N.pl3 | +3.1 +- 2.0 (n 25) | -9.1 +- 1.4 (n 50) | -2.2 +- 1.1 (n 50) | -5.9 +- 1.4 (n 25) | +18.0, -13.3, +0.9, -9.8 (amino (aniline)) |
| Cl | +7.4 +- 2.2 (n 16) | +0.1 +- 1.5 (n 32) | +1.7 +- 1.3 (n 32) | -2.4 +- 1.7 (n 16) | +6.2, +0.4, +1.3, -1.9 (chlorine (chl orobenzene)) |

**The pattern comes out right.** Over the 15 (type, ring position) pairs that have a measured value -- the ipso position left out, because it is the largest effect in every row and would carry the comparison by itself -- the correlation with the measured table is +0.89. Including the ipso position it is +0.72. The sharpest of these is the question the electrostatic field could not answer at all: a pi-donor lowers the para carbon and a pi-acceptor raises it, although the first atom of the donor is negatively charged and that of the acceptor is positive. Here the amino nitrogen gives -5.9 +- 1.4 ppm at the para position, 4 standard deviations from zero, against -9.8 measured, while the sp2 carbon gives +0.7 +- 0.7 against +4.2 measured. The signs are opposite, and the ether oxygen sits on the donor side as well (-1.5 +- 0.9). That is the resonance distinction no model linear in fragment charges can make.

**The sizes are compressed.** A straight line through those 15 pairs has a slope of 0.46, so the model reproduces the shape of the table and about 46 per cent of its size. The root mean square difference from the measured row, per type, is Cl 0.7, C.2 1.8, C.3 2.5, N.pl3 8.1, O.3 14.0 ppm: the

halogen and the two carbons come within a few ppm and the two strong donors are the ones held back. Three things pull that way: the ridge, which shrinks every coefficient; the envelope, which lets a para atom in at 0.41 of itself; and the set, where most rings carry several substituents whose effects the model shares out, while the table is measured on rings carrying one. The type labels are coarser than the table's groups too -- O.3 is every ether and hydroxyl oxygen, not only a methoxy, and C.2 is every sp2 carbon, not only an acetyl -- so the measured column is the nearest published row rather than the same quantity.

The same field set is a flag on the command, so the table above can be read off any per-atom target without the benchmark:

```
cmf local examples/local/shifts_cdcl3.sdf --target shift --align --site ring \
    --rho 0.05 --cif --cofield cofield.cube
```

`benchmarks/results/local_cofield/indicator.png` draws the two co-fields in the ring plane side by side. Both are positive where a substituent sits on the measured carbon and negative at the ortho positions, which is the large part of the effect; what separates the donor from the acceptor is the weaker pattern further out, and that is read off at the points in the table rather than by eye.

**The other site the plan names, the carboxyl carbon, is not rescued by the frame either.** The previous section found no signal on the 912 carboxylic acids: the invariant local kernel was no better than an atom-centred count there, and the difference field of the acid and its anion was far worse. A carboxyl carbon has two oxygens and a carbon of its own, which the atom types separate, so its pose is unique and nothing has to be averaged -- and with it the model loses to the count, -0.101 [-0.164, -0.037], and differs from the invariant kernel by -0.142 [-0.198, -0.088]. The frame was not what was missing. Only the model is measured there: a co-field read off a model that does not predict would be an interpretation of noise, which is why the pKa co-field of the substituted benzoic acids that this item's gate asks for is not drawn.

**What is not done.** The plan asks for a panel in the application that shows the environment with its co-field instead of the molecule. There is none: the co-field comes out of `cmf local --cofield` as a cube file, which a viewer draws, and the application still knows only models whose sample is a molecule.

## 11.9 Which atom is attacked: a site of metabolism

Blocking a metabolic position is one of the standard moves in medicinal chemistry, and the question it asks is about one atom: of the twenty-odd heavy atoms of a drug, which one does a cytochrome P450 attack? That is the shape of target the two previous sections built the machinery for, and plan item MC8 is the measurement of it on drug-like molecules rather than on the small ones those sections use.

**The data set.** `examples/local/metabolism.sdf` holds 636 molecules, 1301 sites among 14 488 heavy atoms, from the benchmark of Zaretzki, Matlock and Swamidass (2013) in the form the FAME.AL project publishes under the MIT licence. The target is one number per heavy atom: one if the atom is a site, zero if it is not. The sites are merged over the P450 isoforms, so the question is the general one -- where is this molecule attacked -- and not which enzyme attacks it. `examples/local/README.md` gives the selection and what was changed.

**The measure is a ranking inside a molecule.** A $q^2$ on a target of zeros and ones is not what a chemist needs. The question is which atom, so the measure is the one the site-of-metabolism literature uses: **in what share of the molecules is a true site among the one, or the two, atoms the model ranks highest**. The ranking happens inside each molecule, so the scale of one

molecule's predictions cannot help or hurt another, and a molecule with no site asks no question. `localenv.top_hits` and `localenv.top_k_accuracy` are that measure.

**The baseline the item names** is a model with no geometry: the same atom types, counted over **bonds** instead of over distance. `localenv.topological_counts` builds it -- the type of the centre, and the types one, two, three and four bonds away -- which is what an atom's topological fingerprint holds. It is the counterpart of `shell_counts`, and the two differ in exactly the thing a three-dimensional model is supposed to add.

`benchmarks/run_metabolism.py` measures all of it on ten folds that hold out whole molecules, with the ridge searched the same way for every row:

| descriptor | the site is first | among the first two | area under the curve | against the count over bonds |
|---|---|---|---|---|
| count over bonds (2D) | 53.6% | 70.9% | 0.859 | -- |
| count over distance (3D) | 57.5% | 73.9% | 0.864 | +3.0% [+0.3%, +5.5%] |
| local field, rho 0.15 | 54.4% | 70.1% | 0.843 | -0.8% [-4.2%, +2.7%] |
| local field, rho 0.25 | 55.0% | 71.1% | 0.853 | +0.2% [-3.3%, +3.8%] |
| local field, rho 0.5 | 55.2% | 69.7% | 0.854 | -1.3% [-4.9%, +2.5%] |
| local field, rho 1 | 55.3% | 69.3% | 0.844 | -1.6% [-5.2%, +2.2%] |
| local field, rho 2 | 53.0% | 66.5% | 0.837 | -4.4% [-8.0%, -0.8%] |
| count and field, rho 0.25 | 60.8% | 76.9% | 0.875 | +6.0% [+3.3%, +8.8%] |
| centre type and field, rho 0.25 | 59.9% | 74.8% | 0.860 | +3.9% [+0.6%, +7.2%] |

**The field adds next to the count, not instead of it.** On its own the local field is no better than the two-dimensional baseline: 71.1% against 70.9%, +0.2% [-3.3%, +3.8%]. Counting the same atom types over distance instead of over bonds does better on its own, 73.9%, +3.0% [+0.3%, +5.5%] -- geometry is worth something here, and it is worth it in the simplest form. The two together reach **76.9%**, which is +3.0% [+0.5%, +5.7%] over the count alone and +6.0% [+3.3%, +8.8%] over the baseline. That is the gate of the item, and it is passed by the pair rather than by the field. The two descriptions hold different things: the count says which atom types are at which distance, the field says how the charge and the shape are arranged there, and neither carries the other.

**Why the field alone is weak here is worth stating,** because it is a property of the descriptor and not of the set. Most of what the count contributes to that mixture is the name of the atom being asked about: replacing the whole count by its first block -- the centre's own atom type, one column per type -- already reaches 74.8%, against the 76.9% of the whole count. The autocorrelation of an environment never names its centre. It is built from distances between pairs of weighted atoms, and the centre is only the atom the envelope weights most. How easily an atom is oxidised depends first of all on what the atom is, so the invariant local field starts without the single most informative

variable and has to recover it indirectly. On the shift set of the first section that costs little, because there every sample is a carbon; here the candidates are carbons, nitrogens, oxygens and sulfurs at once.

**The width is chosen and not taken from the edge of the range.** The field rows rise to rho 0.25, an envelope 2.0 Å wide, and fall on both sides of it. A wider envelope makes every atom of a molecule look alike, because the features are measured from the molecule as a whole and not from the centre; a narrower one leaves too little of the molecule inside the envelope to tell two carbons apart.

**From the command line.** This is the first set in the manual large enough for the exact kernel to be the wrong route: 14 488 atoms give a kernel of 1.7 GB. `--primal` solves the same model in the space of the 255 features instead (the previous section shows the two agree to round-off), and `--rank` reports the ranking measure next to the $q^2$:

```
cmf local examples/local/metabolism.sdf --target som --rho 0.25 \
    --primal --rank --reference
```

That runs in about fifteen seconds and prints both the field model and the atom-centred count, with the share of molecules whose site each puts first and among the first two.

**What is not done.** The mixture of the count and the field, which is the row that passes the gate, is in the benchmark and not in `cmf local`: the command fits one descriptor at a time. An isoform-resolved target is the other open question. The sites here are merged over the isoforms, so the target is a property of the atom; a target resolved by enzyme would ask about the fit of the molecule into one binding site, which is what a field describes best, and that is the test this set cannot give.

## 11.10 Topomer fields

Cramer's topomer idea (topomer CoMFA, 2003), applied to fields. Cut the molecule at the bonds that attach its substituents to a common core and rebuild each substituent in a pose that is a **deterministic function of its topology**: canonical atom numbering, one embedding with a fixed seed, every rotatable torsion set to 180°, the attachment atom at the origin with its open valence along −x, and the remaining roll fixed by the principal axis of the heavy atoms. The same R-group is then always placed in the same position, whatever molecule it came from — a test asserts the coordinates are identical, not merely close.

Two things make this fit CMF better than it fits CoMFA. There is no grid, so a fragment's field is just its atoms and a fragment–fragment kernel is an analytic number, which means the fragment library can be tabulated once and reused. And each attachment point keeps **its own region of space** (the substituent of position $k$ is placed at $x = 100k$ Å, far enough that no Gaussian reaches the next site), so a molecule's kernel is exactly the sum of its per-position overlaps — which is what topomer CoMFA obtains by giving every position its own block of columns. A test checks that identity to 1e-9.

In the application it is *Tools → Replace molecules by their topomer substituents…*, and it is a **transformation of the data set** rather than a hidden switch: the molecules are replaced by their posed substituents, names, activities and the split are kept, and the data-set line says *[topomer substituents of …]*. That is the correct form for it, because a topomer model's co-field and contributions live in the substituent frames and not around the original ligands. Reload the file to get the molecules back.

**Choosing the core matters more than anything else**, which is why the program now chooses it differently. The maximum common substructure of a chemotype mixture is not a scaffold — on GPB it comes out as `CCO`, fused into every ring system, and cuts nothing off at all, losing every molecule. The default is therefore the **Bemis–Murcko framework of the largest chemotype**, the scaffold of one series, with every molecule that framework comes cleanly off forming the series; on GPB that is `C1CCOCC1` and 60 of the 66 molecules. The common substructure is still available for a set you know is congeneric (*maximum common substructure of the series* in the dialog, `--core-mode mcs`), and you can always give the core yourself as SMILES or SMARTS. Check the cuttable fraction the program reports either way: a core fused to the rest of the molecule cannot be removed without opening a ring, and what is left is not a substituent.

## 11.11 What the three routes are worth, measured

Every column below uses the same molecules, fields, α and λ grids and folds.

| Collection | Series | Aligned q² | Topomer q² | The cut costs | The pose costs |
|---|---|---|---|---|---|
| Sutherland, curated frame | 8 | **+0.445** | +0.343 | +0.035 | +0.066 |
| Mittal, automatic frame | 31 | +0.200 | **+0.261** | -0.027 | -0.033 |
| Corrosion inhibitors, neutral | 17 | +0.248 | **+0.307** | +0.024 | -0.083 |
| Corrosion inhibitors, 1 M HCl | 17 | +0.303 | **+0.358** | +0.037 | -0.092 |

The last two columns separate the two things the topomer construction does at once, by way of a control that keeps the same cut substituents at the same separated sites but leaves each one exactly where the alignment put it. **What the pose rule costs depends entirely on the alignment it replaces.** Against the curated Sutherland frame it costs 0.066 q²; against an automatic superposition, or against the frames of the corrosion series, it *gains* 0.033 to 0.092, because a canonical pose is more consistent than an arbitrary embedded conformation. The **cut** costs 0.02 to 0.04 on average and is the expensive half only where the scaffold carries information — separate sites discard the scaffold's own field and the geometry between one substituent and another, which matters when the arms' mutual placement is the pharmacophore (0.21 on thrombin, 0.14 on thermolysin) and costs nothing or even helps when the scaffold is constant and the decorations small (BZR gains 0.102 q², DHFR 0.054).

**Practical guidance.** With a trustworthy curated alignment, keep it: the geometry adds about 0.25 to q² over counts. With an automatic alignment or none, prefer the frame-free routes — topomer for a congeneric series with a clear scaffold whose R-groups come off at acyclic single bonds, and the invariant kernel for anything pooled, multi-scaffold or ring-fused. Always report the count baseline beside the model, so that the value of the geometry is visible rather than assumed.

## 11.12 Heterogeneous series, step by step

When a set holds several chemotypes with no one common substructure, the *Tools* → *Heterogeneous sets* submenu is a four-step workflow:

1. **Find the chemotypes** — Bemis–Murcko frameworks, optionally generic, with small groups merged so that leave-one-chemotype-out does not degenerate into leave-one-molecule-out.

2. **Align within each chemotype** — a superposition that is meaningful, because the members of one cluster do share a substructure.
3. **Place the chemotypes in one frame** — each cluster positioned by maximising field similarity to the others, which is the best that can be done when no common substructure exists.
4. **Leave one chemotype out, with the alignment-free baseline** — the reliable validation, reported beside the count kernel. Random folds overstate $q^2$ on such sets by 0.11 to 0.47 on the five chemotype benchmarks here, which is why this step exists.

# 12 Protein pockets

## 12.1 Getting a pocket on screen

*File → Fetch protein from the PDB...* downloads an entry from RCSB and caches it under `~/.cmfqsar/pdb`, with the dialog pre-filled with the identifier the data set names. *File → Load protein structure...* reads a local PDB or mmCIF file instead. The reader is deliberately independent of RDKit, because crystal structures carry alternate locations, missing elements and non-standard residues that a chemistry toolkit would refuse to sanitise, and none of that matters for drawing.

The free-energy benchmark sets in `examples/openff` and `examples/schindler` give their ligands in the frame of a reference crystal structure and name it per record, so **nothing has to be superimposed**: opening such a set brings its pocket with it from `examples/pockets/`, which holds the residues within 8 Å of any ligand, taken whole, without waters and without the co-crystallised ligand. The *protein site* controls above the 3D view then draw the residues within a chosen distance as wire frame or sticks, with hydrogens, waters and residue labels optional.

## 12.2 Complementarity as numbers

A pocket is a set of atoms, so the same parameterisation that builds a ligand's fields builds the protein's, and the overlap of two Gaussian fields has the same closed form on either side. The *Complementarity* tab (CLI: `cmf_cli.py complement`) turns that into four answers:

- **How complementary a molecule is to the pocket** — overlap integrals between ligand and pocket fields, exact and without a grid, each divided by the two norms so that it is a cosine in $[-1, 1]$ and does not grow with molecular size. Complementarity is not similarity: the electrostatic term carries a minus sign, because a positive ligand region should meet a negative protein region; the hydrophobic term carries a plus; the hydrogen-bond terms pair the donors of one side with the acceptors of the other. Steric overlap is reported separately as *burial*, because bulk against the protein is contact and clash at once, with the number of clashing heavy-atom pairs beside it.
- **Where on the molecule it holds** — the same quantities on a smooth van der Waals surface give a value per vertex, drawn as a green-to-red surface, with a correlation per field over the surface. Electrostatics uses potentials rather than Gaussian densities here, because the potential is what the other side feels.
- **Whether the co-field agrees with the pocket** — the model's own map against the protein, by sign agreement, cosine or correlation. This is the strongest external check a CMF model can be given: the co-field was fitted to activities and never saw the protein.
- **Which residues say so** — the same scores, resolved by receptor residue.

### Which residues say so

An overlap integral is a double sum over atom pairs, so keeping the per-atom column sums instead of the grand total costs nothing and is exact: every term of the score above splits over the residues of the pocket and adds back to it. One cosine becomes an **interaction profile**, a table of residues against interaction types.

This is worth having because a global score cannot say which contact a pose has lost, and neither can the ligand RMSD — a pose can move very little and still give up the one interaction that

mattered. Across a series the profiles also average into a **consensus**: an interaction seen in most of the complexes is a property of the target, while one seen in a single complex is more likely a property of that ligand, which is the distinction a pharmacophore query needs.

On the ten CDK2 ligands of `examples/openff/cdk2.sdf` against their pocket, every molecule reaches 18 of the 53 residues, and the contacts conserved across the whole series are Glu81 donating to an acceptor, Leu83 accepting from a donor, and Ile10 and Leu134 by burial — the hinge and the hydrophobic floor, recovered without being told that either exists.

In the application, tick **by residue** beside *surface scores* before pressing *Compute*. The table under the Complementarity tab then lists every residue within 4.5 Å of the selected molecule, one column per interaction type, sortable by any of them, with the share of the series that shows each residue's interactions in the last column; the header says how much of the whole score those residues carry, the rest being the tail of the Gaussians over the residues behind them. Tick **contributions** beside *residue labels* in the protein-site controls and the 3D view writes each residue's total on its alpha carbon, green where it helps and red where it does not.

From the command line:

```
python cmf_cli.py complement data.sdf --structure pocket.pdb --residues --profile-out profiles.csv
```

`--residues` prints the strongest contacts of each molecule and the conserved pattern of the series, and scores every molecule against that consensus by cosine and by Tanimoto over the thresholded bits; `--reference NAME` compares against one molecule's profile instead of the consensus, `--residue-threshold` sets what counts as present, and `--profile-out` writes one row per molecule and residue.

## 12.3 A macrocyclic host as a pocket

`complement` was written for proteins: a structure with residues, and a score read residue by residue. A cyclodextrin, a cucurbituril or a calixarene is a pocket too, and comes as a molecule file with no residues at all. The program now reads one (plan item SC3): `read_molecule_host` takes an SDF, MOL, MOL2 or PDB file and **names the host's repeating units**, which are its small ring systems -- one glucose of a cyclodextrin, one glycoluril of a cucurbituril -- with every atom outside a ring given to the unit it hangs on, and the macrocycle that joins them left out of the grouping, since it is not a unit. Everything the pocket machinery does per residue is then done per unit.

`cavity_frame` gives the middle of the ring of units and the direction in which they vary least, which is the axis of the channel. `complement --place-in-cavity` uses it: every molecule is put in the cavity, its centroid on the centre and its longest axis along the channel, which is how a guest threads a cyclodextrin, and is then refined there by the local rigid-body placement of `cmfqsar.docking`. Without it the guests are scored where they happen to sit, which for a host read from a file is nowhere near the site.

```
python examples/make_host_structures.py
cmf complement examples/cyclodextrin/beta_cyclodextrin.csv \
    --structure examples/cyclodextrin/hosts/beta_cyclodextrin.sdf --place-in-cavity
```

**The structures.** `examples/make_host_structures.py` embeds a host from its SMILES and keeps the most symmetric of forty conformations -- a cyclodextrin is a ring of identical sugars, so symmetry is the rule, and it uses nothing about the guests. The three it builds have cavity radii of 2.6, 3.0 and 3.9 Å for the six-, seven- and eight-sugar rings, against published values near 2.4, 3.1 and 4.0, so

the cavity it offers is the right size before any guest is scored.

### What it is worth, measured

This is the largest set a complementarity score has been given: one host, one cavity, and 1162 guests with a measured constant, where no protein set in the repository offers more than a few dozen ligands per pocket. `benchmarks/run_host_pocket.py` places every guest and correlates the score with log K.

| host | units | guests | score vs log K | burial vs log K | heavy atoms vs log K | score, size held constant |
|---|---|---|---|---|---|---|
| beta | 7 | 1162 | -0.348 | -0.077 | +0.348 | -0.285 |
| alpha | 6 | 827 | -0.206 | -0.102 | +0.184 | -0.160 |
| gamma | 8 | 292 | +0.025 | -0.022 | +0.383 | -0.072 |

| host | acceptor to donor | donor to acceptor | electrostatic | hydrophobic |
|---|---|---|---|---|
| beta | -0.285 | -0.281 | -0.104 | -0.405 |
| alpha | -0.217 | -0.188 | -0.028 | -0.277 |
| gamma | +0.037 | -0.073 | -0.174 | +0.064 |

**The score does not predict the constant, and on the largest table it points the wrong way.** On beta-cyclodextrin the correlation is -0.348; the guest's heavy-atom count alone gives +0.348, and with the size held constant what the score adds is -0.285. On beta and alpha the hydrophobic term, which is the one a hydrophobic cavity should reward, is the most negative of the four; on gamma, whose cavity is wide enough that a guest of this size need not touch its walls, every correlation is near zero.

**That is the limit of a one-body score, not a fault in the structures.** Binding to a cyclodextrin is driven by what leaves the cavity: the water inside it is poorly coordinated, and a guest that displaces it gains free energy that no overlap between the guest's fields and the host's can contain. A score computed from the two molecules alone has no term for the solvent it displaces, and this measurement is the clearest statement of that the repository has, because nothing else here has a thousand ligands in one pocket to say it with. Read the score as a check on a pose -- does this molecule fit, does it clash -- and not as an affinity.

## 12.4 Docking

*Tools → Dock into the pocket...* places the conformations of a molecule in the site by rigid-body optimisation. It is a convenience for getting a pose into the right frame, not a scoring function; the program's purpose is what happens after the poses exist.

# 13 Three series, worked through

Everything above is a tool. This part uses them, on three series taken from the free-energy benchmark collections because those come with the crystal structure their ligands were posed in, which is what makes the complementarity step possible at all. They were chosen by measuring every set in `examples/openff` and `examples/schindler` that has a pocket and keeping the three best models among different target families; `examples/case_studies.py` reproduces every number below and `benchmarks/results/cases/table.md` records them.

| case | target | n | activity range | α | q²cv | R²pred | worst of 30 shuffles |
|---|---|---|---|---|---|---|---|
| `ptp1b` | protein tyrosine phosphatase 1B | 22 | 3.77 | 3 | +0.499 | +0.147 | **+0.539** |
| `mcl1` | Mcl-1, a Bcl-2 family protein | 25 | 3.05 | 0.2 | +0.548 | +0.581 | +0.098 |
| `cdk8` | cyclin-dependent kinase 8 | 33 | 4.18 | 3 | +0.496 | +0.180 | +0.026 |

The protocol is the same for all three. The attenuation and the ridge are chosen by cross-validated grid search on two thirds of the series, and the other third — every third molecule by activity rank, so that it spans the range instead of sampling one end of it — is held out. The model whose co-fields are then read is refitted on the whole series with those hyper-parameters, because a co-field read from two thirds of a series is a co-field of two thirds of a series.

**One thing to say before any of the three.** What follows is not a chemist's reading of these series. The development plan asks for these cases to be judged by a medicinal chemist, and that judgement has not been made; what is here is every number such a judgement would need, with the places where the program contradicts itself left in rather than tidied away.

## 13.1 The case that does not survive its own check: `ptp1b`

Twenty-two ligands of a phosphatase, spanning 3.77 log units. The model looks good: q²cv +0.499, and on the two thirds it was selected on, +0.824. Then the activities are shuffled thirty times and refitted, and **the best shuffle reaches +0.539** against the real +0.499, for p = 0.065. On twenty-two molecules with five fields, a q² of one half is what chance produces.

So this case is not interpreted, and it is kept for exactly that reason: it is what *Is it chance? y-randomisation* is for, and it is the first thing to run on a series of this size. Everything the program would have said is still computed, and two parts of it are worth noting as warnings rather than findings:

- the fit is R² +0.975 against a cross-validated +0.499. A model that reproduces its own training set almost exactly and half the variance of a molecule it has not seen is memorising;

- the external third gives $R^2$pred +0.147 while the selection fold gave +0.824. The gap between those two numbers is the clearest single sign of a hyper-parameter chosen on too little data.

One measured result does survive, because it needs no model: of the five fields, the one this series does **not** need is the electrostatic one. Removing each field in turn and refitting costs electrostatic -0.009, steric +0.113, hydrophobic -0.016, hba +0.043, hbd +0.025 in $q^2$cv — the steric field carries what there is, and the electrostatic field can go without loss. A phosphatase site is charged; the series does not care.

## 13.2 Shape and lipophilicity in a groove: `mcl1`

Twenty-five ligands of a protein-protein interaction groove. $q^2$cv +0.548, $R^2$pred +0.581 on the 9 held out, and the best of thirty shuffles reaches only +0.098 ($p = 0.032$). This one can be read.

**Which fields.** Removing each in turn: electrostatic +0.032, steric +0.062, hydrophobic +0.071, hba +0.037, hbd -0.077. The hydrophobic and steric fields carry it and the hydrogen-bond donor field is worse than useless — removing it *improves* the model by 0.077. For a groove that binds a helix face, shape and lipophilicity being the answer is the expected answer, and it is also the measured one.

**What the co-fields say.** At the noise level the activities imply (0.559 log units), the share of each co-field that stands more than two standard deviations from zero is electrostatic 2%, steric 23%, hydrophobic 4%, hba 0%, hbd 0%. Only the steric map has a substantial part that the measurement error could not have produced; the electrostatic map, at 2 %, is almost entirely noise and should not be drawn. The pharmacophore read from what survives has 5 features: bulk +44.64; hydrophobic +29.18; positive +4.25; polar -0.71. A large favourable bulk region, a hydrophobic region inside it, and an exclusion volume of almost the same weight (-33.21) — which is the series saying that the groove has a size and that exceeding it costs as much as filling it earns.

**Where one prediction comes from.** The most active ligand (lig_53, measured +7.26, predicted +7.15) decomposes by rings and the pieces between them into CH4 at 37 +2.01, Cl1 at 36 +1.63, ring 2 -1.56. The two largest positive contributions are a small aliphatic piece and a chlorine, and one of the rings contributes *negatively*: against the scaffold baseline, that ring is not what makes this molecule good.

**What the pocket knows.** Scoring all twenty-five ligands against the crystal site (1007 atoms of 61 residues) gives a mean complementarity of +0.201, and its correlation with the measured activity is Spearman +0.254. But the steric overlap alone — burial, how deep in the groove the ligand sits — correlates at **+0.698**. The score that averages five complementarity terms is three times worse than one of its terms. On this target, how far the ligand reaches into the groove is most of the structure-activity relationship, and a composite score hides it.

## 13.3 An acceptor-driven kinase: `cdk8`

Thirty-three ligands spanning 4.18 log units, the widest range of the three. $q^2$cv +0.496, best shuffle +0.026, so the model is real; but $R^2$pred +0.180 on the 11 held out, and a training fit of $R^2$ +0.998 at $\lambda = 0.01$, which is a model interpolating its training set.

**Which fields.** electrostatic +0.043, steric -0.015, hydrophobic -0.024, hba +0.099, hbd -0.061. The hydrogen-bond acceptor field is the one that matters, and three of the five can be removed with no loss. For a kinase, an acceptor-driven model is the expected shape of the answer — the hinge donates — and this is the only one of the three cases where the dominant field is a polar one.

**What survives the noise.** electrostatic 4%, steric 6%, hydrophobic 4%, hba 7%, hbd 0%, at a noise level of 0.697. These are the smallest shares of the three cases: the model predicts, and almost none of its map is distinguishable from what the measurement error could have produced. The pharmacophore has 10 features (bulk +1.34; acceptor_bad -1.18; positive +1.09; positive +1.00), and the second and fifth of them are *unfavourable* acceptor regions — places where an acceptor lowers the activity. A pharmacophore from this series would consist mostly of places to avoid.

**The pocket says nothing.** Complementarity +0.002 with burial 0.014: these ligands barely overlap the trimmed site at all, and the correlation of the score with activity, +0.326, is on a quantity that is almost zero for every ligand. Compare `mcl1`, where burial is 0.588. The difference is not the method but the pocket file: a site trimmed at 8 Å around a shallow, solvent-exposed hinge leaves most of a ligand outside it.

## 13.4 What the three have in common

- **Every one of them loses at least one field without losing accuracy**, and in two of the three the field a chemist would expect to matter is not the one that does.
- **The share of a co-field that survives its own error bar is small**: 23 % at best, 6 % on `cdk8`. Drawing an unmasked co-field from twenty-five molecules shows mostly noise, which is what *The uncertainty of a co-field* exists to prevent.
- **A composite score can be worse than one of its parts** (`mcl1`: +0.254 against +0.698).
- **The check comes first.** One of the three does not beat a shuffle of its own activities, and no amount of interpretation would have revealed that; only the shuffle did.

## 13.5 One command for the alignment: `cmf align`

Alignment was spread over flags of `build` and `predict` and the `hetero` workflow, which made the criteria the program has look like unrelated features. They are one question — where do these molecules go — asked five ways, and `align` asks it:

```
cmf align data.sdf --by mcs    --out aligned.sdf          # on the common substructure
cmf align data.sdf --by field  --reference 0 --out aligned.sdf
cmf align data.sdf --by variance --model m.cmf --max-rmsd 2 --out aligned.sdf
cmf align data.sdf --by pocket --model m.cmf --pocket site.pdb --out aligned.sdf
cmf align data.sdf --by plane  --pose auto --refine plane --reference 0 --out onsurface.sdf
```

| `--by` | what it maximises | what it needs |
|---|---|---|
| `mcs` | the overlap of the maximum common substructure with a reference molecule | a reference, or it takes the first of the most active quarter |
| `field` | the CMF field similarity to a reference molecule, a model's training set, or the set itself | nothing, but `--restarts` decides how hard it tries |
| `variance` | the confidence of the model's own prediction — the criterion of *Alignment by prediction confidence* | a model (`--model`), and `--flexible` adds the torsions |
| `pocket` | that confidence with a complementarity term, $\sigma^2 - w$ [fit − clash] | a model and a structure; see the warning in *Alignment inside the pocket* |

| `--by` | what it maximises | what it needs |
|---|---|---|
| `plane` | nothing, by default: it poses each molecule on a surface $z = 0$ by rule (`--pose flat` or `upright`); with `--pose auto` and `--refine`, the field similarity to one reference pose, moved only as a surface allows | a reference whose input pose is trusted, for `auto` and `--refine`; see *Posed by rule: the surface frame without the slab* |

Every mode prints what it did per molecule and then how far the set moved, and writes nothing unless `--out` is given — an alignment that is not kept is a measurement, and saying so avoids the surprise of a command that silently rewrote a file. `--max-rmsd` limits how far a molecule may travel from the pose it arrived in, which is the guard the confidence criterion needs: without it a molecule is free to leave the series entirely and sit wherever the variance happens to be lowest. `--judge` adds the mean field similarity within the set afterwards, which is the one number that says whether the set now shares a frame.

What `align` does **not** do is tell you whether the result is better. Only *A protocol that can judge an alignment* can, and `benchmarks/run_poses.py` is where that protocol lives.

## 13.6 Co-fields without the application: `cmf cofield`

The co-fields were readable only from the 3D view. `cofield` writes them as Gaussian cube files, which any molecular viewer opens, and prints the numbers that go with them:

```
cmf cofield model.cmf --out-prefix maps --spacing 0.5 --z 2
cmf cofield model.cmf --kind overlap --molecule candidate.sdf --csv agreement.csv
```

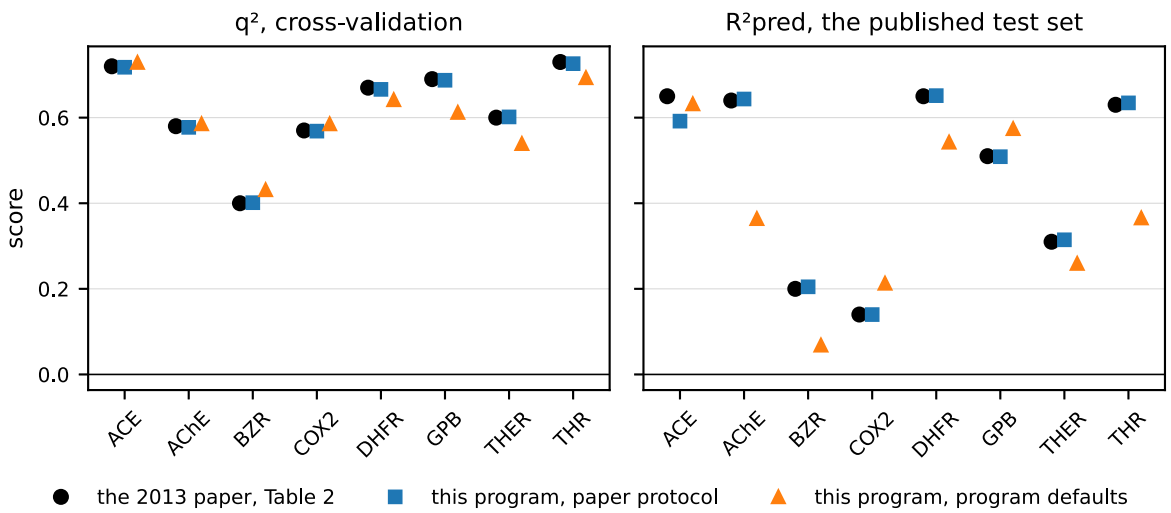
`--kind cofield` is the model's own map, one cube per field. `--kind molecule` is one molecule's field in the model's frame, and `--kind overlap` is that field weighted by the co-field — where the molecule and the model agree, which is the picture that explains a prediction. `--z` masks the map where it lies closer than that many standard deviations to zero (*The uncertainty of a co-field*), and the printed share of the covered grid that survives is the quickest way to see how much of a map is worth looking at:

```
electrostatic    peak     0.6146 (signed    -0.6146)   7% of the covered grid above 2 sd
hydrophobic      peak     0.8206 (signed    +0.8206)   18% of the covered grid above 2 sd
```

## 13.7 The benchmarks from the tool: `cmf benchmark`

`benchmarks/run_all.py` is the one entry point for every measurement in this manual, and `cmf benchmark` is the same program reached through the tool:

```
cmf benchmark --list            # what there is, what it needs, how long it takes
cmf benchmark --check           # do the tables and the documents match the stored results?
cmf benchmark alignment poses   # run these two
```




*The eight Sutherland sets: q² of the cross-validation and R²pred on the published test sets, from Table 2 of the 2013 paper and from this program with the paper's protocol and with its defaults. The paper's numbers are reproduced by the paper's protocol; the defaults differ most on R²pred, where the test sets are small.*

It needs the source checkout, because the benchmarks and their data are not installed with the package; the command says so rather than failing obscurely if they are missing.

# 14 Command-line reference

Everything the application does is available without it, which is what makes the program scriptable for a benchmark or a cluster run. The general form is

```
python cmf_cli.py <subcommand> <dataset> [options]
```

and `python cmf_cli.py <subcommand> --help` prints the full option list; what follows is an overview, not every option.

| Subcommand | What it does |
|---|---|
| `build` | build, cross-validate, test and save a model |
| `build-crystals` | the same on periodic structures (CIF, POSCAR, extended XYZ) |
| `diagnose` | ask a data set what it can support, before modelling it |
| `align` | put a set in one frame: by substructure, by field similarity, by prediction confidence or in a pocket |
| `predict` | predict with a saved model, optionally aligning by confidence first |
| `screen` | score a library against a saved model, reading it in blocks, so its size does not matter |
| `report` | one self-contained HTML page describing a saved model |
| `cofield` | the co-fields of a saved model as cube files and as numbers |
| `benchmark` | run, list or check the benchmarks of the checkout |
| `suggest` | which posed candidate to make next (expected improvement and the rest) |
| `design` | what to put at a position of a parent, ranked by the model |
| `transfer` | compare models by their co-fields: do they ask the same thing? |
| `mixing` | the posterior over the field weights: which fields the series needs |
| `hetero` | the heterogeneous-series workflow: cluster, align, place, leave-one-chemotype-out |
| `topomer` | model a series by its substituents, posed by rule |
| `map` | chemical-space map and activity landscape (kernel PCA + GTM) |
| `explain` | Shapley decomposition of one prediction, for any kernel |
| `mil` | multi-instance learning over conformations |
| `pharmacophore` | read a pharmacophore out of a fitted model's co-fields |
| `difference` | what one molecule's fields have that another's do not |
| `complement` | field complementarity with a protein pocket |
| `compare` | are two sets the same population? (MMD, with a witness) |
| `cluster` | cluster a set by its field kernel (spectral clustering) |
| `atoms` | a target that belongs to the atoms: what a file carries, and how it would be split |
| `local` | a model over atoms: the field around an atom against a count around it |

| Subcommand | What it does |
|---|---|
| `isosteres` | which groups are interchangeable at a position, by the co-field |
| `restriction` | score a ring closure: how much of the field that matters stops moving |
| `unalign` | give every molecule a random pose, the control experiment |

## 14.1 The options of `build`, by group

**Data.** `--activity PROP`, `--activity-file TABLE`, `--test-file FILE`, `--test-fraction F`, `--ensemble-by KEY`.

**Fields.** `--fields electrostatic,steric,hydrophobic,hba,hbd[,prop:name]`, `--cif`, `--cif-scheme sybyl|element|element+aromatic`, `--cif-min N`, `--cif-types LIST`, `--pharm`, `--pharm-min N`, `--pharm-features LIST`, `--charges gasteiger|mmff|file`, `--param rdkit|tripos|tripos-legacy`, `--prop-missing error|zero`, `--increments FILE`, `--learn-increments`, `--learn-increments-alpha A`, `--learn-increments-lam L`, `--learn-increments-rho R`, `--save-increments FILE`.

**Kernel.** `--kernel fields|autocorrelation|surface`, `--ac-sigma S`, `--ac-dmax D`, `--surface-z ZMIN ZMAX ZSTEP`, `--ac-chirality`, `--ac-chirality-shells S`, `--block-weights W`, `--block-groups G`, `--block-rho R`, `--kernel-layer linear|gauss|arccos1|arccos2`, `--gammas LIST`, `--normalize`, `--raw-scale`, `--optimize-h`, `--mkl [align|alignf]`, `--mkl-rho R`.

**Task and learner.** `--task regression|classification|oneclass`, `--method krr|gp|svr|nusvr|svc|nusvc|gpc|ocsvm|ocmean|ocknn|ocgp|ockpca`, `--threshold T`, `--active-below`, `--class-property PROP`, `--oc-k K`, `--oc-components Q`, `--oc-lam L`, `--oc-score mean|mean_over_sd`, `--no-bias`.

**Hyper-parameters.** `--alphas LIST`, `--lams LIST`, `--Cs LIST`, `--epsilons LIST`, `--nus LIST`, `--lam-search grid|brent`, `--alpha-search grid|nelder-mead`, `--nm-start A,L`.

**Conformations.** `--conformations single|ensemble`, `--conformers N`, `--conformer-window A`, `--prune-rmsd A`, `--conformer-align self|core|reference|field`, `--conformer-align-ref I`, `--conformer-weighting uniform|boltzmann|property`, `--kt KCAL`, `--max-conformers N`.

**Protonation and tautomers.** `--ph PH`, `--ph-states N`, `--ph-min-population F`, `--tautomers score|uniform|given`, `--tautomer-forms N`.

**Validation.** `--folds K`, `--cv-no-shuffle`, `--nested`, `--nested-on train|all`, `--outer-folds K`, `--repeats R`, `--seed S`.

**Screening.** `--decoys FILE`, `--align-decoys`, `--decoy-restarts N`, `--decoy-conformers N`.

**Alignment.** `--align-train-variance [MAX_RMSD]`, `--align-train-rounds N`, `--align-train-folds K`.

**Output.** `--save model.cmf`, `--predictions out.csv`, `--json out.json`.

## 14.2 Worked examples

```
# indicator fields, a test split and nested cross-validation
python cmf_cli.py build examples/benzamides_synthetic.sdf --activity pActivity --cif \
       --alphas 0.3,0.5,0.8,1.2 --lams 1e-3,1e-2,0.1,1 --test-fraction 0.25 \
       --nested --save model.cmf

# predict, with the new molecules aligned by prediction confidence first
python cmf_cli.py predict model.cmf new.sdf --align-variance 2.0 \
       --aligned-out aligned.sdf --out predictions.csv

# the published protocol of the 2013 paper on a Sutherland set
python cmf_cli.py build examples/sutherland/THR.sdf --charges file --param tripos-legacy \
       --raw-scale --cv-no-shuffle --lam-search brent --nested --nested-on all

# a frame-free model: the rotation-invariant autocorrelation kernel
python cmf_cli.py build series.sdf --kernel autocorrelation --ac-sigma 0.75 --ac-dmax 12

# a topomer model with the core given explicitly
python cmf_cli.py topomer examples/benzamides_synthetic.sdf --activity pActivity \
       --core "c1ccccc1C(=O)N" --alphas 0.3,0.5,1 --lams 1e-2,0.1,1

# one-class screening against a property-matched background
python cmf_cli.py build actives.sdf --task oneclass --method ockpca \
       --decoys background.sdf --align-decoys --decoy-conformers 5

# a classification model with the Gaussian-process classifier
python cmf_cli.py build examples/sutherland/AChE.sdf --charges file --task classification \
       --threshold 6.84 --method gpc --Cs 1,10,100
```

# 15 Using it as a library

The library is the layer the application and the command-line tool are built on, and it is small enough to use directly in a notebook.

```
from cmfqsar import FieldSet, CMFModel, load_dataset
from cmfqsar.fields import AtomTable, build_fields, cif_types_in
from cmfqsar.model import grid_search

ds = load_dataset("aligned.sdf", activity_prop="pIC50")
fs = FieldSet.create(cif_types=cif_types_in(ds.mols, "sybyl", 5))
mf = [build_fields(m, fs, n) for m, n in zip(ds.mols, ds.names)]
train = AtomTable.from_molecules(mf)
model = CMFModel(fieldset=fs, method="krr")
grid_search(train, ds.activity, model, alphas=[0.3, 0.5, 1.0], lams=[1e-3, 1e-2, 0.1, 1])
model.fit(train, ds.activity, cv_folds=10, reuse_kernels=True)
print(model.stats.summary())
```

Decomposing one prediction, which works for the nonlinear kernels as well:

```
from cmfqsar.shapley import atom_shapley, field_shapley, scaffold_decomposition

res = atom_shapley(model, mf[0])                       # one value per atom
res = field_shapley(model, mf[0], interactions=True)   # one value per field type
res = scaffold_decomposition(model, mf[0], ds.mols[0], ds.mols[1:20])   # per substituent
print(res.prediction, res.baseline, res.table(), res.top_interactions(5))
```

Three more tasks that a notebook user is likely to want outside the application. Each script runs as written on a file from `examples/`.

A prediction interval with a coverage guarantee. The calibration uses the cross-validated errors that `fit` has already computed, so nothing is recomputed; see *How certain is one prediction? Conformal intervals* for what the guarantee means.

```
from cmfqsar import FieldSet, CMFModel, load_dataset
from cmfqsar.fields import AtomTable, build_fields
from cmfqsar.conformal import coverage

ds = load_dataset("examples/steroids_cbg.sdf", activity_prop="CBG_pK")   # 21 train, 10 test
fs = FieldSet.create()
mf = [build_fields(m, fs, n) for m, n in zip(ds.mols, ds.names)]
tr, te = ds.train_indices, ds.test_indices
model = CMFModel(fieldset=fs, method="krr", params={"lam": 0.01})
model.fit(AtomTable.from_molecules([mf[i] for i in tr]), ds.activity[tr], cv_folds=10)
model.calibrate_conformal(level=0.9)     # from the cross-validated errors fit() has already made
print(model.conformal_.describe())
yp, lo, hi = model.predict_interval(AtomTable.from_molecules([mf[i] for i in te]))
print(coverage(ds.activity[te], lo, hi))
```

A model on a series that has no alignment. A SMILES table is enough: `load_dataset` gives every molecule its own 3D structure, in no common frame, and the rotation-invariant autocorrelation kernel of *Working without an alignment* does not need one. The last line shows which atoms the prediction for one molecule came from.

```
from cmfqsar import FieldSet, CMFModel, load_dataset
from cmfqsar.fields import AtomTable, build_fields
from cmfqsar.model import grid_search

ds = load_dataset("examples/cyclodextrin/gamma_cyclodextrin.csv", activity_prop="activity")
fs = FieldSet.create(alpha=1.0)
mf = [build_fields(m, fs, n) for m, n in zip(ds.mols, ds.names)]
train = AtomTable.from_molecules(mf)
model = CMFModel(fieldset=fs, method="krr", normalize=True,
                 kernel_source="autocorrelation", ac_sigma=0.75, ac_d_max=12.0)
grid_search(train, ds.activity, model, alphas=[1.0], lams=[1e-3, 1e-2, 0.1, 1])
model.fit(train, ds.activity, cv_folds=10)
print(model.stats.summary())
print(model.atom_contributions(mf[0]))      # one value per atom of the first molecule
```

Two endpoints on one kernel. The multi-task model of *Several endpoints at once* is a function of the kernel matrix, so it takes the kernel of any model, aligned or not. Here every guest carries its binding constant to beta-cyclodextrin and some guests carry the alpha-cyclodextrin constant as well; the two are fitted together, tied by one correlation chosen by cross-validation.

```
import numpy as np
from cmfqsar import FieldSet, CMFModel, load_dataset
from cmfqsar.fields import AtomTable, build_fields
from cmfqsar import multitask as MT

ds = load_dataset("examples/cyclodextrin/beta_with_alpha.csv", activity_prop="activity")
alpha = np.array([float(m.GetProp("alpha_logK")) if m.HasProp("alpha_logK") and
                  m.GetProp("alpha_logK").strip() else np.nan for m in ds.mols])
fs = FieldSet.create(alpha=1.0)
mf = [build_fields(m, fs, n) for m, n in zip(ds.mols, ds.names)]
model = CMFModel(fieldset=fs, method="krr", normalize=True, kernel_source="autocorrelation")
K = model.training_kernel(AtomTable.from_molecules(mf))    # one kernel for both endpoints
obs = MT.Observations.from_columns(np.column_stack([ds.activity, alpha]), names=("beta", "alpha"))
rho, lam, score, per_task = MT.search(K, obs, rhos=[0.0, 0.5, 0.9], lams=[0.01, 0.1, 1.0])
print(rho, lam, per_task)                    # cross-validated q2 of each endpoint
fit = MT.fit(K, obs, rho=rho, lam=lam)
print(fit.predict_selectivity(K[:5], 0, 1))  # beta minus alpha for the first five guests
```

Four notebooks under `docs/tutorials/` take the library through the same ground step by step, with plots and 3D views: a first model on an aligned series, a series with no alignment, the checks that say how far a prediction can be trusted, and two endpoints with their selectivity. Each runs on the example data of the repository, locally or in Google Colab from the link at its top.

The function-by-function reference is a separate file, `docs/api.md`. It is generated from the docstrings of the source by `python docs/make_api.py`, so it lists every public class, method and function with its signature, and it is as current as the last time that command was run. This chapter says what the modules are for; that file says what each function takes and returns.

The modules, one line each. The package has more modules than the application has panels: many of them exist only for the command-line tool and for use as a library.

Data, fields and kernels:

```
cmfqsar/dataio.py        reading SDF / MOL2 / SMILES, model persistence, exports
cmfqsar/fields.py        field parameterisation; FieldSet / MoleculeFields / AtomTable
cmfqsar/atomtypes.py     Sybyl-like atom typing and van der Waals parameters
cmfqsar/increments.py    atomic increments of the original CMF implementation (2013)
cmfqsar/learnincrements.py  increments fitted to the data instead of taken from the table
cmfqsar/pharmtypes.py    six pharmacophore features as indicator fields
cmfqsar/probes.py        GRID-like probe fields written as sums of Gaussians
cmfqsar/anisotropic.py   atom functions with a direction (Gaussian times a linear factor)
cmfqsar/exposure.py      solvent exposure as a weight on the fields
cmfqsar/descriptors.py   descriptors that are not fields: one number per molecule
cmfqsar/orca.py          quantum-chemical properties from ORCA, as fields or descriptors
cmfqsar/cdft.py          conceptual-DFT fields on a grid, and their Gaussian fits
cmfqsar/kernels.py       analytic field kernels, vectorised
cmfqsar/fastkernel.py    a compiled, threaded atom-pair loop for the kernels
cmfqsar/mixing.py        a posterior distribution over the field weights
cmfqsar/perfield.py      a separate attenuation for every field, chosen by the evidence
```

Models, their parameters and their validation:

```
cmfqsar/model.py         KRR / SVR / GP, cross-validation, grid search, nested CV, statistics
cmfqsar/evidence.py      choosing α and λ by the marginal likelihood instead of by cross-validation
cmfqsar/neldermead.py    the Nelder-Mead minimiser of the original R scripts
cmfqsar/ranking.py       the ranking SVM: learning the order of the molecules
cmfqsar/multitask.py     two endpoints at once; selectivity as a co-field
cmfqsar/gpc.py           Laplace Gaussian-process classification
cmfqsar/oneclass.py      centroid, k-NN, GP and kernel-PCA one-class descriptions
cmfqsar/pu.py            learning from positives and unlabelled molecules
cmfqsar/fusion.py        one score from the similarities to several known actives
cmfqsar/validation.py    y-randomisation, applicability domain and the error of q²
cmfqsar/conformal.py     prediction intervals with a coverage guarantee
cmfqsar/errorpred.py     which error estimate knows when a prediction is bad
cmfqsar/mmd.py           the two-sample test: is this library like the training set?
cmfqsar/contract.py      what has to match between fitting and predicting, checked
cmfqsar/diagnose.py      what to ask of a data set before modelling it (`cmf diagnose`)
```

Reading a model:

```
cmfqsar/cofields.py      co-fields and molecular fields on grids, isosurfaces, cube export
cmfqsar/shapley.py       Shapley decomposition over atoms, substituents, field types
cmfqsar/pharmacophore.py typed features with tolerance spheres, from the co-field
cmfqsar/fielddiff.py     field differences against a reference molecule
cmfqsar/transfer.py      comparing two models by their co-fields
cmfqsar/spectral.py      clustering a set by its kernel rather than by its scaffolds
cmfqsar/gtm.py           chemical-space maps and activity landscapes
cmfqsar/report.py        the HTML report of a model
```

Alignment and poses:

```
cmfqsar/align.py         3D embedding, MCS alignment, field-similarity refinement, unalign
cmfqsar/coalign.py       co-alignment of a training set by expectation-maximisation over poses
cmfqsar/varalign.py      alignment by prediction confidence (GP posterior variance)
cmfqsar/flexalign.py     the same search with the torsions free
cmfqsar/fieldgraph.py    field-point graphs: alignment without a common substructure
cmfqsar/poses.py         a protocol that judges an alignment against reference poses
cmfqsar/stereo.py        handedness as a property of a stereocentre
```

Working without an alignment:

```
cmfqsar/invariant.py     rotation-invariant field autocorrelation kernels
cmfqsar/periodic.py      the periodic autocorrelation kernel for crystals
cmfqsar/topomer.py       topomer fields: substituents posed by rule
cmfqsar/chemotypes.py    scaffold clusters, placement, grouped folds, the count kernel
```

Conformations and ionisation states:

```
cmfqsar/conformers.py    conformational ensembles: generation, weighting, grouping
cmfqsar/mil.py           multi-instance learning over conformations
cmfqsar/protonation.py   protonation ensembles: a molecule as a weighted mean of its states
cmfqsar/tautomers.py     tautomer ensembles
```

Screening and design:

```
cmfqsar/decoys.py        posing a background set for screening evaluation
cmfqsar/screen.py        screening a library that does not fit in memory
cmfqsar/active.py        which molecule to make next
cmfqsar/design.py        what to put at a position, ranked by the model
cmfqsar/gradients.py     which way to move an atom: the derivative of a prediction
```

Protein pockets:

```
cmfqsar/protein.py       PDB / mmCIF reading, binding-site selection, frame check
cmfqsar/complementarity.py  pocket fields, complementarity scores, co-field agreement
cmfqsar/pocketalign.py   aligning a molecule inside the pocket
cmfqsar/docking.py       rigid-body placement of conformations in the pocket
```

The two front ends:

```
cmfqsar/cli.py           the `cmf` command-line tool
cmfqsar/gui/             the PySide6 application
```

# 16 The example data sets

| Set | Contents |
|---|---|
| `examples/benzamides_synthetic.sdf` | 109 aligned N-phenylbenzamides with a *synthetic* activity: the overlap of their fields with a hidden pseudo-receptor co-field plus noise (σ = 0.25). A model has to recover those zones; $q^2 \approx 0.87$ against a noise ceiling of 0.94. The set to learn the program on, because the right answer is known. |
| `examples/steroids_cbg.sdf` | the classic CoMFA benchmark: 31 steroids (Cramer, Patterson & Bunce 1988) with corticosteroid-binding-globulin affinities, in the corrected Gasteiger-group geometries, re-superimposed on the steroid skeleton of testosterone. Customary split 1–21 / 22–31, leave-one-out $q^2 \approx 0.79$. |
| `examples/sutherland/*.sdf` | the eight literature sets of the 2013 paper (Sutherland, O'Brien & Weaver 2004): ACE, AChE, BZR, COX-2, DHFR, GPB, THER, THR, with the aligned MMFF94s conformations as distributed, the published split, and the original Sybyl types and MNDO ESP charges as per-atom properties. Choose *Partial charges: file*. |
| `examples/selectivity/ampa_glycine.sdf` | the 57 AMPA / glycine-NMDA antagonists of the 2003 selectivity-fields paper (Baskin, Tikhonova, Palyulin & Zefirov) with `pKi_AMPA`, `pKi_glycine`, their difference `delta_AMPA_glycine` and the published split (52 / 5), superimposed on the tricyclic core; the set to try *Model several endpoints (selectivity co-field)…* on. Use Gasteiger charges: the file charges of sixteen compounds are unreliable. `docked_ampa.sdf` and `docked_glycine.sdf` hold the same compounds docked in the two receptor models, `receptor_ampa.pdb` and `receptor_glycine.pdb`. |
| `examples/mittal/*.sdf` | 38 of the 40 benchmark sets of Mittal, McKinnon & Sorich (2009), built here from their 2D supplementary data by ETKDG and an MCS superposition, with each molecule's fit quality recorded in `align_rmsd`. The collection to use when you want to know how a method behaves on an *automatic* alignment. |
| `examples/openff/*.sdf` | 21 congeneric series, 513 ligands, from the Open Force Field protein–ligand benchmark, each posed in its target's crystal structure; activity tag `pAffinity`. |
| `examples/pockets/*.pdb` | the binding sites of those series, trimmed and already in the ligands' frame, so they open together with the data. |
| `examples/thrombin_atom_properties.sdf` | a worked example of `prop:<name>` fields: per-atom quantum properties carried in an SDF. |

`docs/benchmark.md` compares the program with Table 2 of the 2013 paper on the eight literature sets. `benchmarks/run_sutherland.py` reruns that comparison, `benchmarks/run_topomer.py` measures the alignment against the topomer rule (`--mittal` for the automatically aligned collection), `benchmarks/run_alignment.py` compares the value of a curated alignment with that of an automatic one, and `benchmarks/run_corrosion.py` does the same for corrosion-inhibitor series.

# 17 When something goes wrong

**The 3D view is black, or the program will not start.** The viewer needs working OpenGL through PyOpenGL and pyqtgraph. On a remote desktop or in a virtual machine, software rendering is usually the fix (on Windows, a Mesa `opengl32.dll` beside the program; on Linux, `LIBGL_ALWAYS_SOFTWARE=1`). Everything except the 3D view works regardless, and the command-line tool needs no graphics at all.

**q² is high but the test predictions are poor.** Three usual causes, in order of frequency: the hyper-parameters were chosen on the same cross-validation you are quoting (run *External CV*); the set holds several chemotypes and random folds are overstating it (use leave-one-chemotype-out); or the alignment carries information that will not transfer (compare against the count baseline).

**q² is negative.** The model predicts worse than the mean: check that the activity column is the right one and on a logarithmic scale, and that the molecules really share a frame. For $R^2_{pred}$ on a narrow test set it may be an artefact of the variance convention, and the program reports both.

**The optimum sits at the edge of the α or λ grid.** Extend the grid. The *Hyper-parameter search* tab exists to make this visible.

**A per-atom property is rejected.** The hydrogen count almost certainly changed when the file was loaded. Write the property for all atoms including hydrogens, or pass `--prop-missing zero` deliberately.

**The topomer transformation drops most of the molecules.** The core is fused to the rest of them. Look at the cuttable fraction in the log and give a core joined to its R-groups by acyclic single bonds.

**A decoy set gives suspiciously good enrichment.** Check that the decoys are property-matched and posed in the training frame; cross-target decoys are separable by almost anything.

**Co-fields are unavailable.** A nonlinear kernel layer or the autocorrelation kernel source is selected, and neither is linear in the field, so the co-field does not exist. Use the *Shapley* tab instead.

**It is slow.** Kernel cost grows with the square of the total atom count, so the expensive sets are those with many large molecules. Cut the α grid first, since each α needs its own kernel, and remember that the count kernel is 10 to 34 times faster than the Gaussian route for a baseline.

# Appendix A — Field parameterisation

Two parameterisations are available (*Increments* in the model panel, `--param`):

| Field | `rdkit`, the default | `tripos`, the original CMF scripts |
|---|---|---|
| electrostatic | partial charge: Gasteiger, MMFF94, or read from the file | the same |
| steric | Tripos vdW well depth and width $\alpha/R^2$_vdW, parameters by element | parameters by Sybyl atom type, Tripos 5.2 table |
| hydrophobic | Wildman–Crippen atomic logP contribution | in-house one-atom fragment increments by Sybyl type and attached hydrogens |
| H-bond acceptor / donor | approximate Abraham-type increments, donor weight on the polar hydrogen | in-house increments from Abraham's A and B constants, donor weight on the heavy atom |
| indicator field | 1 for atoms of the given type | the same |
| `prop:<name>` | the value of a per-atom property carried by the file | the same |

`tripos-legacy` reproduces the hydrogen-counting bug of the original R scripts, which is needed to match the published numbers exactly and should not be used for new work.

## A.1 Increments fitted to the data

The hydrophobic field and the two hydrogen-bond fields are built from one number per atom type: the increment of a Sybyl type carrying a given number of hydrogens. Those numbers come from regressions of logP and of Abraham's solute constants, not from the activity being modelled, and the two parameterisations in the table above differ by 0.20 in q2 on one Sutherland set. That difference says the increments are not settled, so the program can fit them instead (`cmfqsar/learnincrements.py`, `build --learn-increments`).

**Fitting them is cheap, because the kernel is a quadratic form in the increments.** Every atom of a type carries the same weight, so

$$K(A,B) = \sum_{t,u} c_t c_u M_{tu}(A,B)$$

where `M` sums the Gaussian overlap integral over the atom pairs whose types are `t` and `u` and does not depend on the increments at all. `M` is built once per attenuation, every candidate kernel is one contraction, and the derivative of the cross-validated error can be carried back to the increments exactly, through the cosine normalisation and the fold solves. The program minimises the same cross-validated squared error it selects models on, with the published values as the starting point and a penalty on the distance from them. Tests compare every link of that chain with finite differences, and compare the quadratic form with the kernel the program builds the ordinary way.

Two properties of the parameterisation are worth knowing. A cosine-normalised kernel does not change when all the increments of a field are multiplied by the same number, so only their relative sizes can be learned; the fitted values are reported at the length of the published ones, which

changes no kernel. And the penalty cannot be chosen on the cross-validation being minimised, which always prefers the freest fit, so it is chosen on training molecules that the fitting does not see.

## What it is worth, measured

`benchmarks/run_increments.py` measures four models on the eight Sutherland sets, on their published training and test split, with the same five fields, the same attenuation and ridge grids and the same ten-fold cross-validation: the published increments; increments fitted on this set's own training molecules; increments fitted on each of the other seven sets and applied here; and the best of those seven.

| Model | mean q2 (training cross-validation) | mean R2 on the test set |
|---|---|---|
| published increments | +0.596 | **+0.383** |
| fitted on this set | **+0.816** | +0.310 |
| fitted on another set (mean of seven) | +0.593 | +0.363 |
| the best of those seven | +0.646 | +0.449 |

**Fitting the increments improves the number it is fitted on and does not improve prediction.** The cross-validated q2 rises from +0.596 to +0.816, which is what minimising it must do. On the test set, which the fitting never saw, R2 goes from +0.383 to +0.310; fitted increments predict better than the published ones on 2 of the 8 sets. The gap between the two movements is the selection bias of reading a q2 off the cross-validation that chose the parameters.

**What is learned does not transfer.** Increments fitted on one set and applied to another give +0.363 against +0.383 for the published table; transferred increments beat the published ones on 3 of the 8 sets. The best of the seven donors reaches +0.449, which is above the table, but that donor is picked with the test set already in hand. The fitted vectors do not agree with each other either: the mean cosine between the increments fitted on two different sets is +0.89 for the hydrophobic field, +0.72 for the acceptor field and +0.54 for the donor field, where every fit started from the same table.

The "best of seven" column is the most generous reading there is, and it is not a rule anyone could follow: the best donor is picked with the test set already in hand. Read it as an upper bound on what choosing a donor set could ever give.

**So the increments are a per-set nuisance parameter**, which is the answer the plan asked for. The published table is a reasonable prior and should stay the default; fitting the increments is worth doing to see how much a set wants to move them, not to make a model that predicts better. The fit moves the two hydrogen-bond fields much further than the hydrophobic one (+0.99, +0.86 and +0.70 mean cosine with the published values), which is also where the two parameterisations of Appendix A differ most: they put the donor weight on different atoms. How far a set moves its increments does not say how much it will lose, though -- over the eight sets the correlation between the two is -0.25, which is nothing.

`--learn-increments` is therefore not a default. It prints a warning that the q2 of the same cross-validation is optimistic, and `--save-increments` writes the table so that it can be read back with `--increments FILE` and used on another series -- which is how the transfer above was measured.

# Appendix B — Files and tags the program understands

| Tag or convention | Meaning |
|---|---|
| any numeric SD property | candidate activity; the loader guesses and the dialog confirms |
| `subset`, `set`, `split` | the train/test split, as `train`/`test` or 1/2 |
| `atom.dprop.<name>` | a per-atom double property, written and read by RDKit |
| `atom.prop.sybyl_type` | per-atom Sybyl types, used automatically for indicator fields |
| a molecule property with one number per atom | the same per-atom property in quantum-chemistry dump form |
| `.cmf` | a saved model (joblib): field set, kernels, coefficients, training molecules, settings |
| Gaussian cube | the export format for co-fields, molecular fields, Shapley fields and difference maps |
| `REMARK 400` in a pocket PDB | the provenance of a trimmed binding site |

# Appendix C — Glossary

The terms this manual uses in a particular sense, with the chapter that treats each one.

| term | what it means here |
|---|---|
| **alignment**, or **frame** | the common coordinate system the molecules are placed in before their fields are compared. The overlap kernel needs one and is only as good as it is; the count, autocorrelation, surface and periodic kernels do not. *Preparing the data*; *Working without an alignment* |
| **applicability domain** | how close a molecule outside the training set is to it, measured in the kernel the model itself uses, with a threshold taken from the training set's own spread. Reported per molecule as the **AD** column and carried into the exported predictions. *Where does the model apply?* |
| **attenuation**, α | the width parameter of a field's Gaussians: α large makes a field that falls away within an atom's own volume, α small one that reaches across the molecule. The single most consequential setting after the alignment. *The fields*; *Attenuation, regularisation and the grids* |
| **autocorrelation kernel** | geometry without a frame: the smoothed distribution of field-weight products over interatomic distance, invariant to rotation and translation by construction. *The rotation-invariant autocorrelation kernel* |
| **BEDROC, RIE** | screening figures that weight the top of a ranked list, where AUC saturates. *Decoys and the figures that matter* |
| **Bemis–Murcko framework** | the ring systems of a molecule together with the bonds that join them. `cmf diagnose` measures the common framework to ask whether one frame could exist at all. *Before you model* |
| **block weights** | one coefficient per block of channels — a block being one pair of field types — applied before the invariant kernels are normalised, so a weight of zero removes a block exactly. *Mixing coefficients over the channel blocks* |
| **CHELPG**, **Hirshfeld**, **Löwdin**, **Mulliken** charges | four ways of dividing a molecule's electron density among its atoms, all of which ORCA can report and any of which can become the electrostatic field's weights. They disagree, sometimes substantially, which is why the charge model is part of what must match between fitting and predicting. *Quantum-chemical properties with ORCA* |
| **chemotype** | a group of molecules sharing a scaffold. Splitting a benchmark by chemotype rather than at random is the test of whether a model carries beyond the series it was fitted on. *The example data sets* |
| **co-field**, $C_f(\mathbf{r})$ | the model as a function of space: the training molecules' fields summed with the fitted coefficients, one per field type. Every prediction is the overlap of a molecule's own field with it. *The same prediction, as a field*; *Co-fields* |
| **co-pattern** | what the invariant kernels offer in place of a co-field — the weight of every channel of the fitted linear model, named by its field pair and distance, ranked by `top_channels`. *Crystals*; *Molecules on a surface* |
| **contributions** | the decomposition of one prediction over the training molecules, and of a training molecule's share over its atoms. *Contributions* |

| term | what it means here |
|---|---|
| **count kernel** | the pose-free baseline: a kernel on atom-type counts, with no geometry at all. Worth running before any aligned model, because it sometimes wins. *The pose-free count kernel* |
| **decoys** | presumed-inactive molecules used **only** to evaluate a screening model, never to fit it; property-matched decoys are the fair kind. *Decoys and the figures that matter* |
| `cmf diagnose` | the command that runs the five baselines — aligned, counts, autocorrelation, topomer, descriptors — on a data set before any model is built, and says which route to take. *Before you model* |
| **dual coefficients**, **a**, and **bias** $b$ | what fitting produces: one coefficient per training molecule and one constant, from which every later prediction is a weighted sum of kernel values. *Fitting: what is learned from the training set* |
| **ECFP4** | a circular fingerprint of the atoms' environments out to two bonds: the standard 2D similarity measure, and the baseline the field similarity of this program is compared against on screening tasks, where it does about as well. *Decoys and the figures that matter* |
| **ETKDG** | RDKit's distance-geometry conformer generator, which builds the 3D structure whenever the input is 2D and supplies the starting conformations for the conformer ensembles, the flexible alignment and the topomer rebuilds. Its poses are arbitrary until they are aligned. *What the program reads*; *Conformations: the fourth dimension* |
| **external cross-validation**, $q^2_{ex}$ | the whole model-building procedure, hyper-parameter search included, repeated inside each outer fold; the only figure that is not optimistic about the search. *Validation* |
| **field**, $X_f(\mathbf{r})$ | a continuous function of space: one Gaussian per atom, weighted by that atom's charge, bulk, lipophilicity, H-bond strength, indicator value or file-borne property. The object CMF models. *The fields* |
| **field scaling**, $c_f$ | the factor that puts a normalised field kernel's mean diagonal at 1 over the training set, so the fields enter the mixture on equal terms. Fixed on the training set and stored. *Normalisation and scaling* |
| **field set** | the list of field types a model uses, with their attenuations, charge model and parameterisation. It must match exactly between fitting and predicting. *What has to match between fitting and predicting* |
| **flexible alignment** | alignment that turns the rotatable bonds as well as moving the molecule, with the force field as a prior on the conformation and several generated starting conformations. The best pose method measured here: 0.86 Å median against the crystal poses. *Judging an alignment, and aligning flexibly* |
| **Fukui index**, $f^+$, $f^-$, $f^0$ | a per-atom measure of where a molecule gives or takes electron density, computed from the N+1 and N−1 states by ORCA and used as a `prop:` field. The **dual descriptor** is $f^+ - f^-$. *Quantum-chemical properties with ORCA* |
| **h**, the **mixing coefficient** $h_f$ | the weight of one field's normalised kernel in the joint kernel. Set by hand, optimised, or derived by kernel alignment. *Mixing the fields* |
| **increments** | the table of per-atom parameters a field is built from — the `rdkit` default or the `tripos` tables of the original CMF scripts. Not a setting of the method but of the chemistry it reads. *Appendix A* |
| **indicator field** | a field whose weight is 1 for atoms of a given type and 0 elsewhere, so that its co-field says where atoms of that type help. *Choosing the fields* |

| term | what it means here |
|---|---|
| **joblib** | the serialisation library a saved `.cmf` model is written with. A `.cmf` file is therefore a Python object graph, not an interchange format: it is read by this program and not meant to be read by another. *Saving and reusing* |
| **joint kernel**, $K(A,B)$ | the sum of the normalised, scaled field kernels with their mixing coefficients: the single similarity the learner sees. *Mixing the fields* |
| **Kabsch superposition** | the least-squares rotation and translation that best superimposes two sets of matched points. It is the second half of an MCS alignment -- the substructure search finds the correspondence, this places the molecule -- and the pose benchmark measures how far the result lands from the crystal pose. *Alignment*; *Judging an alignment, and aligning flexibly* |
| **kernel layer** | a nonlinear function applied to the joint kernel — Gaussian or arc-cosine — after which there is no co-field, because the prediction is no longer linear in the fields. *Nonlinear kernel layers and multiple kernel learning* |
| **MCS alignment** | superimposing molecules on their maximum common substructure. The conventional frame, and the baseline every pose method here is measured against. *Alignment* |
| **MKL** | multiple kernel learning: deriving the mixing coefficients from the kernels and the activities by centred kernel alignment rather than setting them. *Nonlinear kernel layers and multiple kernel learning* |
| **nested cross-validation** | choosing the hyper-parameters inside each training fold and scoring outside it, so that the reported figure has not seen the molecules the choice was made for. *Validation* |
| **normalisation** | dividing a field kernel by the geometric mean of the two self-kernels, so every molecule has unit self-similarity and a large molecule cannot look important merely for being large. *Normalisation and scaling* |
| **one-class model** | a model fitted on actives alone, which asks whether a molecule resembles them rather than how active it is. *One-class models: screening with actives alone* |
| **overlap integral**, $K_f(A,B)$ | the integral over all space of the product of two molecules' fields, which for Gaussians is a closed-form double sum over atom pairs. The kernel CMF is built on. *The overlap integral* |
| **periodic kernel** | the autocorrelation descriptor for a crystal: the pair histogram of one cell with its periodic images, per atom, with the uniform background removed and the cutoff tapered. *Crystals: the periodic autocorrelation kernel* |
| **pharmacophore** | a set of typed spheres extracted from the lobes of a co-field, which is the model's own statement of what a molecule should carry and where. *A pharmacophore out of the co-field* |
| **probe field** | a GRID-like interaction energy with a chosen probe, expanded in Gaussians so that it keeps the closed-form kernel and the co-field. *GRID-like probe fields* |
| `prop:` **field** | a field whose per-atom weights are read from the file or computed by ORCA rather than from a parameter table. *Charges and per-atom properties* |
| **$q^2$, $R^2_{pred}$, RMSE** | the cross-validated, test-set and fit figures of merit. $q^2$ is not a test: it is what the search optimises, and a model can be chosen for it. *Validation*; *Statistics reported* |

| term | what it means here |
|---|---|
| **selectivity co-field** | the difference of two endpoints' co-fields in a multi-task model, which draws where a molecule should differ to be selective rather than merely potent. *Several endpoints at once* |
| **Shapley value** | the share of a prediction attributed to each field or atom by averaging over all subsets, with the pairwise interaction indices beside it. *Shapley values* |
| **solvent exposure weighting** | multiplying each atomic weight by its relative solvent accessibility. Measured here and found not to help: ligands of this size have no interior. *Weighting the fields by solvent exposure* |
| **surface kernel** | the in-plane invariant kernel for molecules on a surface: distances in the plane, resolved by the heights of both atoms, invariant to lateral position and rotation about the normal and to nothing else. *Molecules on a surface* |
| **topomer field** | a field built from substituents rebuilt in a pose that is a deterministic function of their topology, so the same R-group always lands in the same place and no alignment is needed. *Topomer fields* |
| **y-randomisation** | refitting the whole procedure on shuffled activities many times; if the shuffled models score as well as the real one, the real one is chance. *Is it chance? y-randomisation* |

# Appendix D — Bibliography

References are grouped by what they are to this program: the papers that define the method it implements, those that apply it to one-class classification and virtual screening, and the work it takes methods or benchmark data from. Journal names follow the ACS abbreviations; a DOI is given wherever the work has one.

## D.1 The method


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## D.2 One-class classification and virtual screening


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